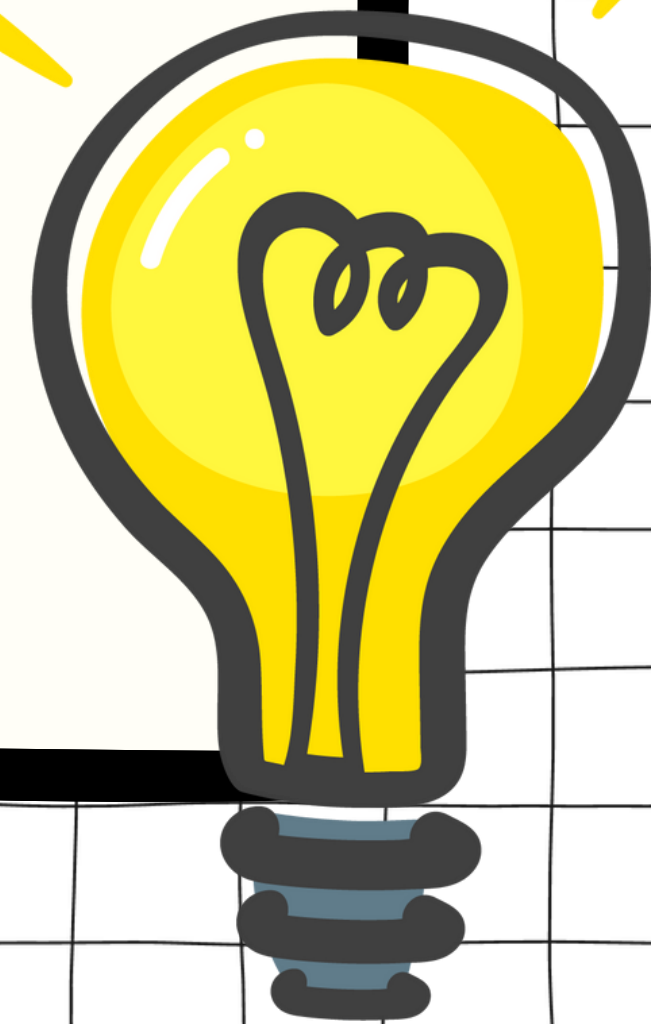


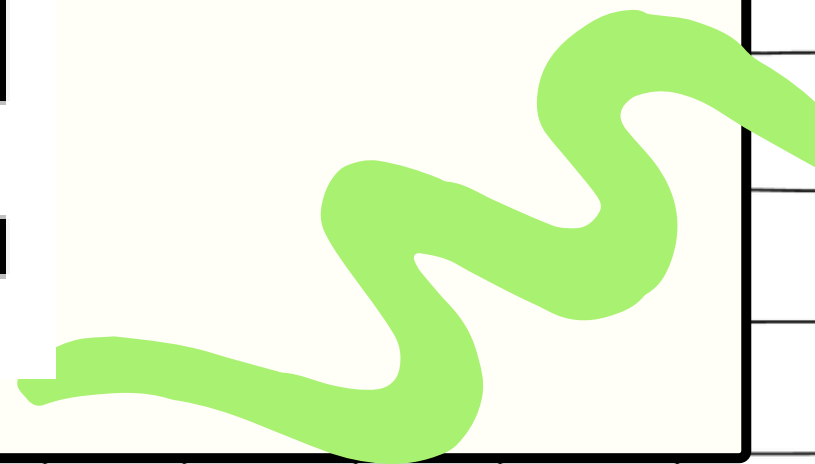
Automation, Instrumentation and Control System

IV. Agricultural and Biosystems Automation, Instrumentation and Control System The registered and licensed Agricultural and Biosystems Engineer can perform the competencies under the following sub-topics:	Percentage Weight and Distribution	Number of Items and Distribution	Easy (30%)		Moderate (50%)	Difficult (20%)		
			Remembering Recall information Recognizing, listing, describing, retrieving, naming, finding	Understanding Explain, Interpret, summarize, examine paraphrase, classify	Applying Using the information in another familiar situation Implementing carrying out, using, executing	Analyzing Understanding, Comparing, organizing, deconstructing, interrogating, finding	Evaluating hypothesizing, critiquing, experimenting, judging	Creating Generating Designing, constructing, planning, producing, inventing
			32%	100				
	4	12						
1. Apply agricultural and biosystems sensors such as, but is not limited to, computer models for control and automation in the agricultural and fishery production industry, biological systems, automated controls, precision farming systems, agricultural safety and controlled-environment agriculture.	2	6		1	4			1
2. Apply metrology equipment such as weighing scale and other metrology equipment for agriculture and biosystems engineering sector.	2	6		2	4			

Information and Communication Technology



Scan me



Pre-Test # 1



It deals with the development and testing of software and software systems.

- a. Computer Science**
- b. Information and Communication Technology**
- c. Information Technology**
- d. All of the above**

Computer Science

It deals with the development and testing of software and software systems.

Information Technology

It deals with the implementation, support, and management of computers and information systems.

Information and Communication Technology

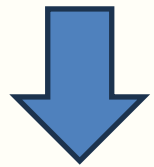
It has all aspects of IT along with it focuses on Communication sector.

ICT includes:

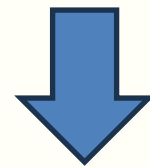
- Capturing technologies (*camcorders*)
- Storage technologies (*CD-ROMs, DVDs, Pen drives, etc.*)
- Processing technologies (*application software*)
- Communication technologies (*local area networks*)
- Display technologies (*computer monitors, LCDs*)

Types of ICT Tools

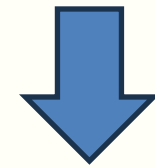
Traditional ICTs



Radio



Television

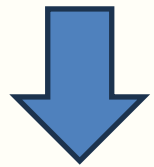


Print Media

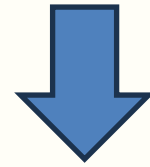


Types of ICT Tools

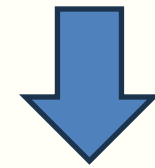
Modern ICTs



Internet



Portal



Mobile



Pre-Test # 2

...

It refers to the physical and visible components of the system such as a monitor, CPU, keyboard and mouse.

- a. Commands**
- b. Hardware**
- c. Peopleware**
- d. Software**

HARDWARE



VS

SOFTWARE



It refers to the **physical and visible components** of the system.

It tells the hardware what to do and how to do it

HARDWARE vs SOFTWARE

physical elements of a computer or electronic system

has four main categories: input devices, output devices, secondary storage devices and internal components

tangible

developed using electronic and other materials

when damaged, it can be replaced with a new component

starts functioning once the software is loaded into the system

a collection of instructions that tell the computer how to perform a task

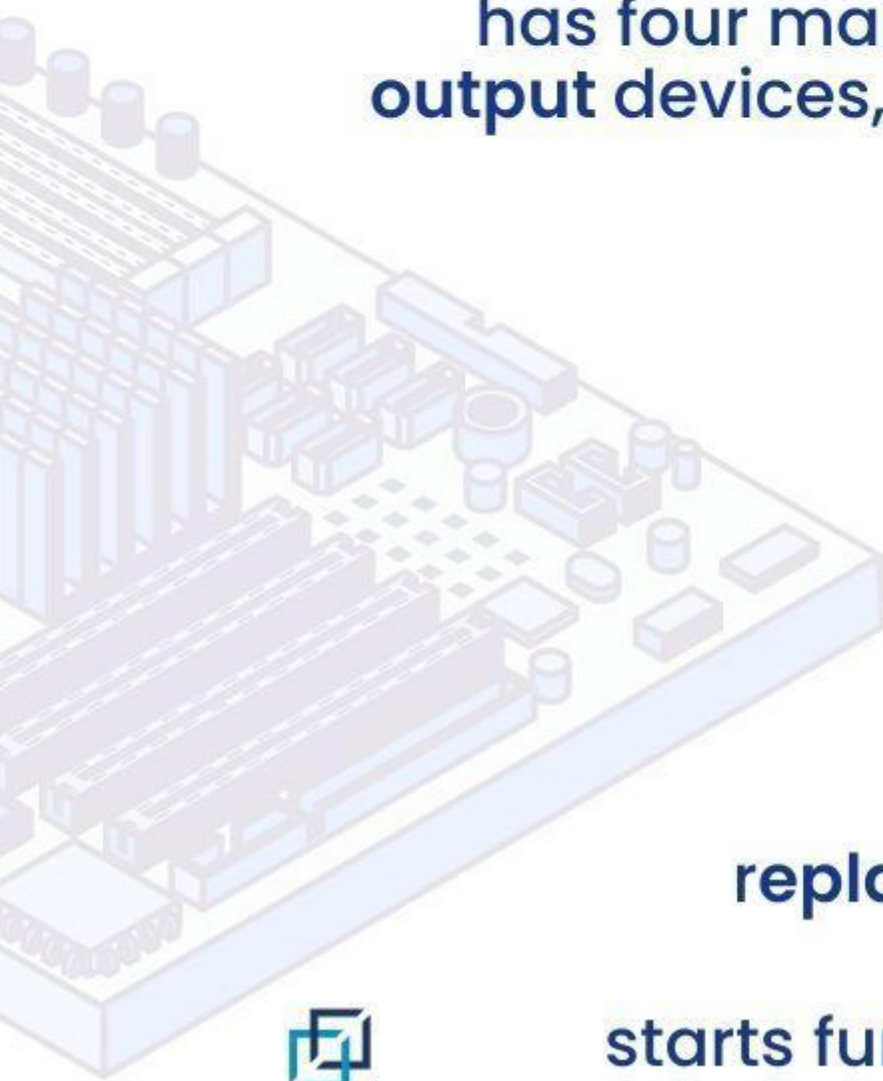
mainly divided into system software and application software

intangible

developed by writing instruction using a programming language

when damaged, it can be reinstall using a backup copy

should be installed into the computer to function

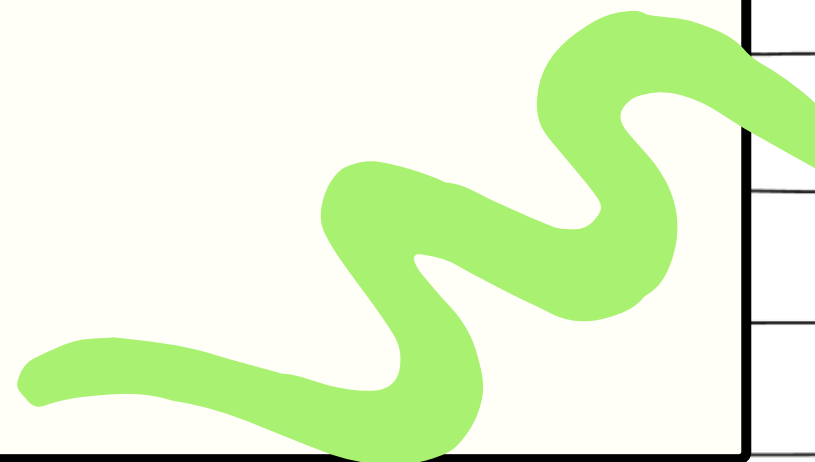


Pre-Test # 3



Hardware is categorized into four main categories: input devices, output devices, processing devices, and storage devices. Which of the following is not a peripheral device?

- a. input devices**
- b. output device**
- c. processing device**
- d. storage device**



Four categories of Hardware

Input devices : for raw data input

Processing devices: To process raw data instructions into information

Output devices: To disseminate data and information

Storage devices: For data and information retention

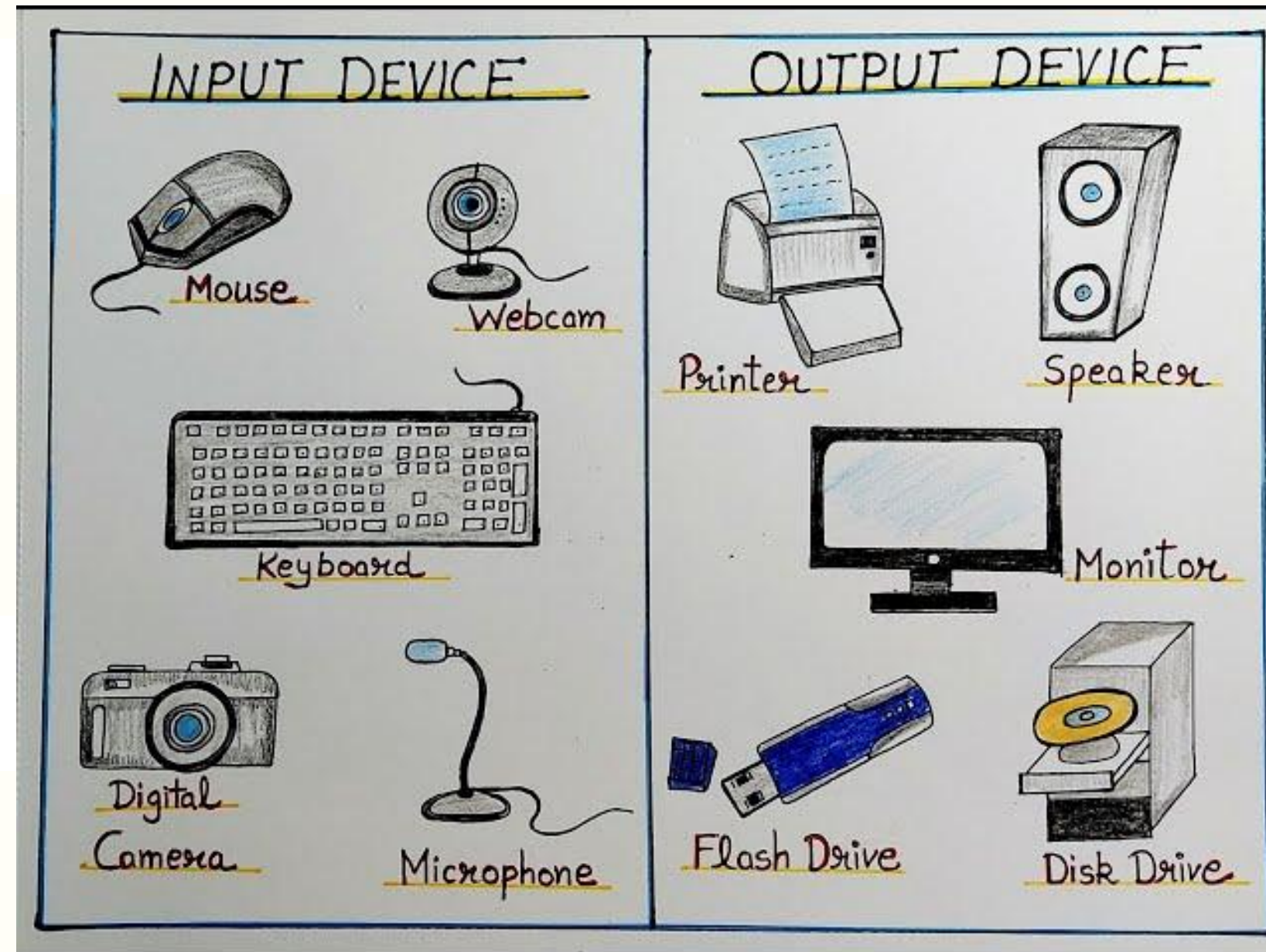
Pre-Test # 4



Which of the following is an output device?

- a. microphone**
- b. mouse**
- c. scanner**
- d. speaker**

Input devices are used to enter data or information into a computer.



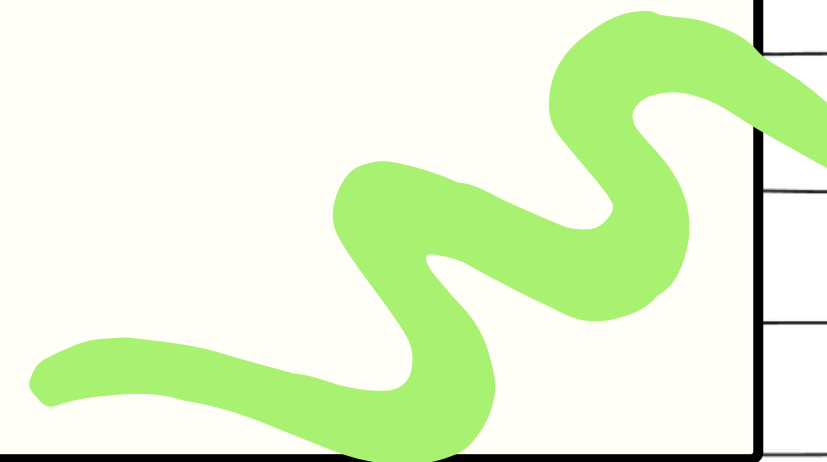
Output devices are used to present processed data or information from a computer.

Pre-Test # 5



It acts as the control center and brain of a computer, responsible for executing instructions and performing calculations.

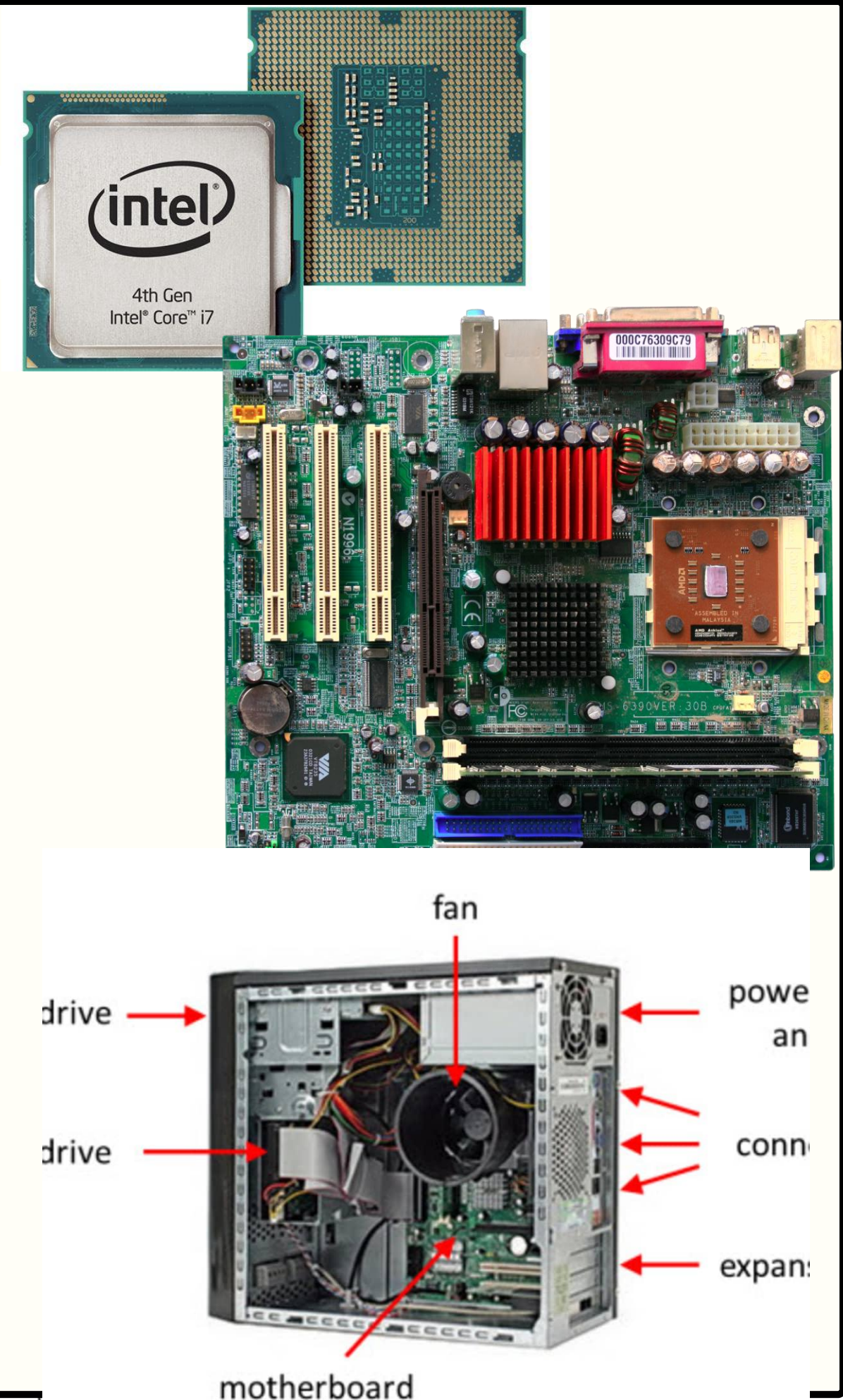
- a. Central Processing Unit**
- b. Motherboard**
- c. Processing Unit**
- d. System Unit**



The **CPU** interprets, processes and executes instructions from the hardware and software programs running on the device.

The **motherboard** is the “backbone” of the computer responsible for communication between components and transmission of information

A **system unit** is the part of a computer that houses the primary devices that perform operations and produce results for complex calculations.



Pre-Test # 6

...

It refers to a memory that cannot be change by a program or user. It retains its memory even after the computer is turned off.

- a. Cache**
- b. RAM**
- c. ROM**
- d. SSD**

Types of Memory

Random Access Memory is memory that stores the data that you're currently working with, but it's volatile, meaning that as soon as it loses power, that data disappears.

Read-Only Memory refers to permanent memory. It's non-volatile, so when it loses power, the data remains.

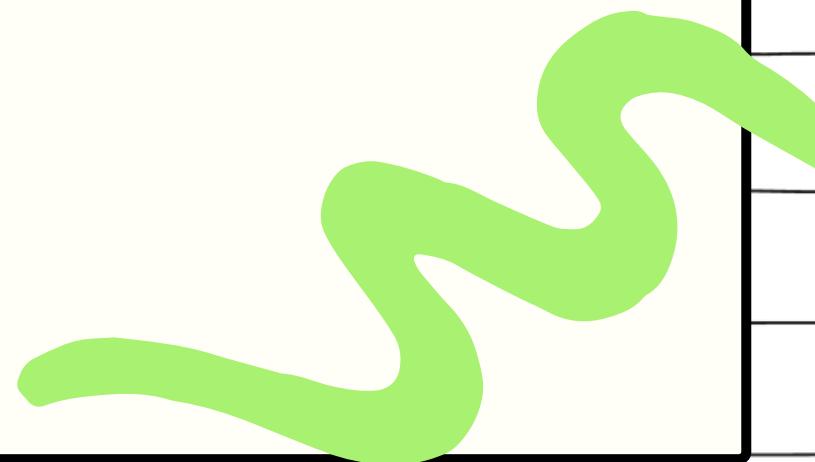
Cache refers to small-capacity memory which allows quick access to data.

Pre-Test # 7



This port allows devices to communicate without a physical connection.

- a. HDMI Port**
- b. Infrared Port**
- c. Network Port**
- d. VGA Port**



Port

is a hardware piece on a computer where an electrical connector can be plugged to link the device to external devices

All Types of Computer Ports

www.eTechnophiles.com



HDMI



Modem



Display



VGA



Ethernet



USB-A



DVI



PS/2



Audio



POWER LINE



USB-B



SERIAL



FIREWIRE



eSATA



GAME



USB-C



PARALLEL



MINI DISPLAY



S VIDEO



THUNDERBOLT

Types of Port

USB: connect all kinds of external USB devices, like external hard disk, mouse, keyboard, etc.

Serial port: aka a communication port, used for connection of external devices like a modem, mouse, or keyboard (basically in older PCs).

Parallel ports: used for connecting scanners and printers. It can send several bits at the same time as it uses parallel communication.

Types of Port

Video Graphic Array (VGA ports): connect the monitor to a computer's video card.

Sockets: connects microphones and speakers to the sound card of the computer.

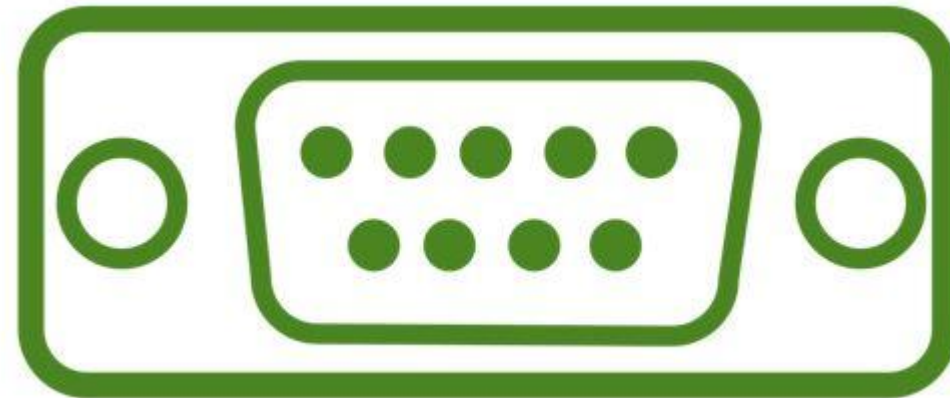
Infrared port: wireless communication interface that is used to send and receive infrared signals from other devices ranging 5-10ft.

High-Definition Multimedia Interface (HDMI) ports: most frequently used HD signal for transferring both high-definition audio and video over a single cable.

Types of Port



USB Port



Serial Port



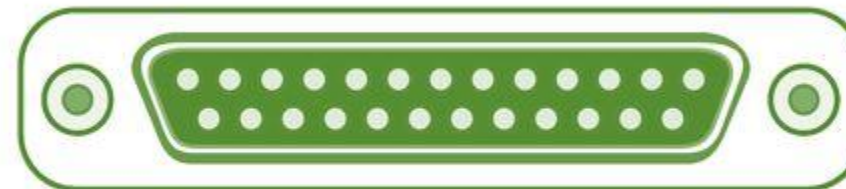
Sockets



Infrared Port



VGA Port



Parallel Port



HDMI Port

Pre-Test # 8

...

A type of proprietary software that is initially shared by the owner for trial use at little or no cost.

- a. Freeware**
- b. Open ware**
- c. Open source**
- d. Shareware**

Freeware: software that requires no paid licenses, no restrictions on how many times you can download or open the program, and no expiration date.

Shareware: software has limited functionality or incomplete documentation until the user sends payment to the software developer.

Opensource: any program whose source code is made available for use or modification as users or other developers see fit



Freeware



Shareware



Freeware



Shareware



Opensource



Opensource

Other Terms to Note

Malware: any program or file that is intentionally harmful to a computer, network or server.

Copyright: type of intellectual property that protects original works of authorship as soon as an author fixes the work in a tangible form of expression.

Product Identification Number: number that comes with legally purchased software as an authenticity validation

Pre-Test # 9

...

A small-scale network that revolves around one person or device.

- a. CAN**
- b. LAN**
- c. PAN**
- d. WAN**

Types of Network

Local Area Network: allows users to connect within a short distance in a common area.

Personal Area Network: a small-scale network that revolves around one person or device.

Campus Area Network: a group of interconnected local area networks operating within a limited geographical area.

Wide Area Network: a larger network that covers a city or region

Types of Network

Internet: a world network of connected computers, connected through WAN and LAN

Intranet: a small network, closed-version of internet where only certain people (authorized) are granted access (members of the organization.)

Extranet: a controlled private network that allows access to authorized set of customers – normally to a subset of the information accessible from an organization's intranet.

World Wide Web: sites all over the world that are connected by links between them (hyperlinks) forming a web

Pre-Test # 10

...

These software programs are designed to run a computer's application programs and hardware.

- a. Application software**
- b. Driver software**
- c. Middleware**
- d. System Software**

Categories of Software



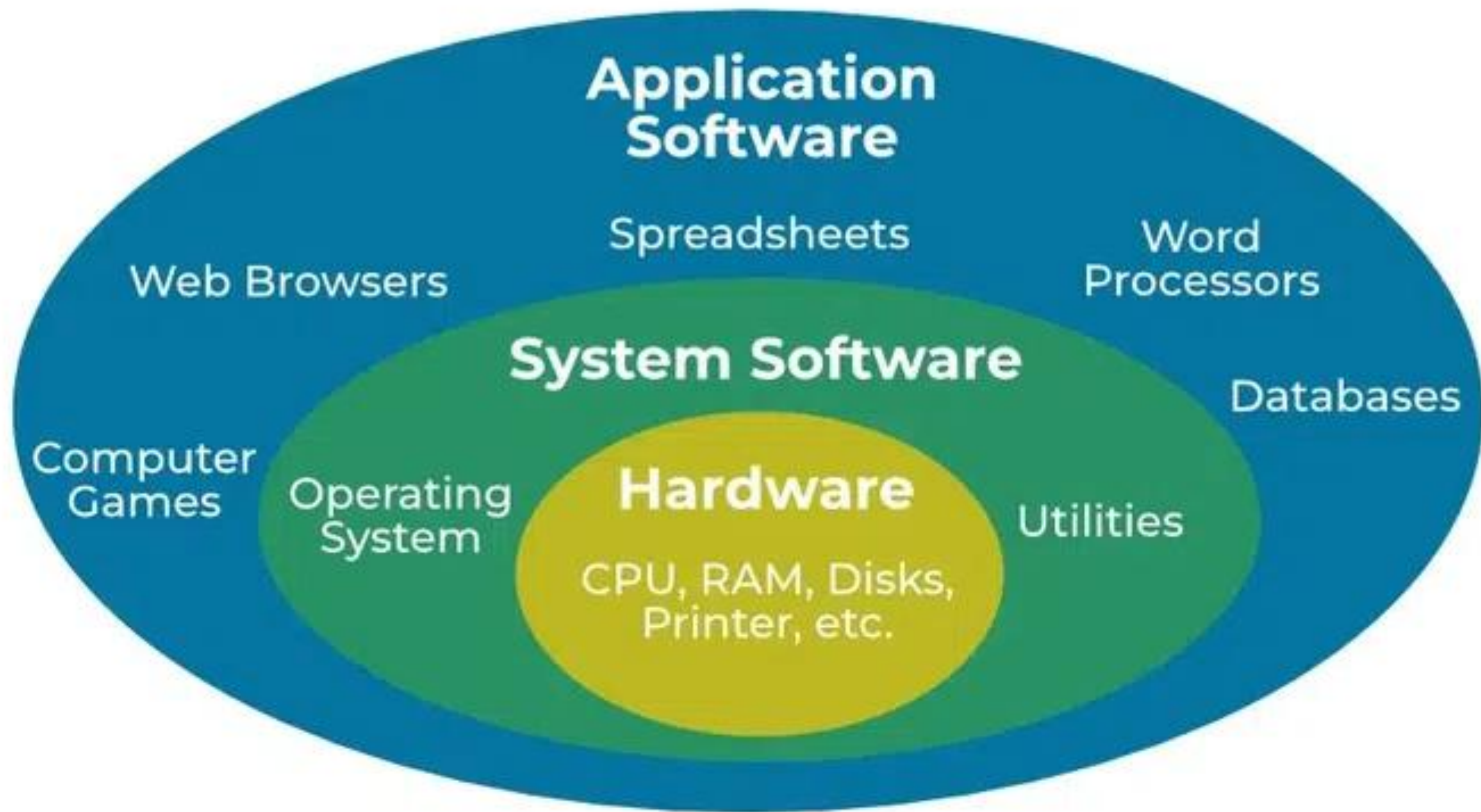
Application Software

fulfills a specific need or performs tasks



System Software

is designed to run a computer's hardware and provides a platform for applications to run on top of.



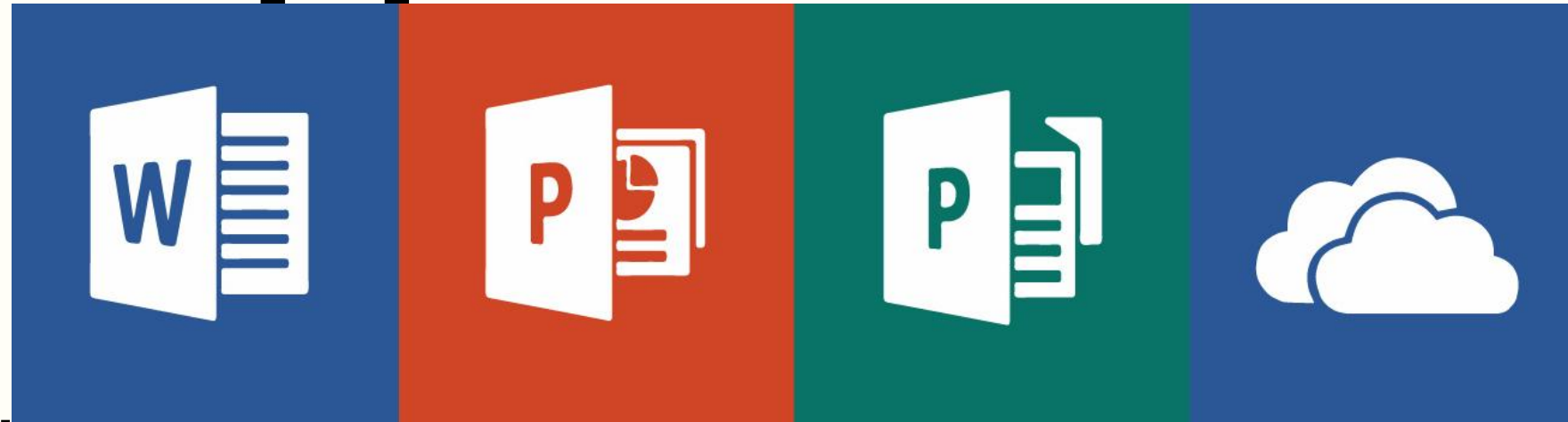
Pre-Test # 11

...

This Microsoft application provides real-time collaboration and file sharing with colleagues.

- a. Access**
- b. Onedrive**
- c. Outlook**
- d. Teams**

Microsoft Applications



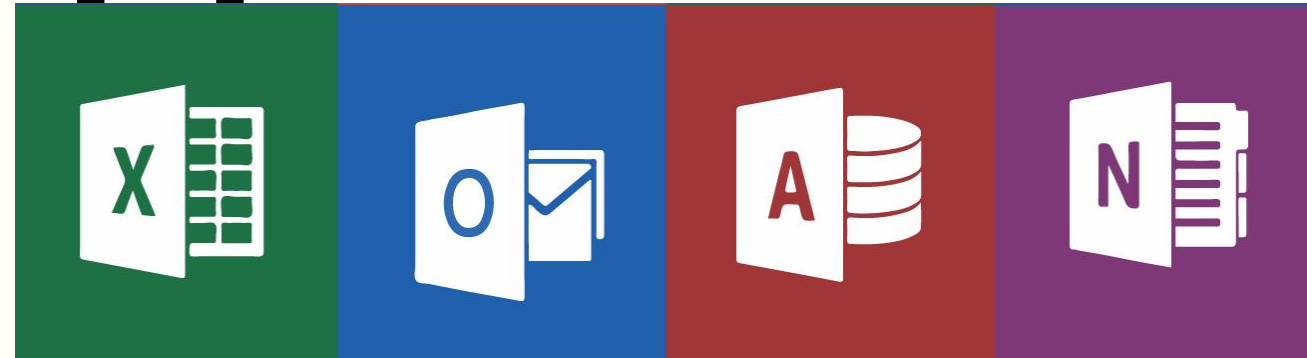
OneDrive provides cloud-based storage for important documents and files.

Outlook provides tools for email communication, calendaring, contacts and task management.

Teams provides real-time collaboration and file sharing with colleagues.

Word provides tools for creating and sharing professional word processing documents.

Microsoft Applications



Excel provides tools for creating spreadsheets, charts, and graphs.

PowerPoint provides tools for creating presentations.

Access provides tools for creation and maintenance of records in database form.

OneNote provides tools for free-form note taking and information gathering.

Visio provides tools to create professional diagrams such as business process maps and technical diagrams.

Pre-Test # 12

...

In Microsoft Excel, this function is used to count the number of cells within a range that meet multiple criteria.

- a. COUNTIF**
- b. COUNTIFS**
- c. DCOUNT**
- d. DCOUNTA**

Microsoft Excel is a spreadsheet program that allows users to organize, format, and calculate data in a spreadsheet.



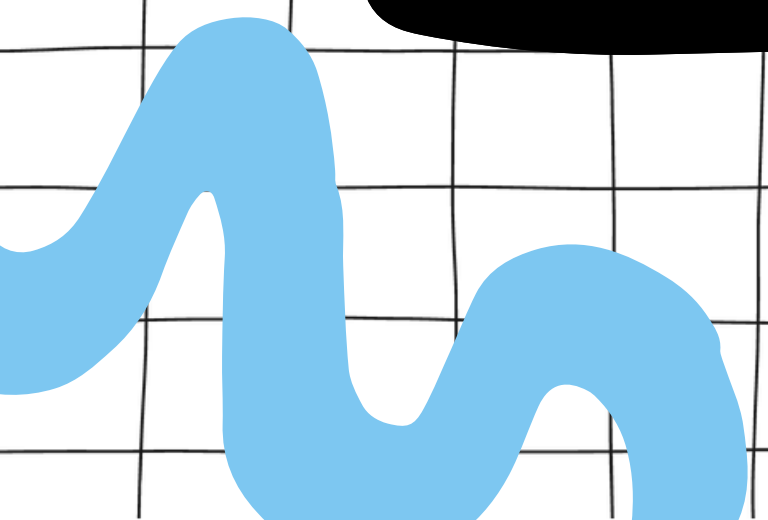
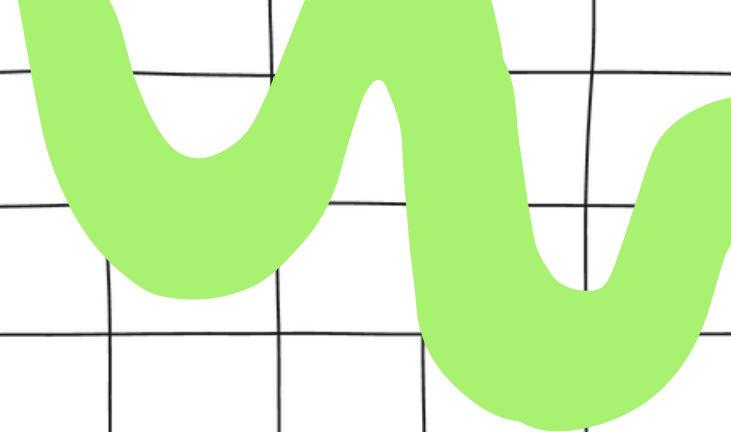
COUNTIF function

Statistical: Counts the number of cells within a range that meet the given criteria

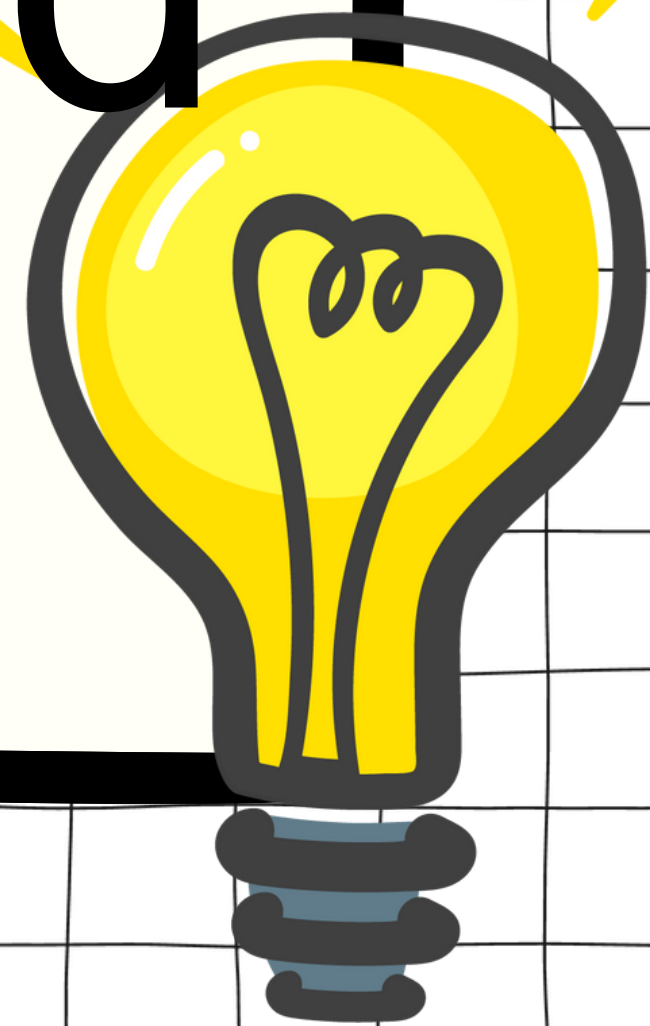
COUNTIFS function

Statistical: Counts the number of cells within a range that meet multiple criteria

<https://support.microsoft.com/en-us/office/excel-functions-alphabetical-b3944572-255d-4efb-bb96-c6d90033e188>



Computer-Aided Design



Pre-Test # 13

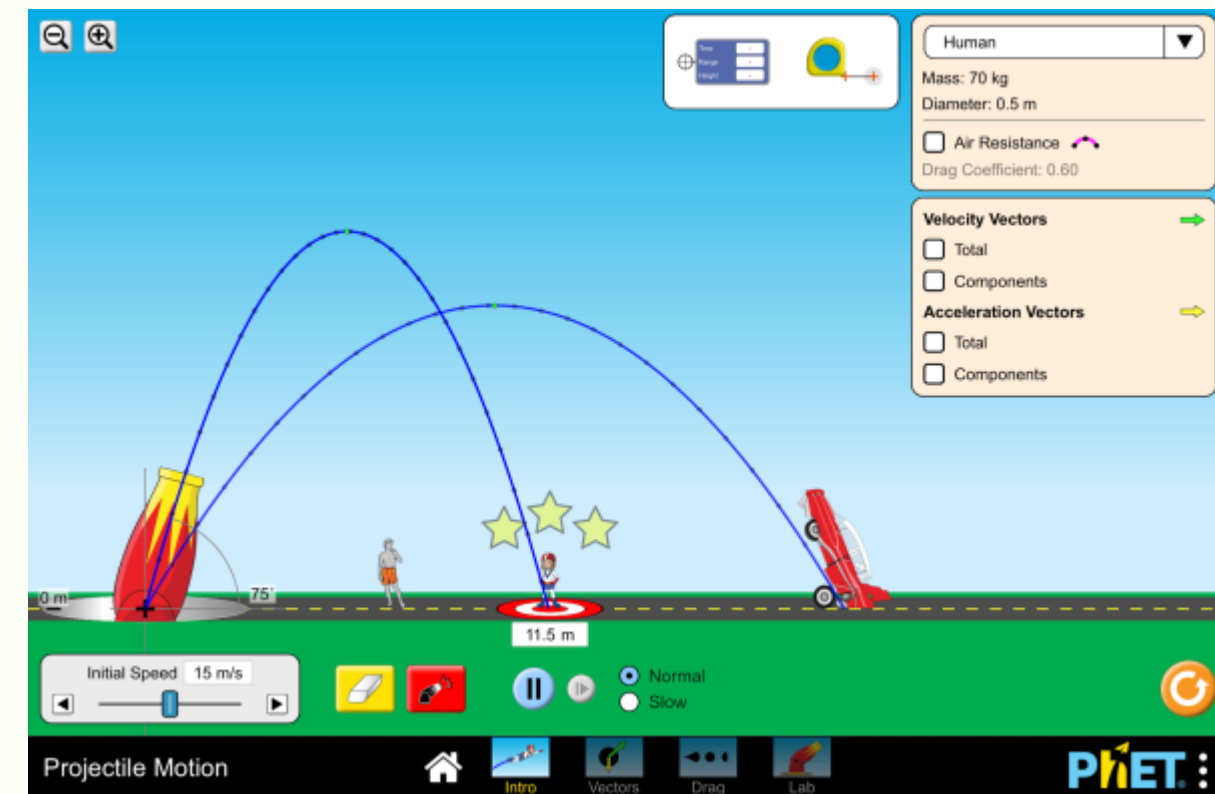
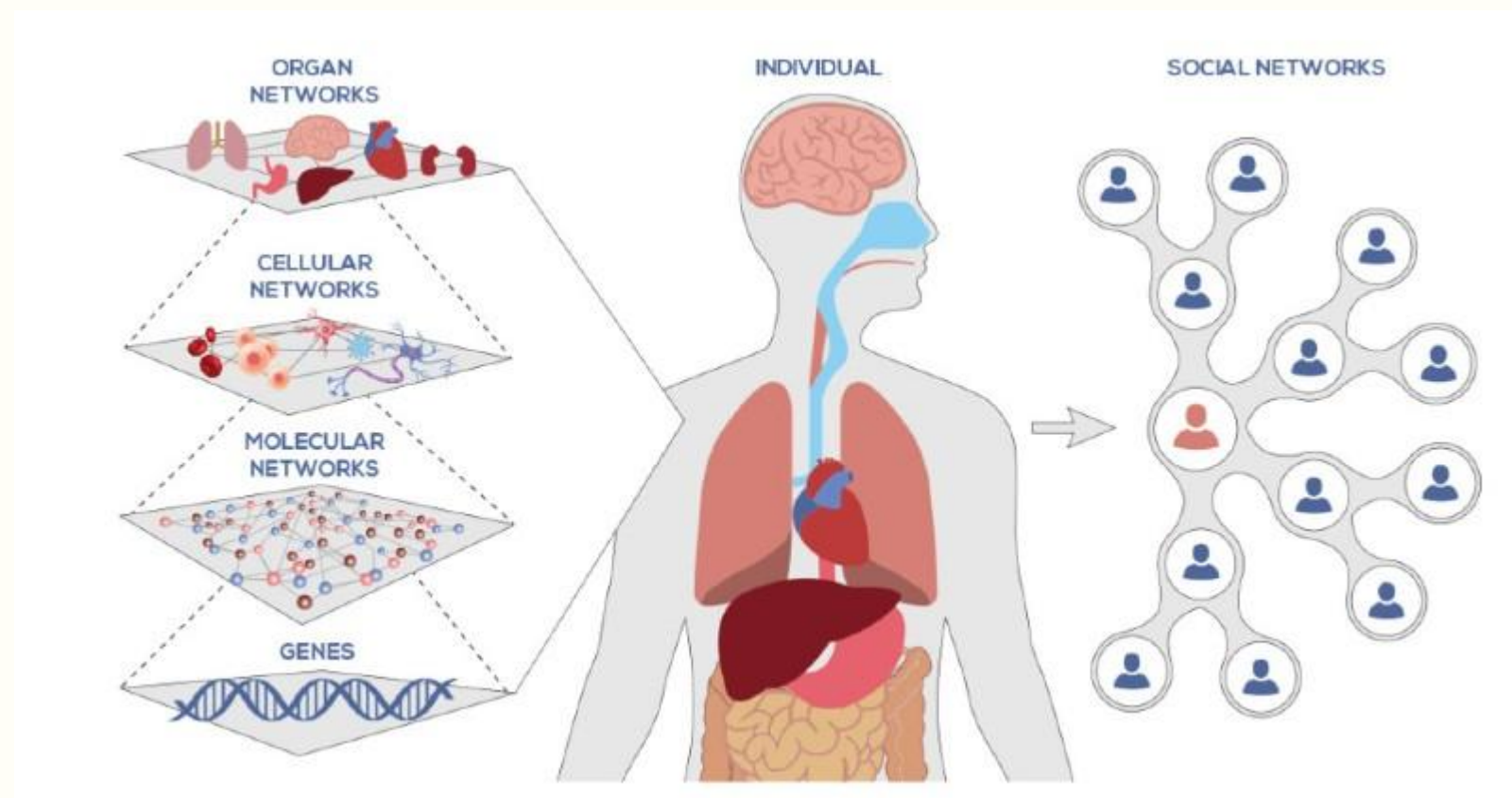
...

Which of the following is not a type of computer model?

- a. Climate models**
- b. Crop Simulation models**
- c. Dynamic models**
- d. None of the above**

Computer Modeling

A **computer model** is a translation of objects or phenomena from the real world into mathematical equations.



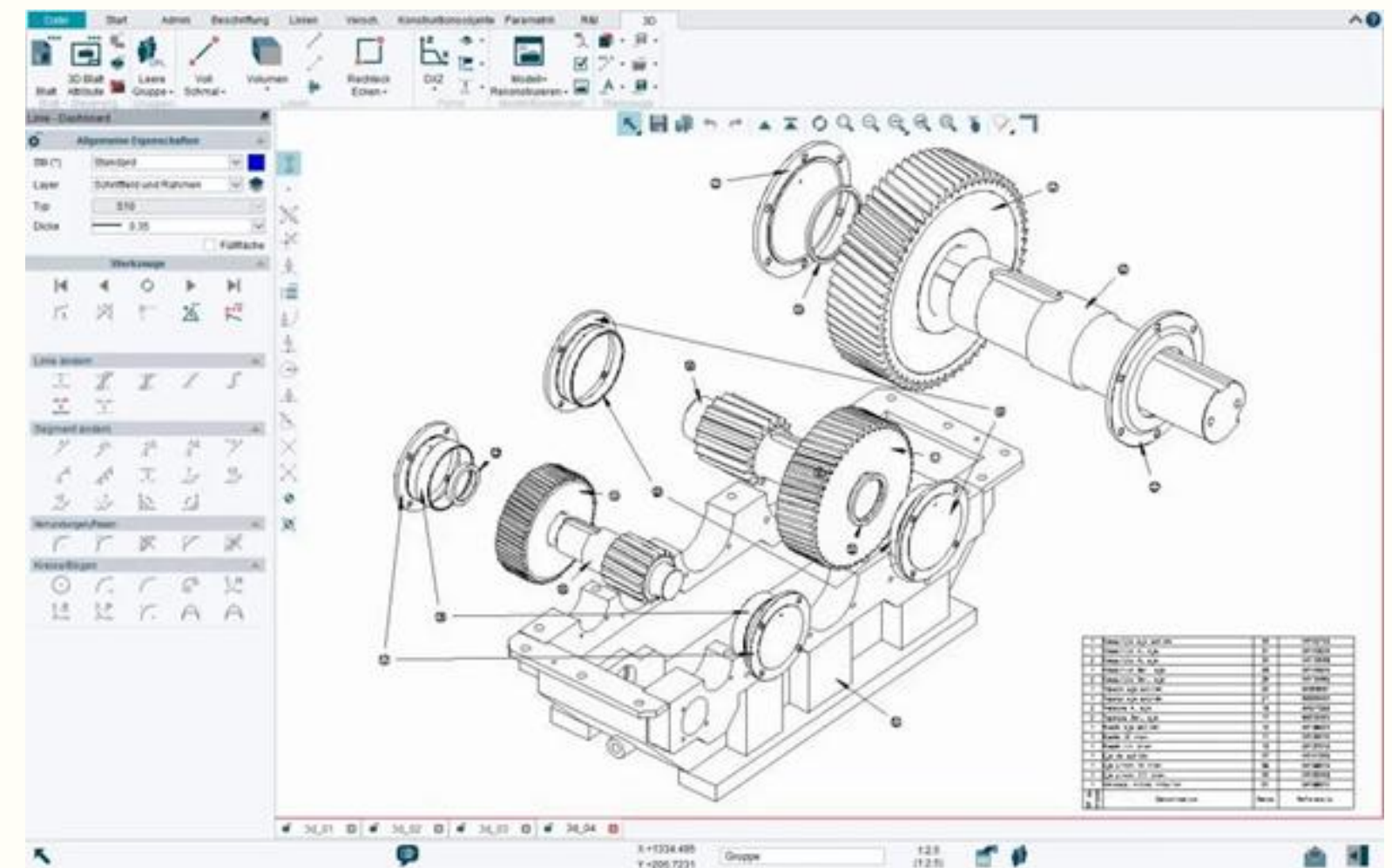
Computer simulation is the use of a computer to represent the dynamic responses of one system by the behavior of another system modeled after it.

What is CAD?

The process of digitally creating design simulations of real-world products in 2D or 3D, complete with scale, precision, and physics properties, to optimize and perfect the design before manufacturing.

Advantages

- To draw to scale
- Layout the drawing conveniently
- Flexible to organize drawing information
- Establish drafting standards



Pre-Test # 14



Which of the following is an opensource CAD software?

- a. AutoCAD**
- b. Microstation**
- c. ProgeCAD**
- d. QCAD**

What is CAD?

- Most popular CAD software like AutoCad, ProgeCAD, Microstation are **high-priced** for individuals. Alternatively, individuals can try free opensource CAD drafting software QCAD, LibreCAD and OpenSCAD.
- Every new release of the CAD software, operator has to **update their skills**.
- Improper use of blocks and layers make updating and modification of the drawings a cumbersome task for another person

Pre-Test # 15

...

A system that locates points in space according to distances from three intersecting axes.

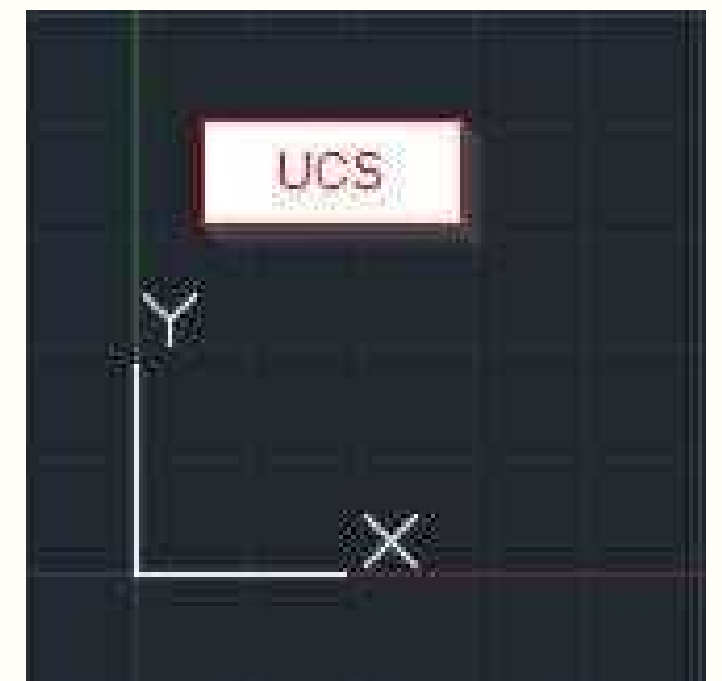
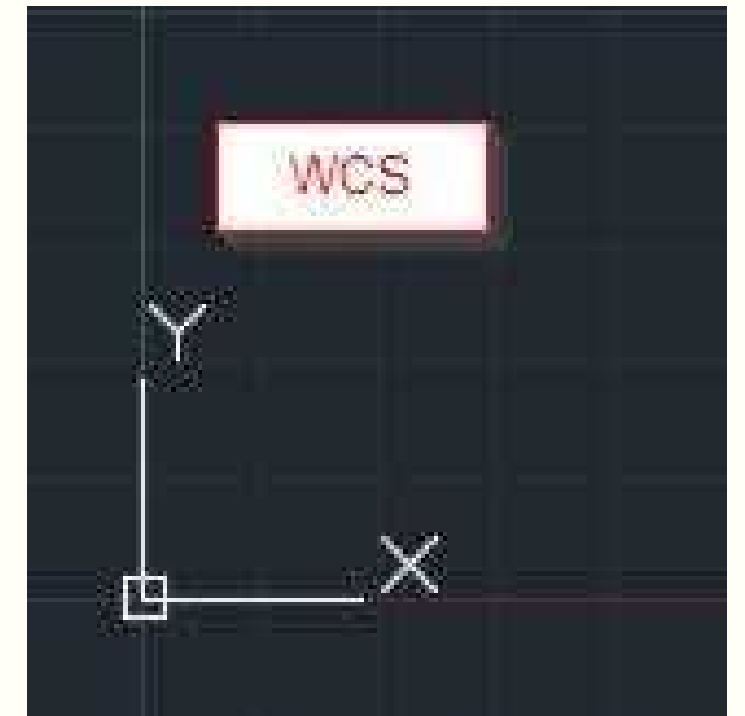
- a. Cartesian Coordinate System**
- b. Universal Coordinate System**
- c. User Coordinate System**
- d. World Coordinate System**

Basic Terms

World Coordinate System (WCS): is the imaginary plane that is parallel to the ground. It is the default coordinate system.

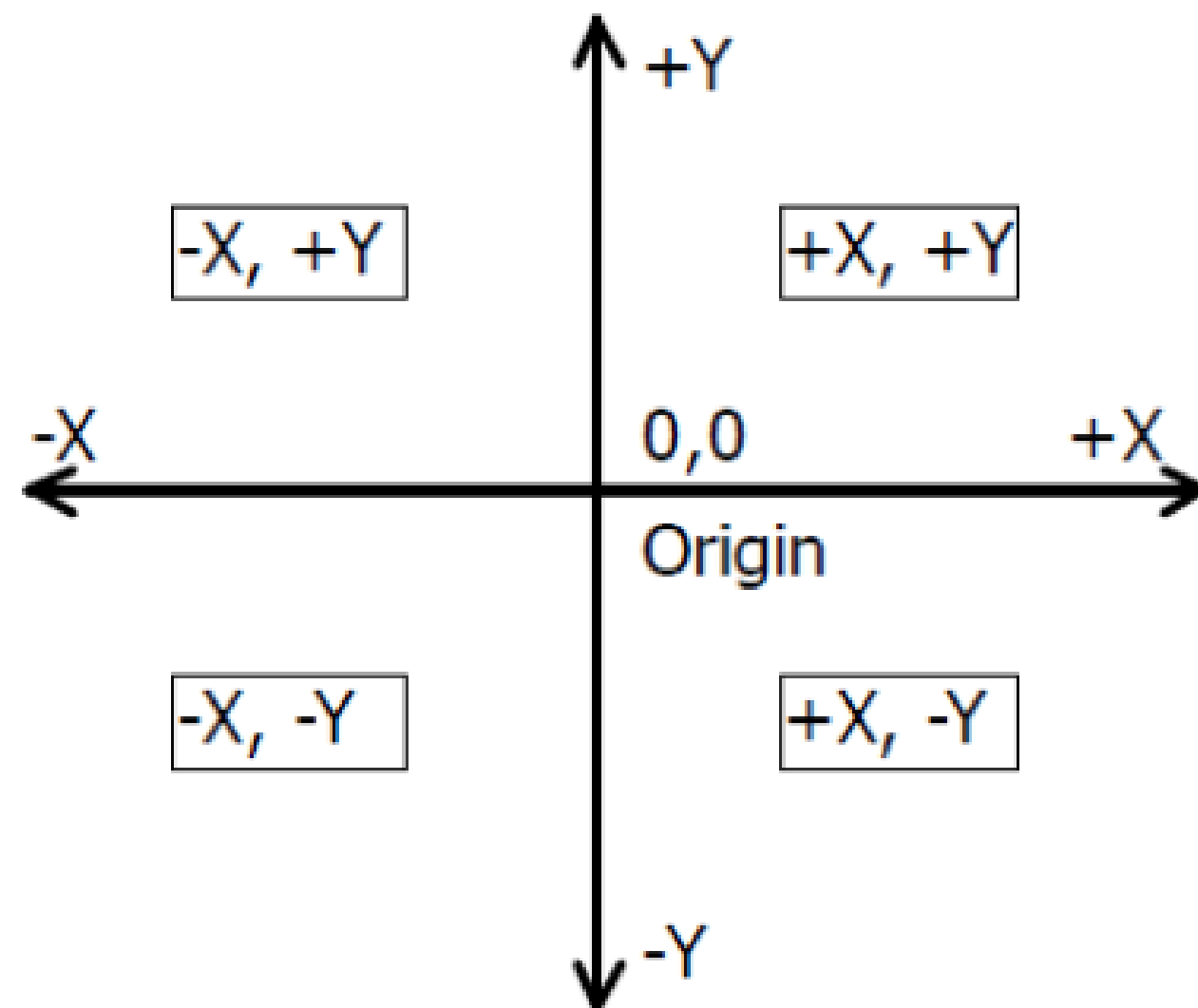
Modifications made to the World Coordinate System (WCS) result in a **User Coordinate System (UCS)**. It is the plane that you work on. It enables the user to draw 3 dimensional objects.

Cartesian Coordinate System: A system that locates points in space according to distances from three intersecting axes



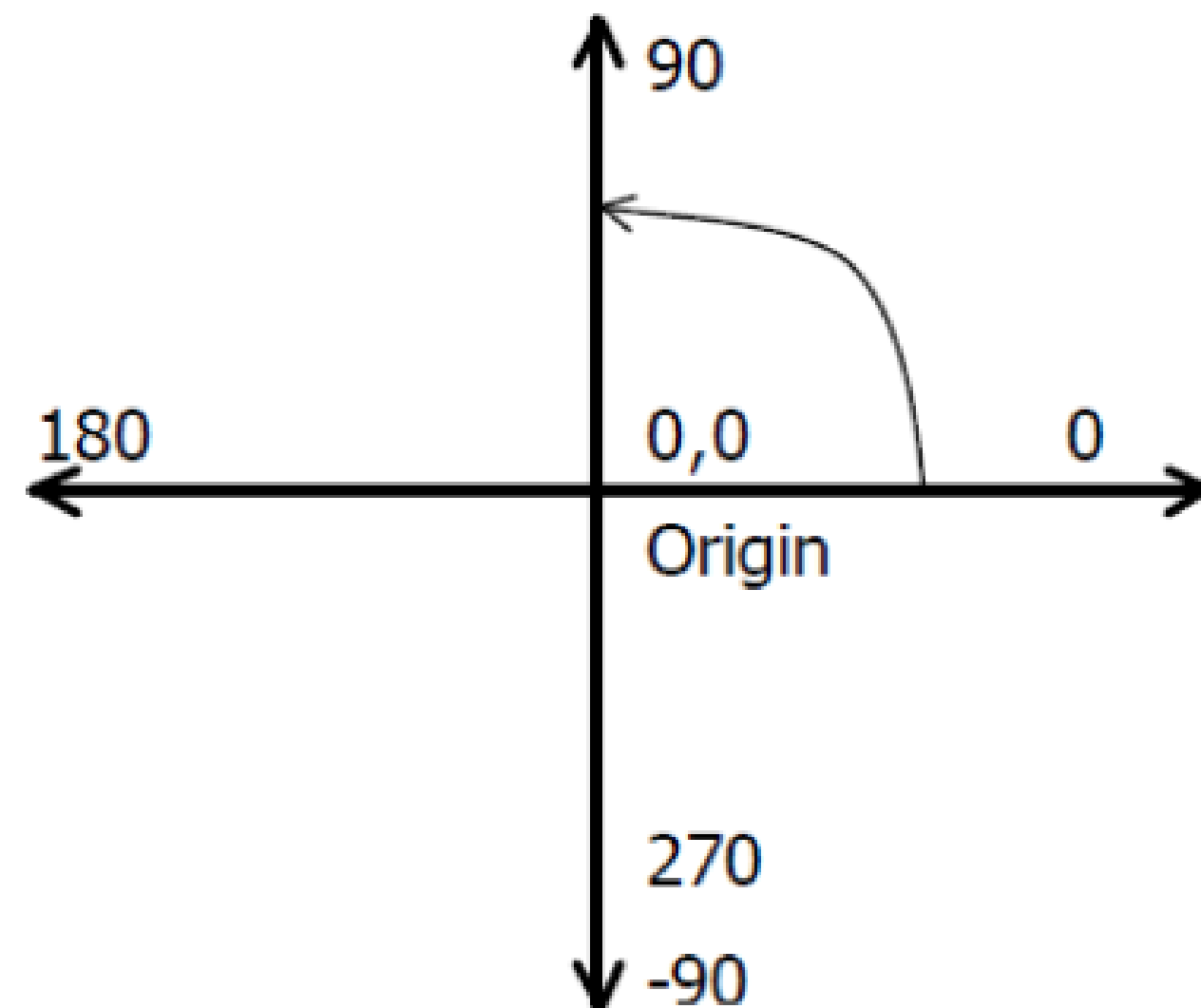
Cartesian Coordinate System

Locations are measured along the x axis (horizontal) and y axis (vertical).



Polar Coordinate System

Locations are measured as distance and angle in degrees, with 360 degrees in all.



Absolute coordinates are measured from the origin or 0,0 point.
Relative coordinates are measured from the last defined point.

Conversion Between the Two:

From Polar to Cartesian:

$$x = r \cos(\theta), \quad y = r \sin(\theta)$$

From Cartesian to Polar:

$$r = \sqrt{x^2 + y^2}, \quad \theta = \tan^{-1} \left(\frac{y}{x} \right)$$

Example:

Polar coordinates:

$$(r, \theta) = (5, 53.13^\circ)$$

Cartesian coordinates:

$$(x, y) = (3, 4)$$

Pre-Test # 16

...

A common file format recognized by other CADD systems.

- a. Drawing Exchange File**
- b. Drawing File Format**
- c. Drawing Standard File**
- d. Drawing Template**

Basic Terms

Defpoints: The points used to specify the dimension location and the center point of the dimension text.

Drawing exchange file (DXF): A common file format recognized by other CADD systems

Drawing Standards file (DWS): A file used to check the standards of another file using AutoCAD standards checking tools

Drawing template (DWT): A file that contains standard drawing settings and objects for use in new drawings

Pre-Test # 17

...

In AutoCAD, the one-key shortcut “E” enables the user to:

- a. Edit**
- b. Ellipse**
- c. Erase**
- d. Extend**

AutoCAD one-key shortcuts



Pre-Test # 18

...

In AutoCAD, what shortcut key should you press to open the Design Center Palette?

- a. Ctrl + 1**
- b. Ctrl + 2**
- c. Ctrl + 3**
- d. Ctrl + 4**

AutoCAD shortcuts

Ctrl+0 (zero) Clean Screen

Ctrl+1 Property Palette

Ctrl+2 Design Center Palette

Ctrl+3 Tool Palette

Ctrl+4 Sheet Set Palette

Ctrl+6 DBConnect Manager

Ctrl+7 Markup Set Manager Palette

Ctrl+8 Quick Calc

Ctrl+9 Command Line

F1 Display Help

F2 Toggle text screen

F3 Toggle object snap mode

F4 Toggle 3DOsnap

F5 Toggle Isoplane

F6 Toggle Dynamic UCS

F7 Toggle grid mode

F8 Toggle ortho mode

F9 Toggle snap mode

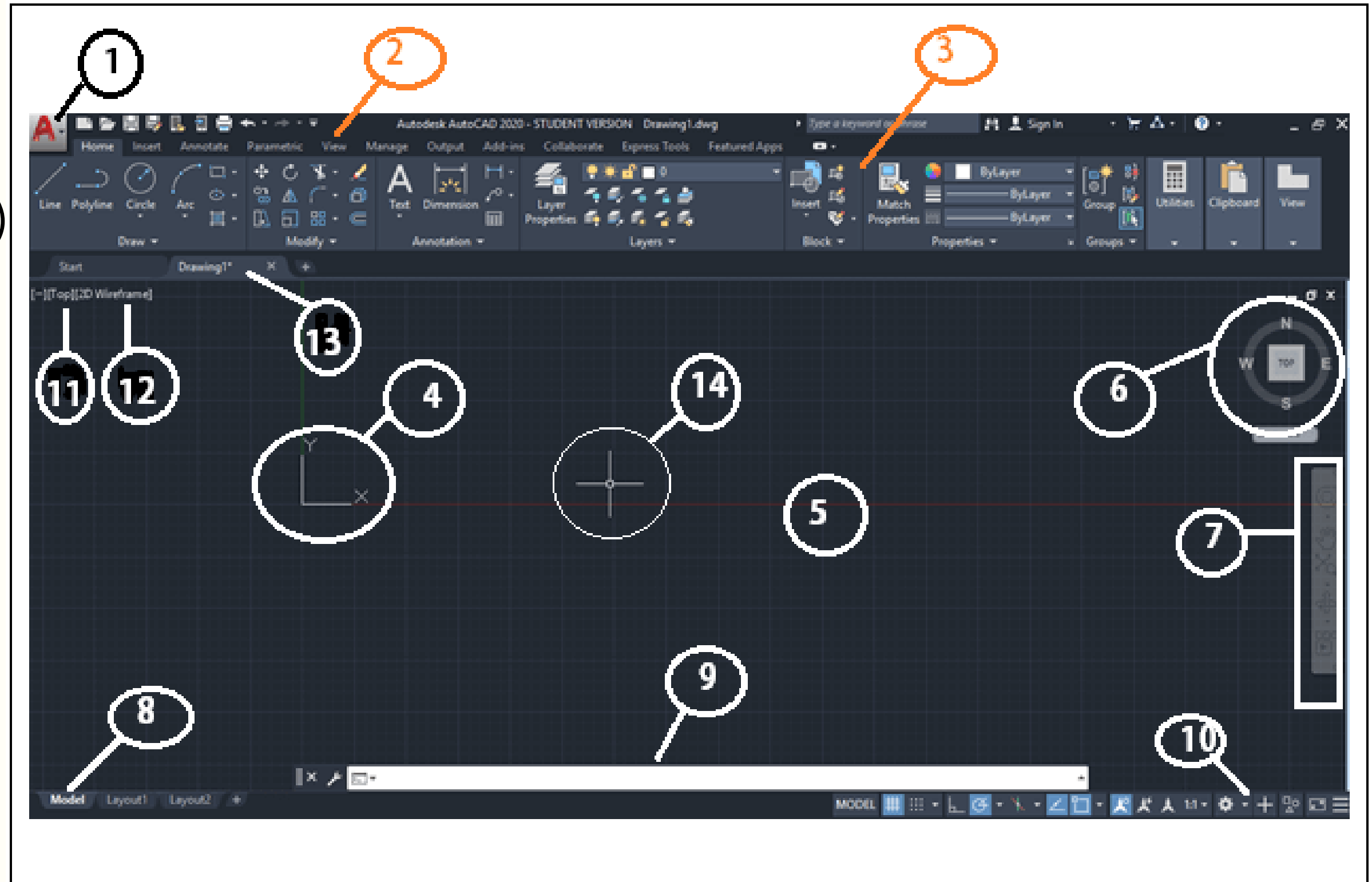
F10 Toggle polar mode

F11 Toggle object snap tracking

F12 Toggle dynamic input mode

AutoCAD interface and ribbons

1. Application menu/button
2. Quick Access Toolbar
3. Ribbon Panel
4. User Coordinate System (UCS)
5. Model Space / Work Space / Drawing Window
6. View Cube
7. Navigation Bar
8. Model Layout Tab
9. Command Line Window
10. Status Bar
11. View Control
12. Visual Style Control
13. File Tab
14. Mouse Cursor



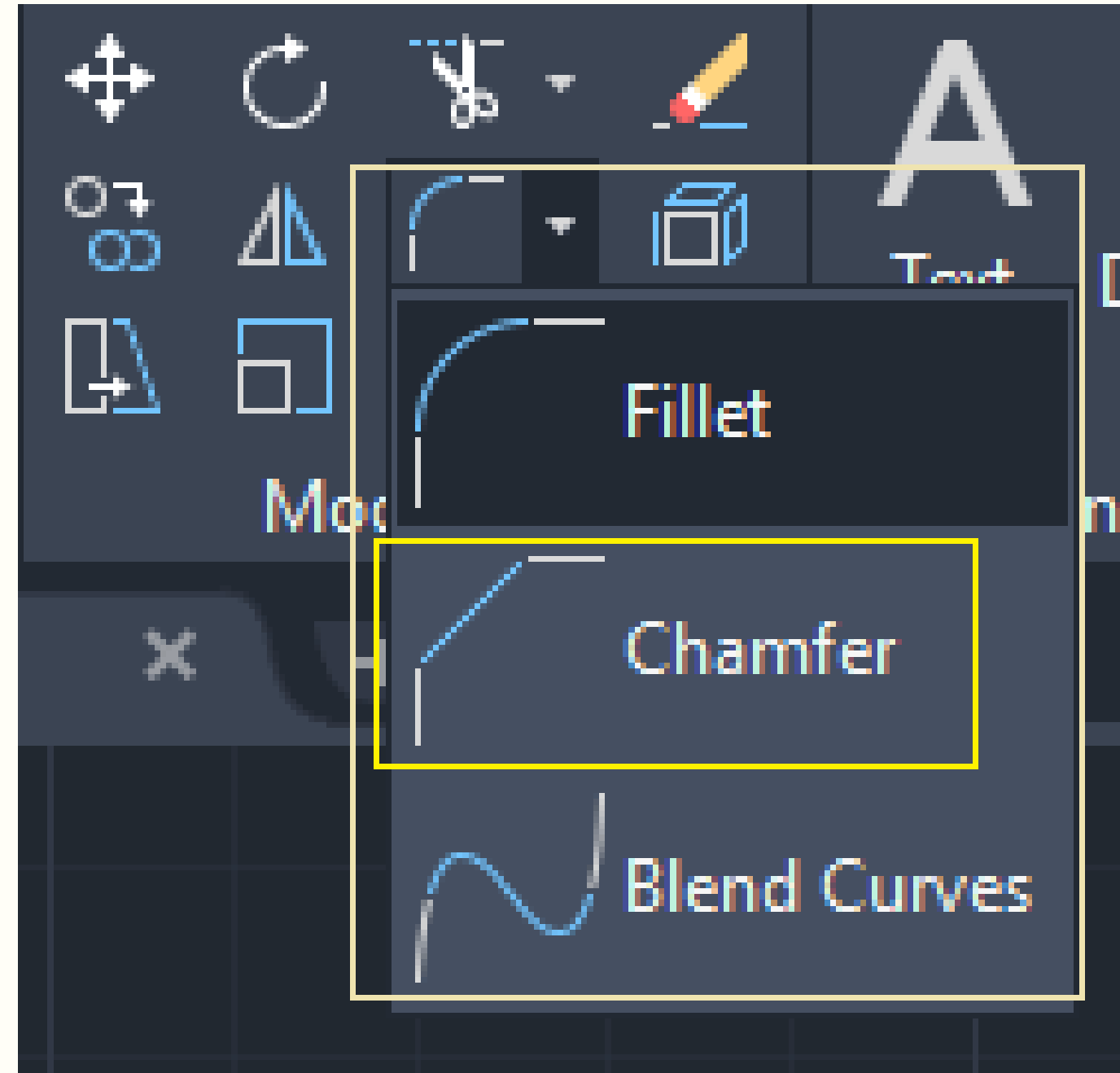
Pre-Test # 19

...

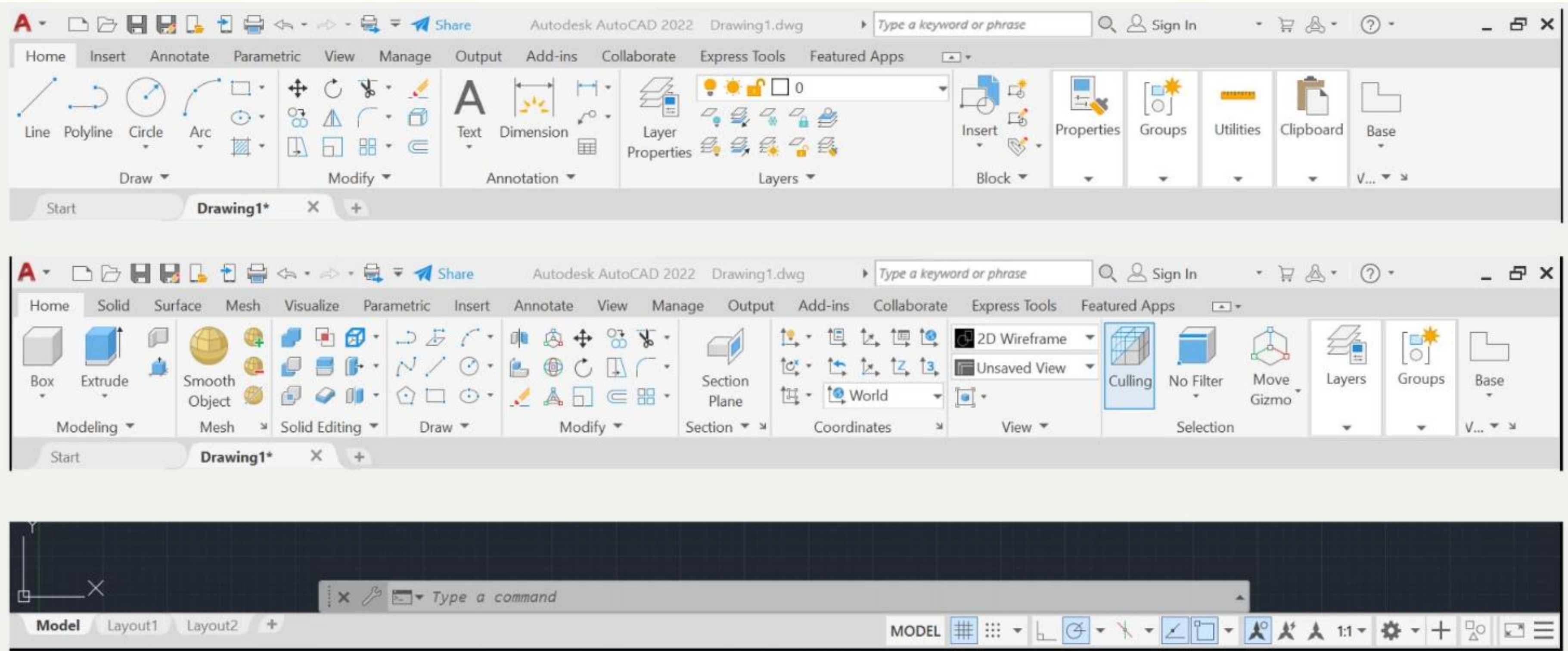
In AutoCAD, what shortcut key should you press to create slanted edges?

- a. CHA**
- b. FIL**
- c. SLA**
- d. SLAE**

Some AutoCAD ribbons



Some AutoCAD ribbons



Pre-Test # 20

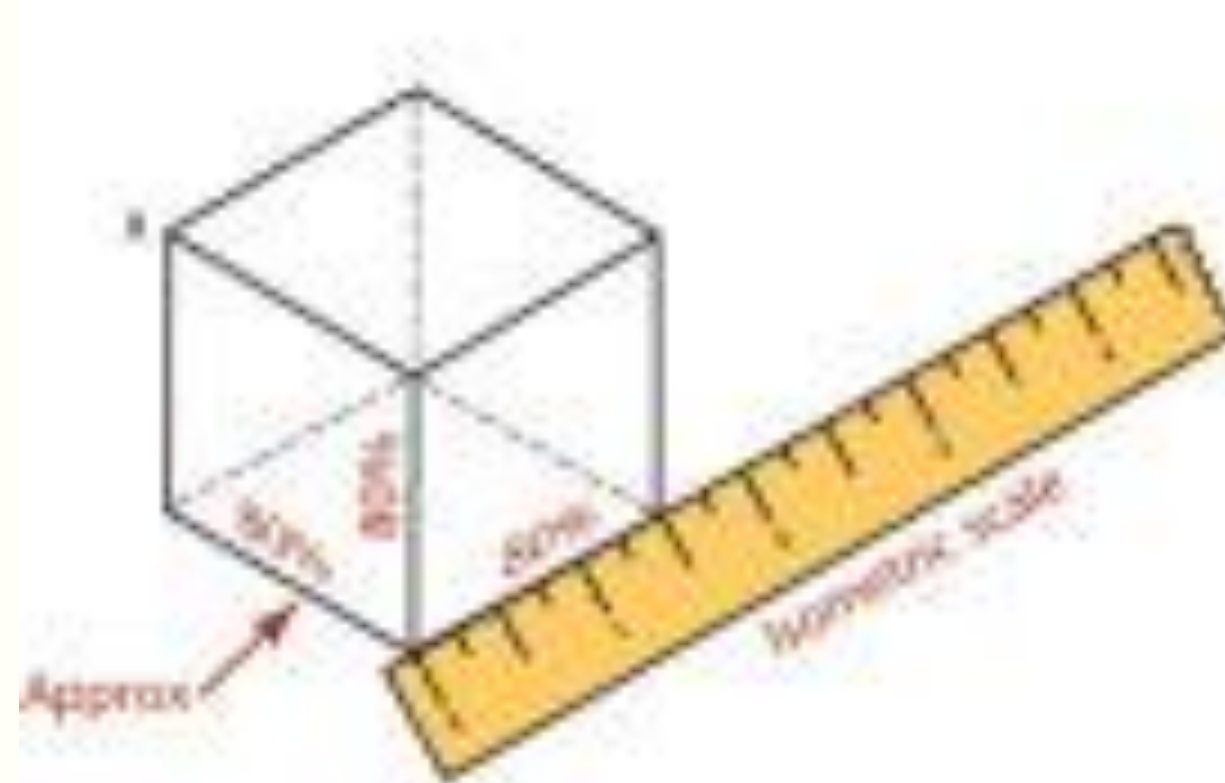


A view drawn using Isometric Scale.

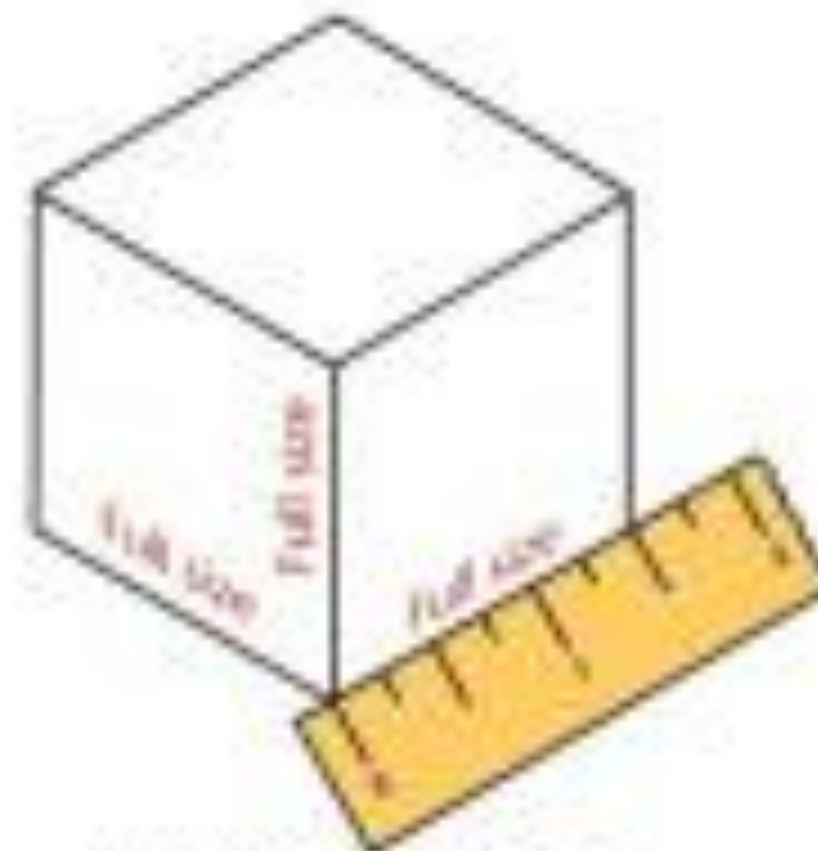
- a. Isometric Projection**
- b. Isometric View**
- c. Orthographic Projection**
- d. Orthographic View**

Isometric view vs. Isometric Projections

In **isometric view**, the object is represented in drawing with actual dimensions using normal scale and in **isometric projection** objects are symbolized in illustrations with reduced or isometric scale.



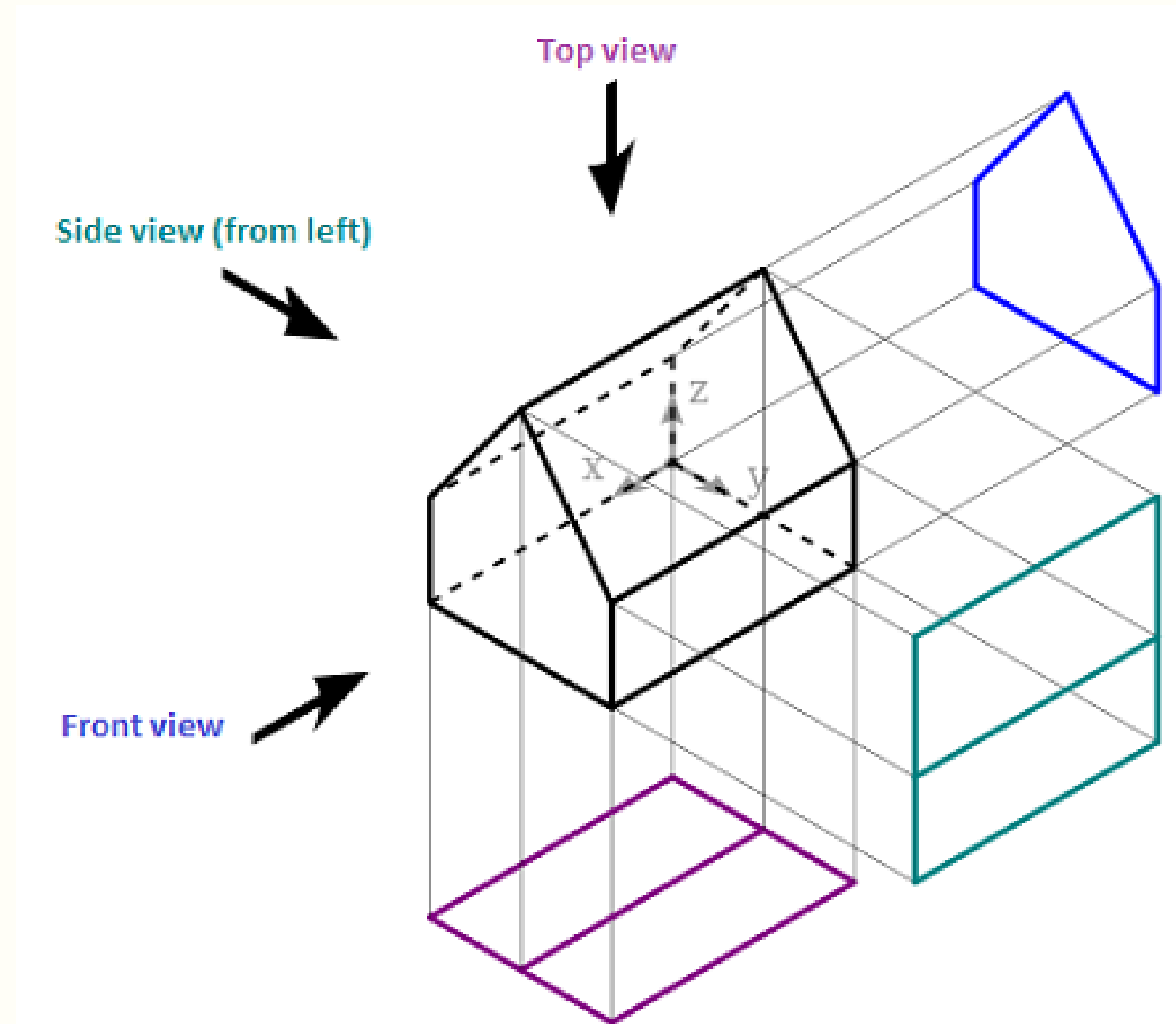
(a) Isometric projection



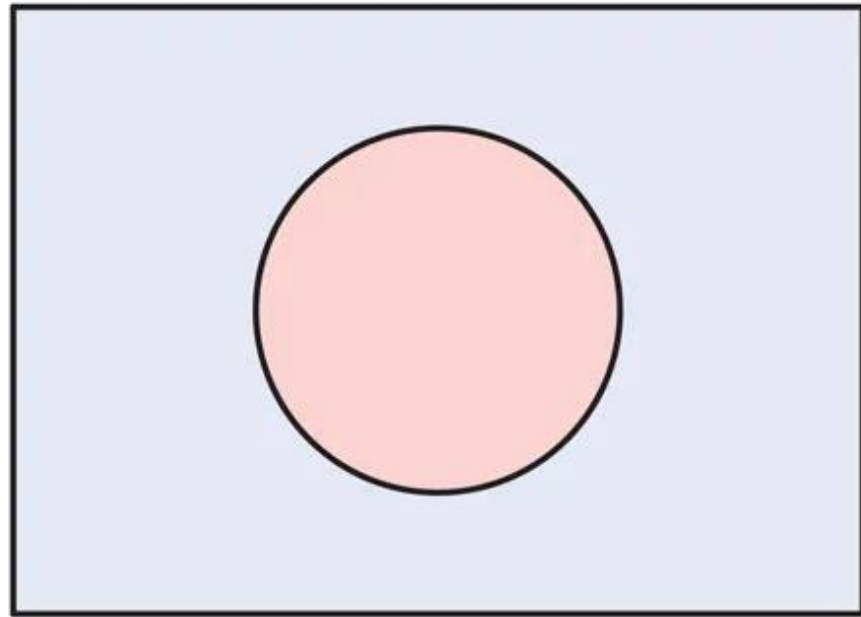
(b) Isometric drawing

Isometric vs Orthographic Drawings

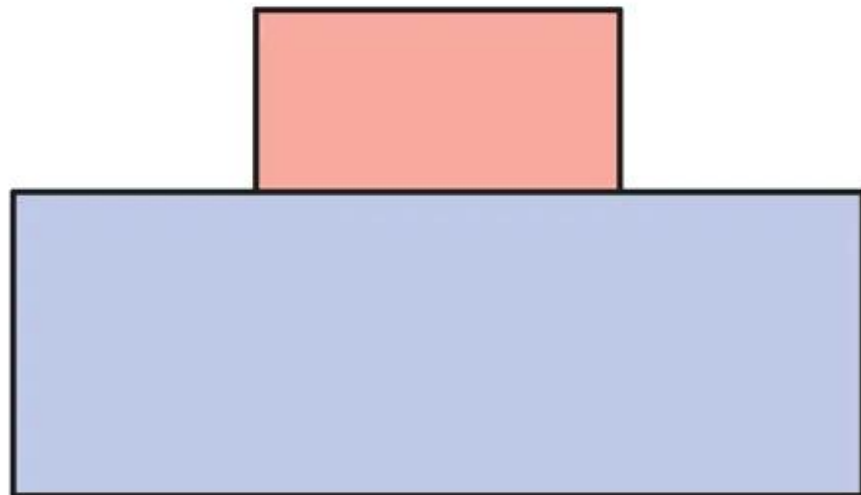
Isometric, or pictorial drawings represent an object in a three-dimensional fashion by showing 3 surfaces of the object in one drawing. **Orthographic, or plan view drawings** represent an object in a two-dimensional fashion by showing each surface of the object in its actual shape.



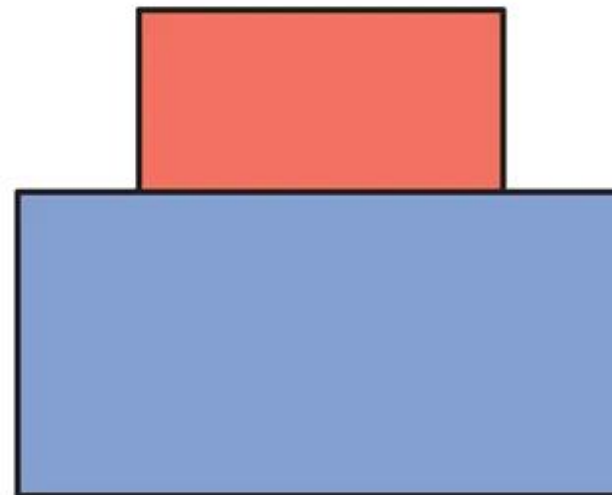
Orthographic and Isometric projections



top view

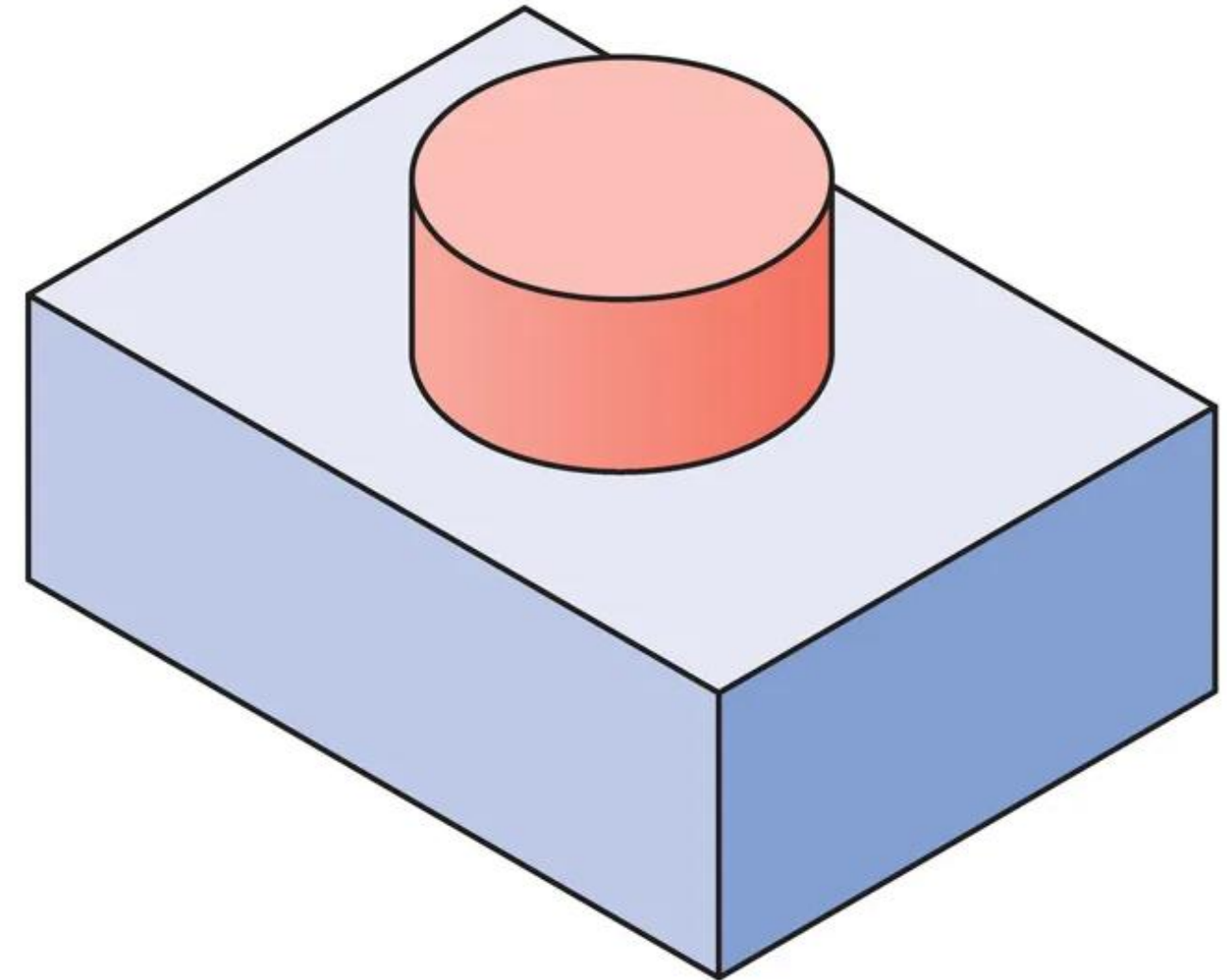


front view



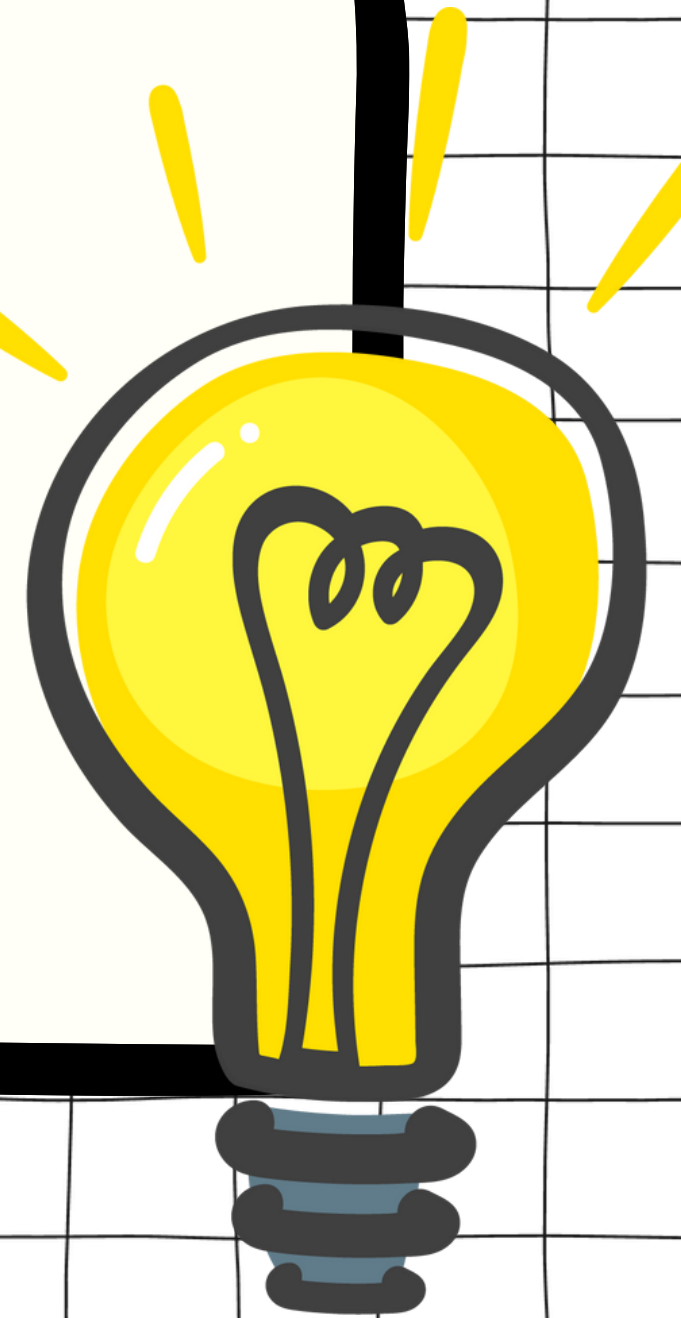
side view

2-dimensional orthographic projection



3-dimensional isometric projection

Computer Programming



Pre-Test # 21

...

**The major computer operations include all,
EXCEPT:**

- a. Input**
- b. Output**
- c. Processing**
- d. Storage**

Computer Programming

the process of writing code to facilitate specific actions in a computer, application or software program, and instructs them on how to perform.

Major Computer Operations:

- Input
- Processing
- Output



Pre-Test # 22

...

It is a diagrammatic representation of the step-by-step approach to solving a task.

- a. Algorithm**
- b. Flowchart**
- c. Pseudocode**
- d. All of the above**

A **flowchart** is a schematic representation of an algorithm or a stepwise process, showing the steps as boxes of various kinds, and their order by connecting these with arrows.

Algorithm is a step-by-step recipe for processing data.

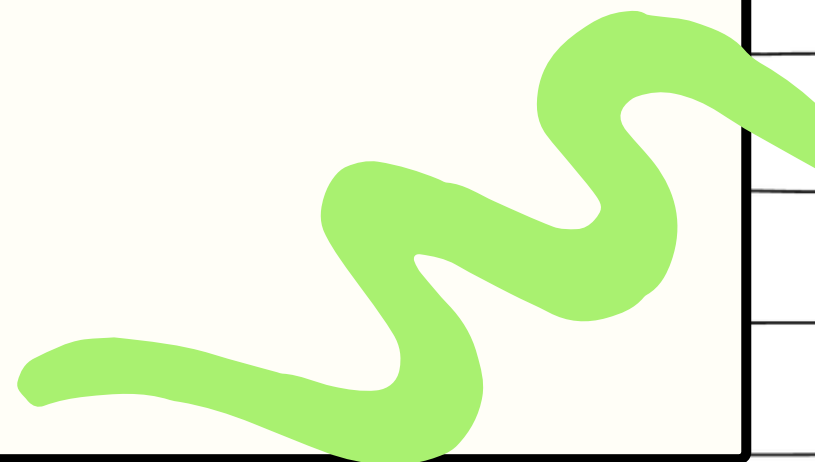
Pseudocode is a step-by-step description of an algorithm in code -like structure using plain English text.

Pre-Test # 23

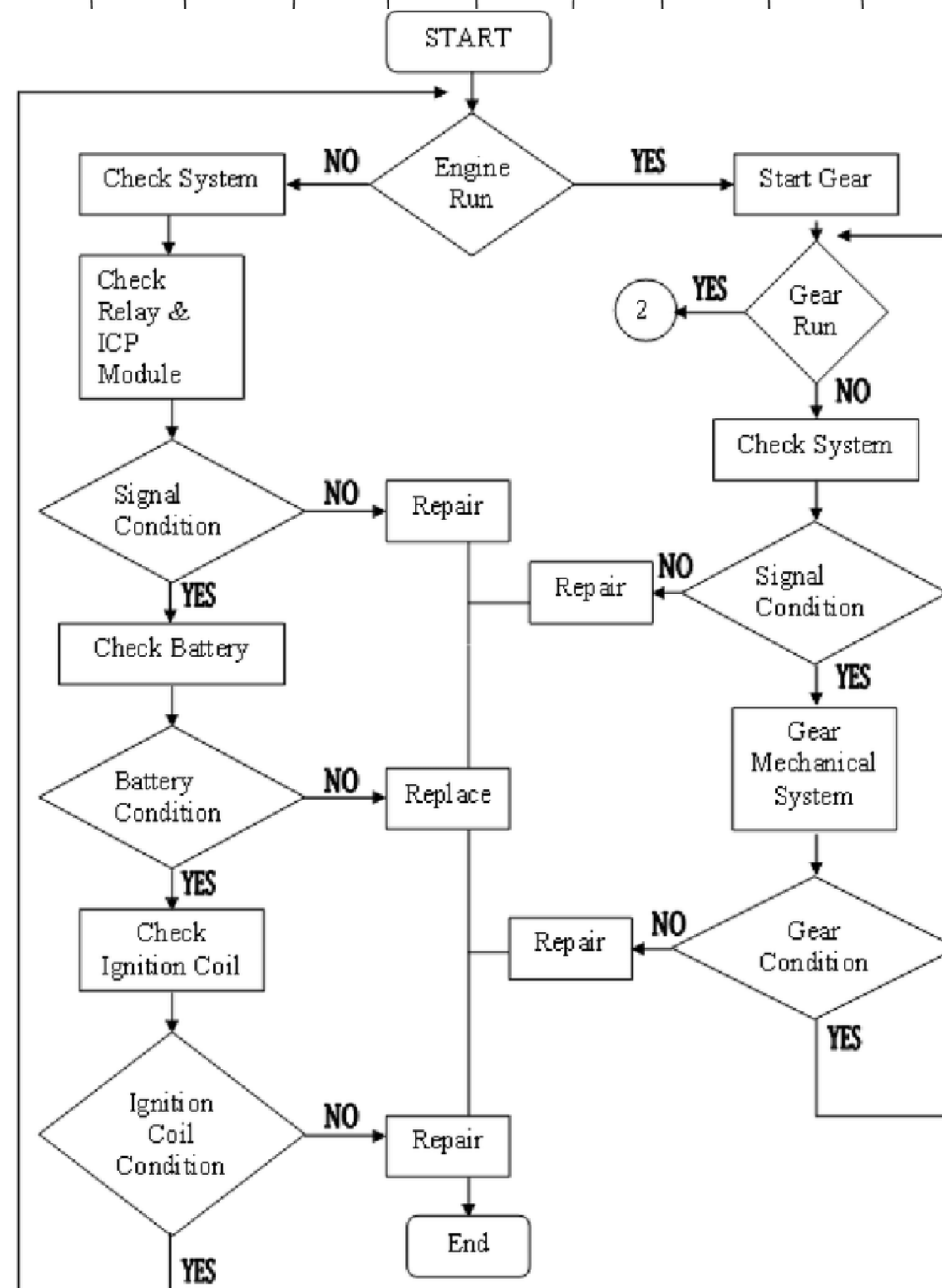


In a flowchart, the decision symbol is in the shape of what polygon?

- a. Diamond**
- b. Hourglass**
- c. Rectangle**
- d. Triangle**



Symbol	Name	Function
	Start/end	An oval represents a start or end point
	Arrows	A line is a connector that shows relationships between the representative shapes
	Input/Output	A parallelogram represents input or output
	Process	A rectagle represents a process
	Decision	A diamond indicates a decision



Pre-Test # 24

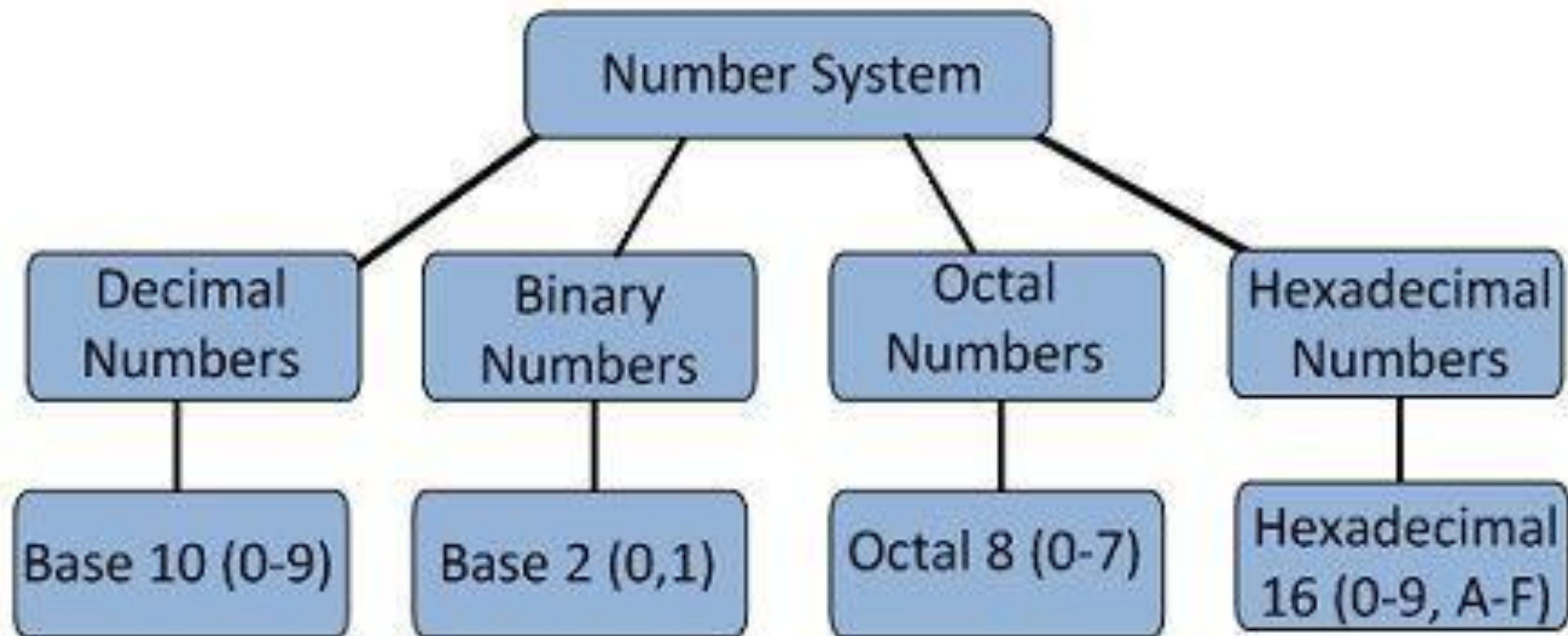
...

It is used extensively in digital electronics and computing because it is easy to represent them using switches, where 0 represents "off" and 1 represents "on."

- a. Binary numbers**
- b. Decimal numbers**
- c. Hexadecimal numbers**
- d. Octal numbers**

Number system

writing system for denoting numbers logically using digits or symbols. All numbers can be formed using the digits 0 to 9.



Pre-Test # 25

...

Convert 123_8 to binary number.

- a. 1010001_2
- b. 1010011_2
- c. 1010111_2
- d. 1110010_2

Pre-Test # 26

...

The binary number 11111_2 when converted to a hexadecimal number is _____.

- a. $2G_{16}$
- b. $5G_{16}$
- c. $3F_{16}$
- d. $7F_{16}$

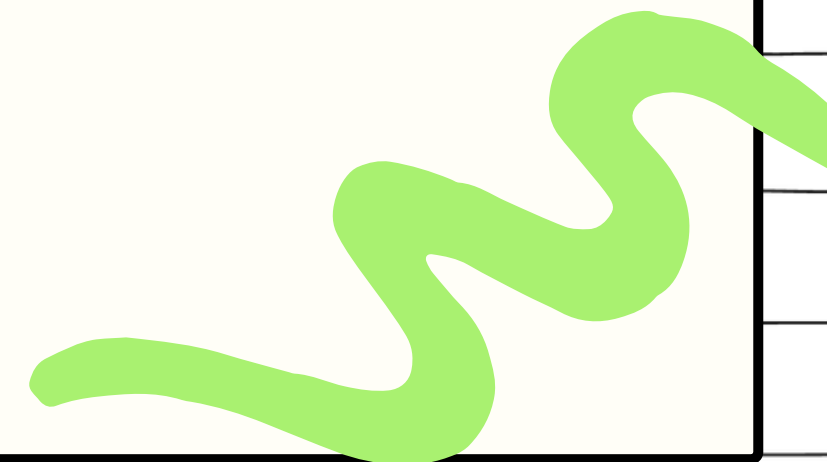
Pre-Test # 27

...

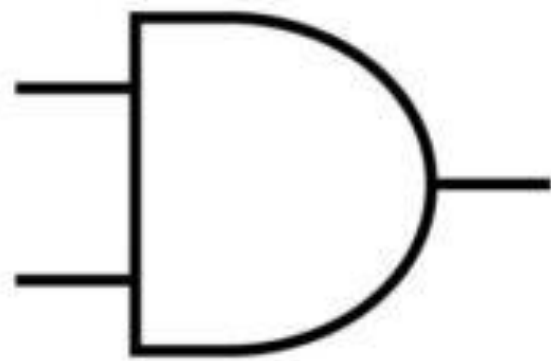
Identify the type of logic gate:



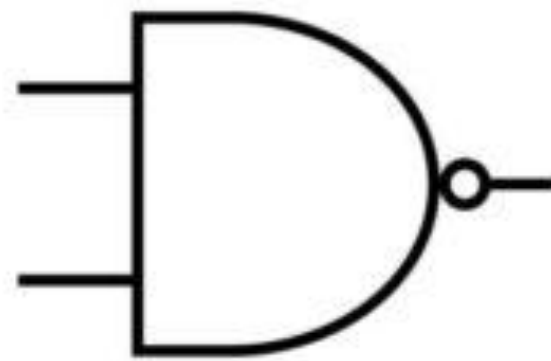
- a. AND
- b. NAND
- c. XNOR
- d. XOR



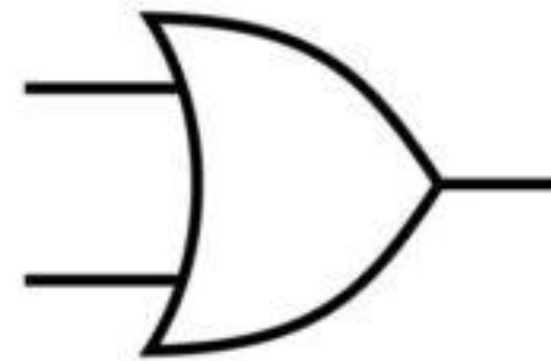
LOGIC GATE SYMBOLS



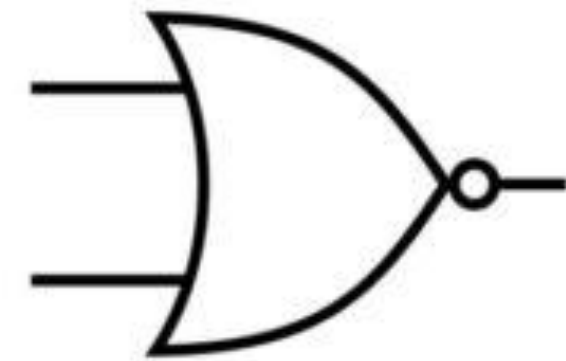
AND



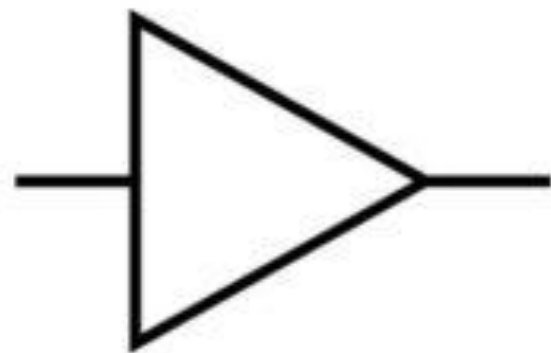
NAND



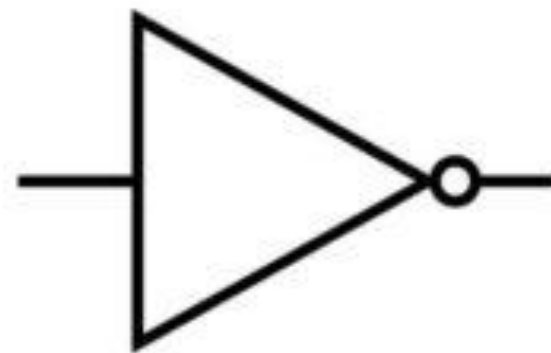
OR



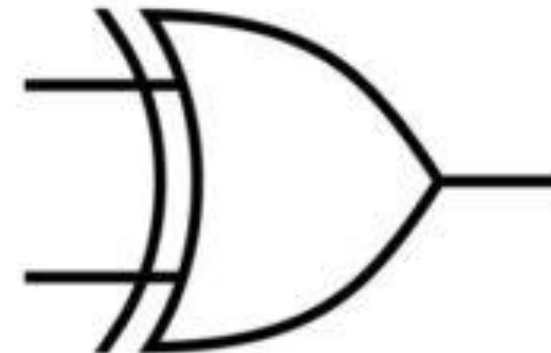
NOR



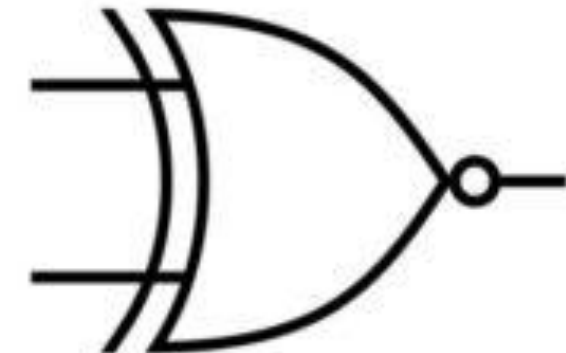
BUFFER



NOT



XOR



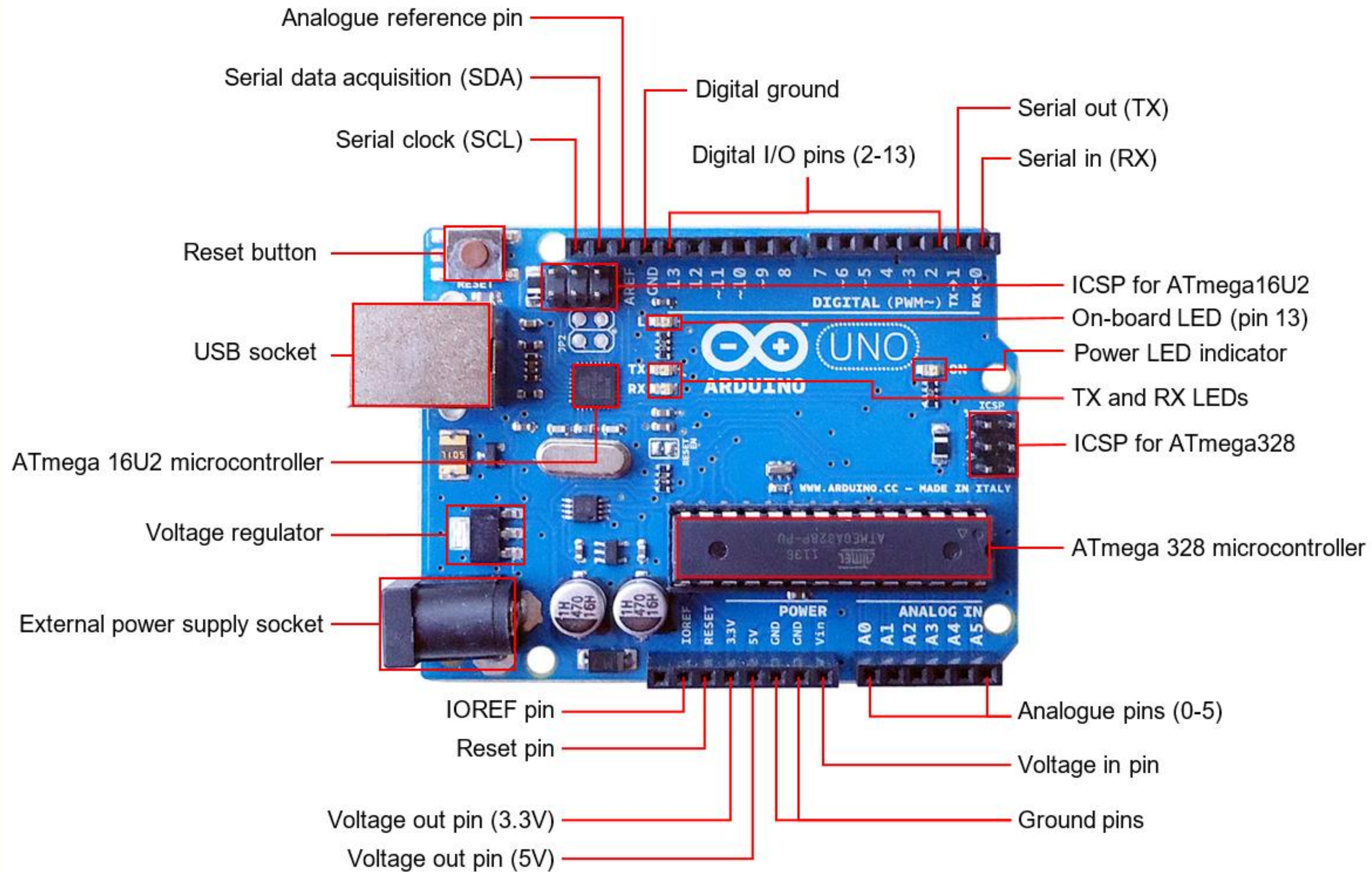
XNOR

Pre-Test # 28

...

How many analog pins are there in the Arduino Uno board?

- a. 3**
- b. 6**
- c. 9**
- d. 12**



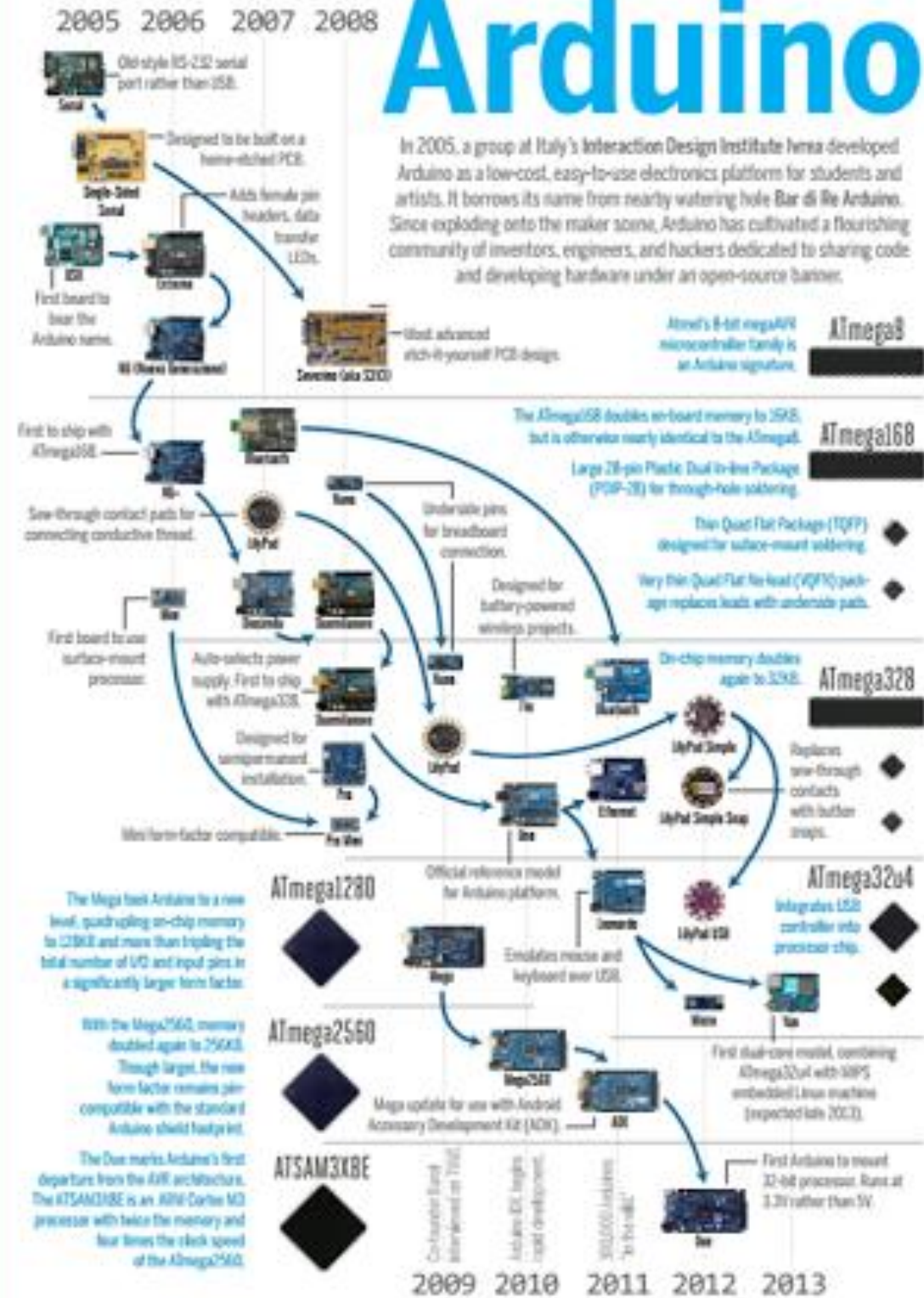
Pre-Test # 29

...

Which of the following DOES NOT belong to the Arduino family?

- a. Gemma**
- b. Giga**
- c. Mega**
- d. Nano**

Evolution of Arduino



Make:

Arduino Due	Arduino Leonardo	Arduino Yun
Arduino Mega 2560	Arduino LilyPad	Arduino Industrial 101
Arduino Esplora	Arduino Uno	Arduino 101
Arduino Pro Mini	Arduino Mini	Arduino Tian
Arduino BT (Bluetooth)	Arduino Nano	Arduino Zero
Arduino Ethernet	Arduino Pro	Arduino CTC 101
Arduino Mega ADK	Arduino Fio	Arduino YUN Mini
Arduino Mega 2560	Arduino LilyPad	Arduino Materia 101
Arduino MKR WAN 1300	Arduino Uno wifi REV 2	Arduino Gemma
Arduino M0 PRO	Arduino M0	Arduino MKR GSM 1400

Pre-Test # 30

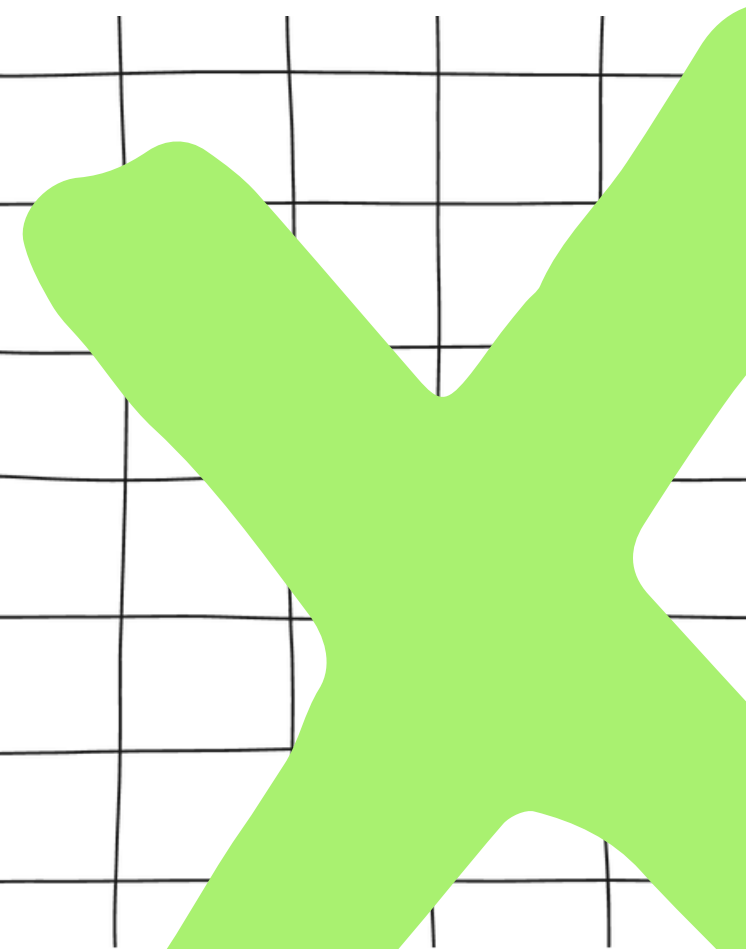
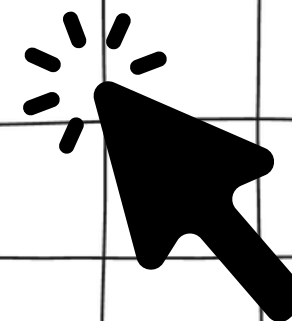
...

Arduino is an open-source platform that uses a programming language based on ____.

- a. C#**
- b. C++**
- c. Nano**
- d. Phyton**

Compiled Review Materials

https://drive.google.com/drive/folders/1QVP9cD6yhTQVdjkQwRMBItzvBaQWmvEL?usp=drive_link

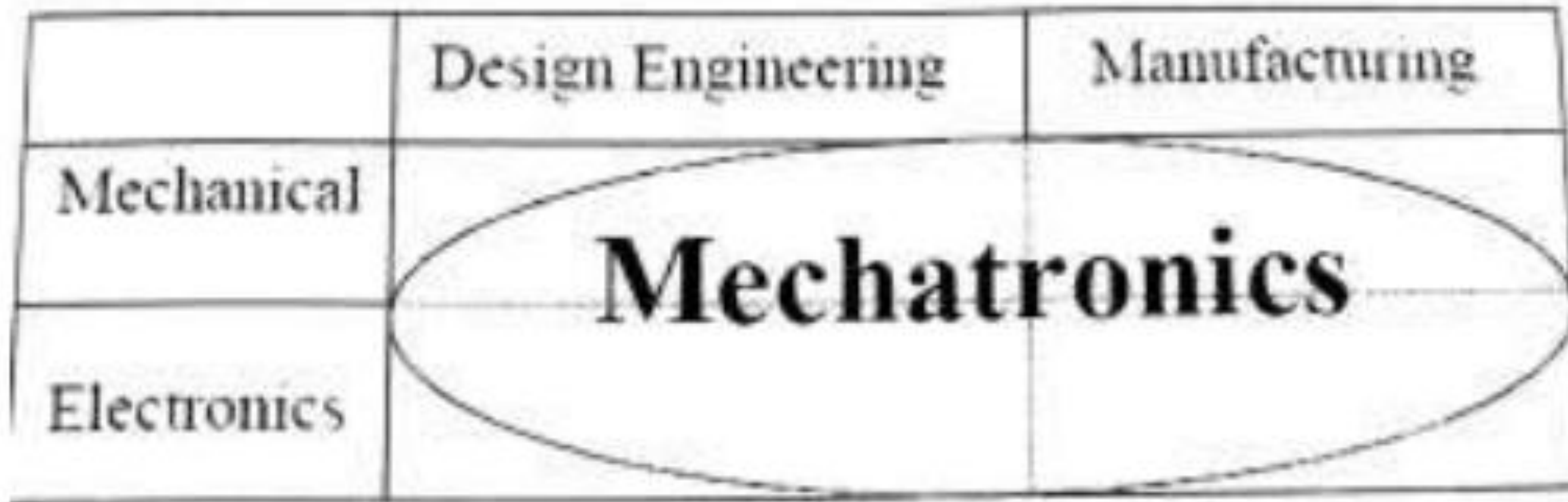


INTRODUCTION

Mechatronics involves the design and analysis of mechanical systems, such as sensors, actuators, and mechanical structures. This can include anything from robotic arms and automated machinery to consumer products like cameras with auto-focus mechanisms.

Mechatronics is an integration of mechanical, electronics, computer, and controls engineering. The key mechatronic areas **include sensors and instrumentation systems, drives and actuators, and computer-aided engineering design.**

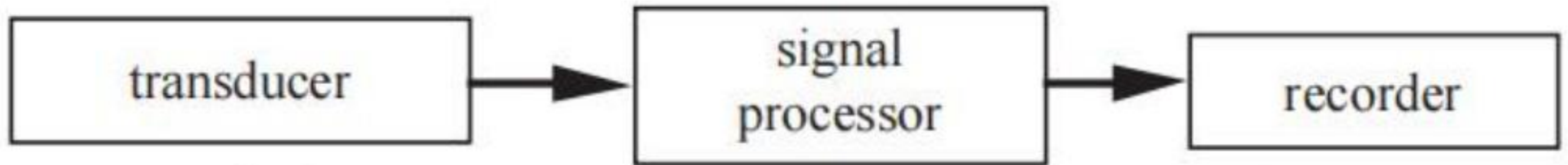
INTRODUCTION



Mechatronics is multi-disciplinary.

MEASUREMENT SYSTEMS

Measurement is an important subsystem of a mechatronics system. Its main function is to collect the information on system status and to feed it to the micro-processor(s) for controlling the whole system.



Elements of a measurement system.

MEASUREMENT SYSTEMS

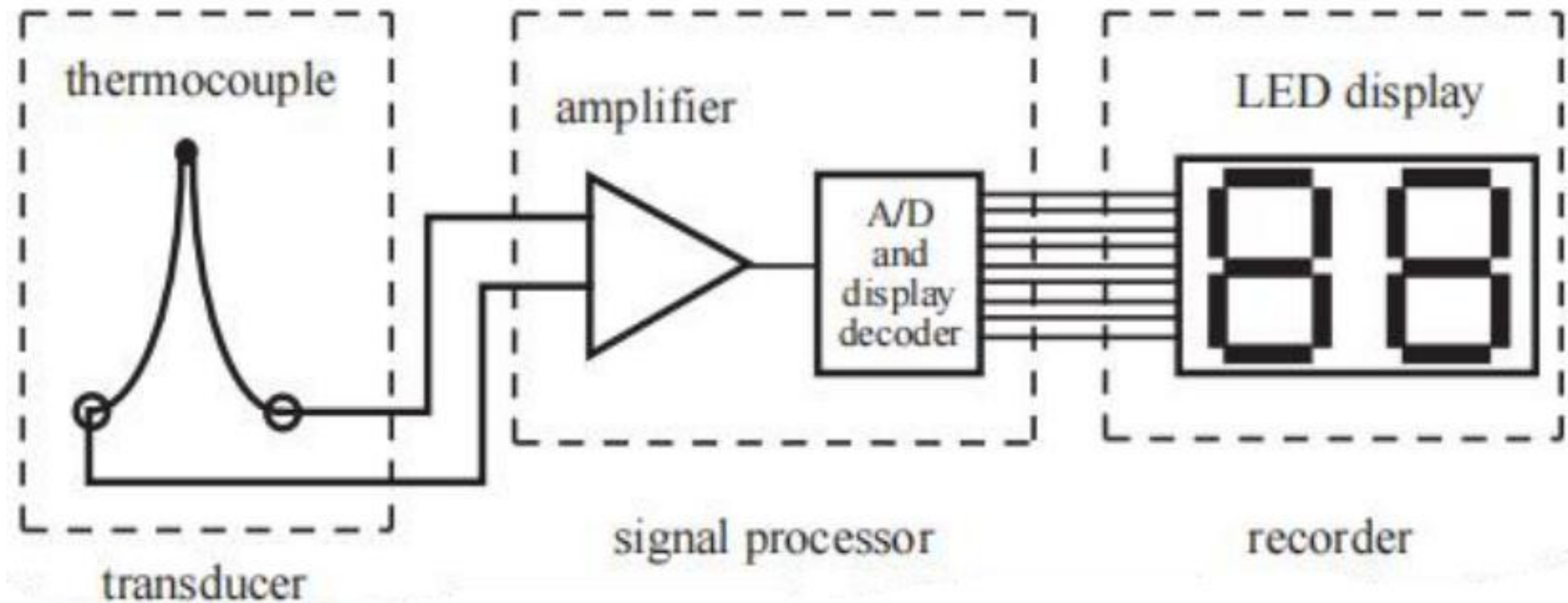
Transducer is a sensing device that converts a physical input into an output, usually a voltage.

Signal processor performs filtering, amplification, or other signal conditioning on the transducer output.

NOTE: The term sensor is often used to refer to the transducer or to the combination of transducer and signal processor.

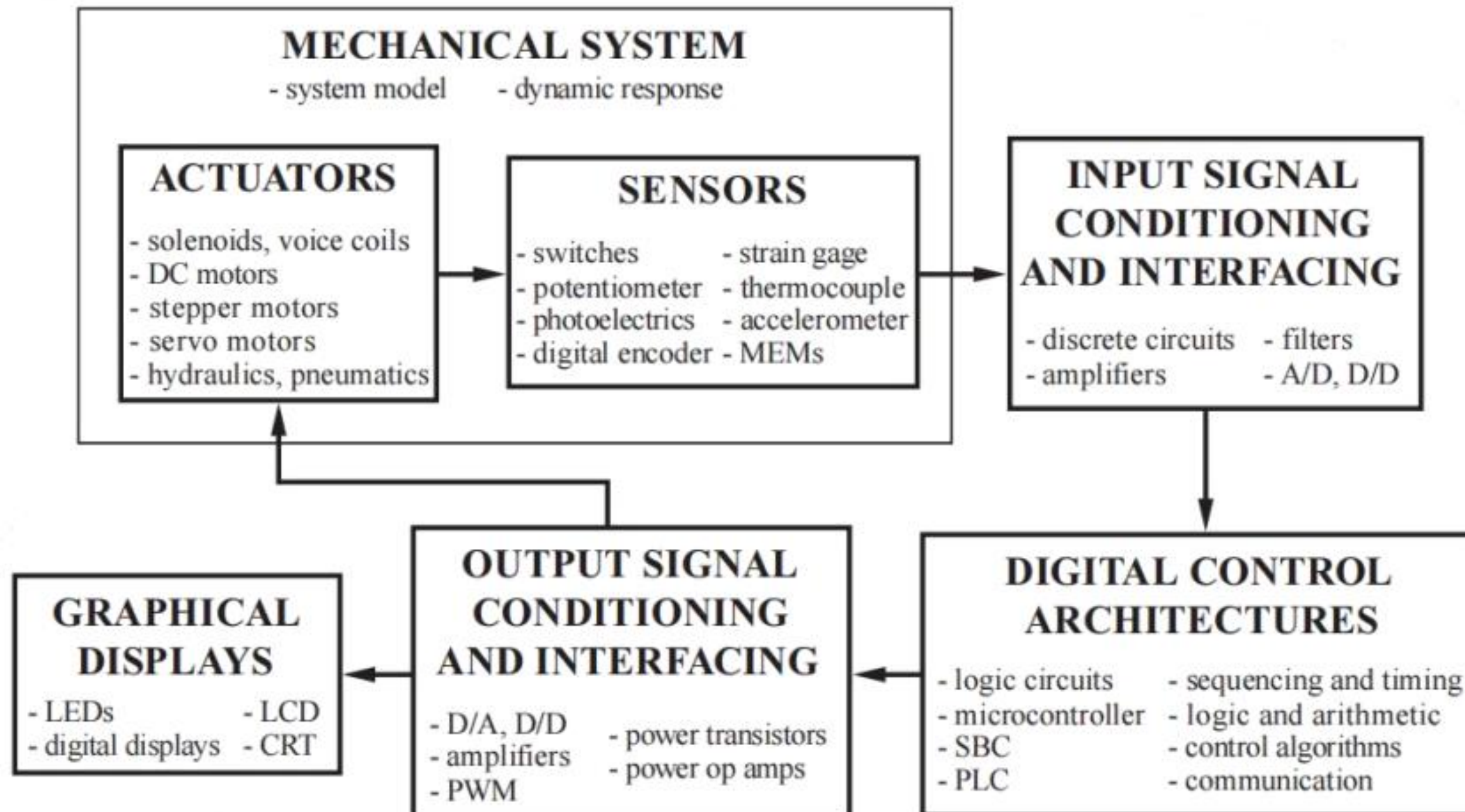
Recorder is an instrument, a computer, a hard-copy device, or simply a display that maintains the sensor data for online monitoring or subsequent processing

MEASUREMENT SYSTEMS



Digital Thermometer

ELECTRIC CIRCUITS AND COMPONENTS



ELECTRIC CIRCUITS AND COMPONENTS

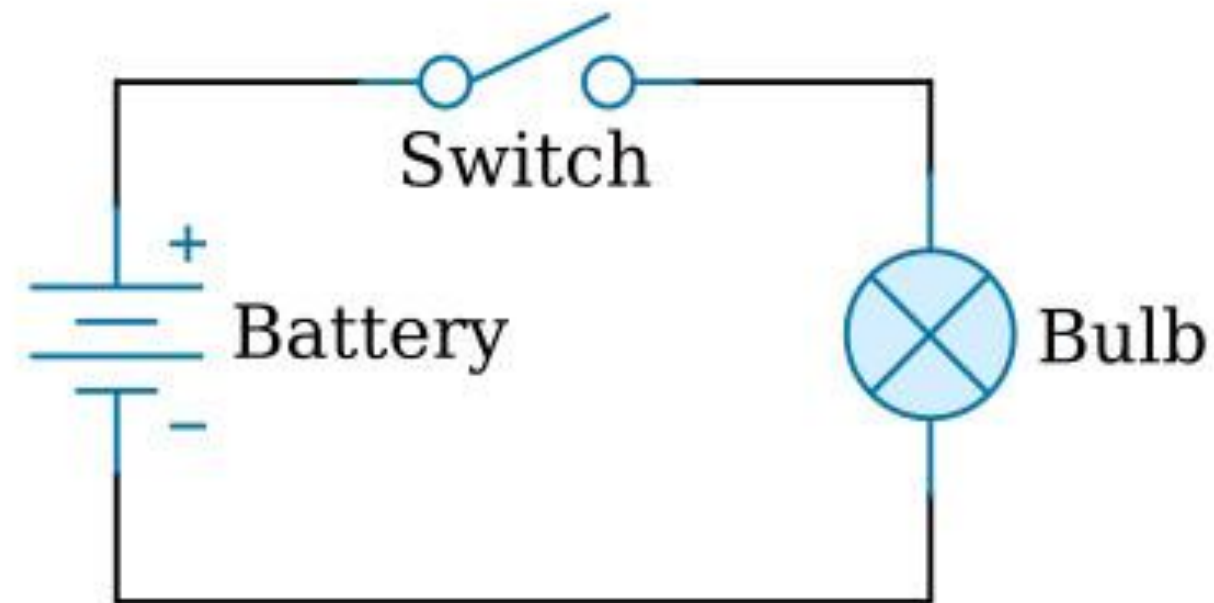
When electrons move, they produce an electrical current, and we can do useful things with the energized electrons. The resulting movement of electrons is the **current**.

The reason they move is that we impose an electrical field that imparts energy by doing work on the electrons. A measure of the electric field's potential is called **voltage**.

ELECTRIC CIRCUITS AND COMPONENTS

Voltage is sometimes referred to as **electromotive force**, or *emf*.

An electrical circuit is a closed loop consisting of several conductors connecting electrical components. Conductors may be interrupted by components called **switches**.



BASIC ELECTRICAL ELEMENTS

A. Passive Electrical Elements

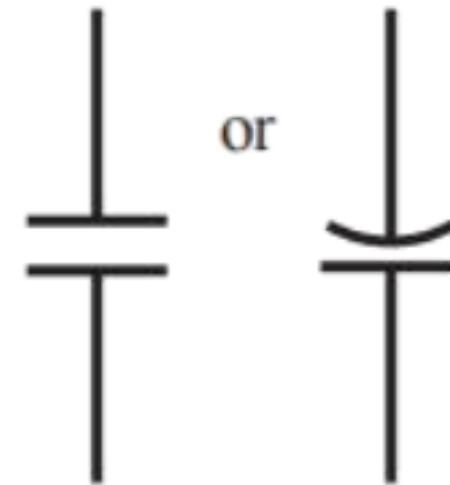
-require no additional power supply

1. resistor (R)
2. capacitor (C)
3. inductor (L)

resistor
(R)



capacitor
(C)



inductor
(L)



voltage
source
(V)



current
source
(I)



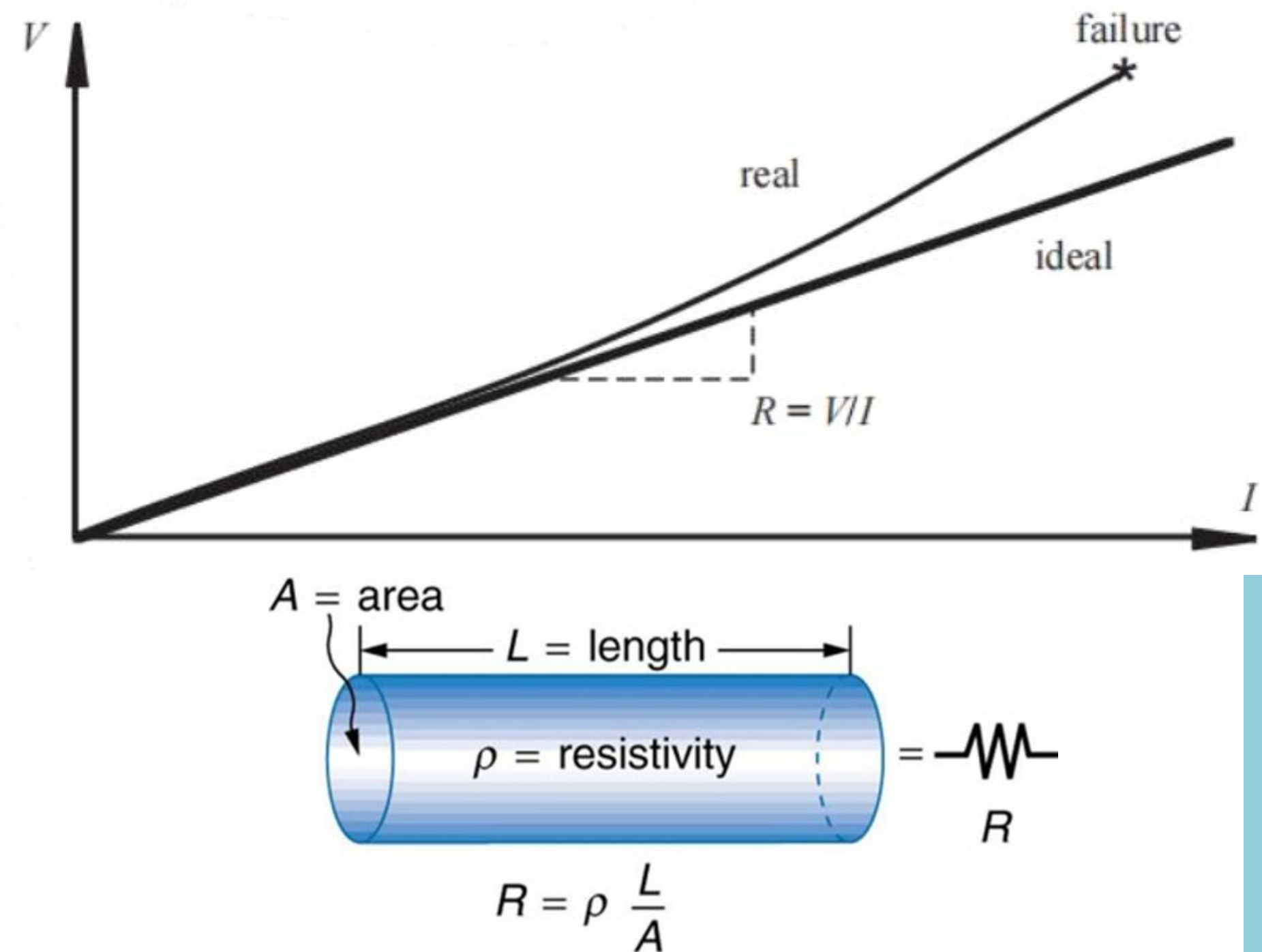
B. Active Devices

-like integrated circuits (ICs)

BASIC ELECTRICAL ELEMENTS

Resistor is a dissipative element that converts electrical energy into heat. Ohm's law defines the voltage-current characteristic of an ideal resistor.

The unit of resistance is the **ohm** (Ω). Resistance is a material property whose value is the slope of the resistor's voltage-current curve.



BASIC ELECTRICAL ELEMENTS

$$R = \frac{\rho L}{A}$$

ρ = resistivity

L = length

A = cross sectional area

Conductor Material	Resistivity (Ohm meters @ 20 °C)
Silver	1.64×10^{-8}
Copper	1.72×10^{-8}
Aluminum	2.83×10^{-8}
Tungsten	5.50×10^{-8}
Nickel	7.80×10^{-8}
Iron	12.0×10^{-8}
Constantan	49.0×10^{-8}
Nichrome II	110×10^{-8}

BASIC ELECTRICAL ELEMENTS

Determine the resistance of an aluminum wire 1.0 mm in diameter and 20 m long.

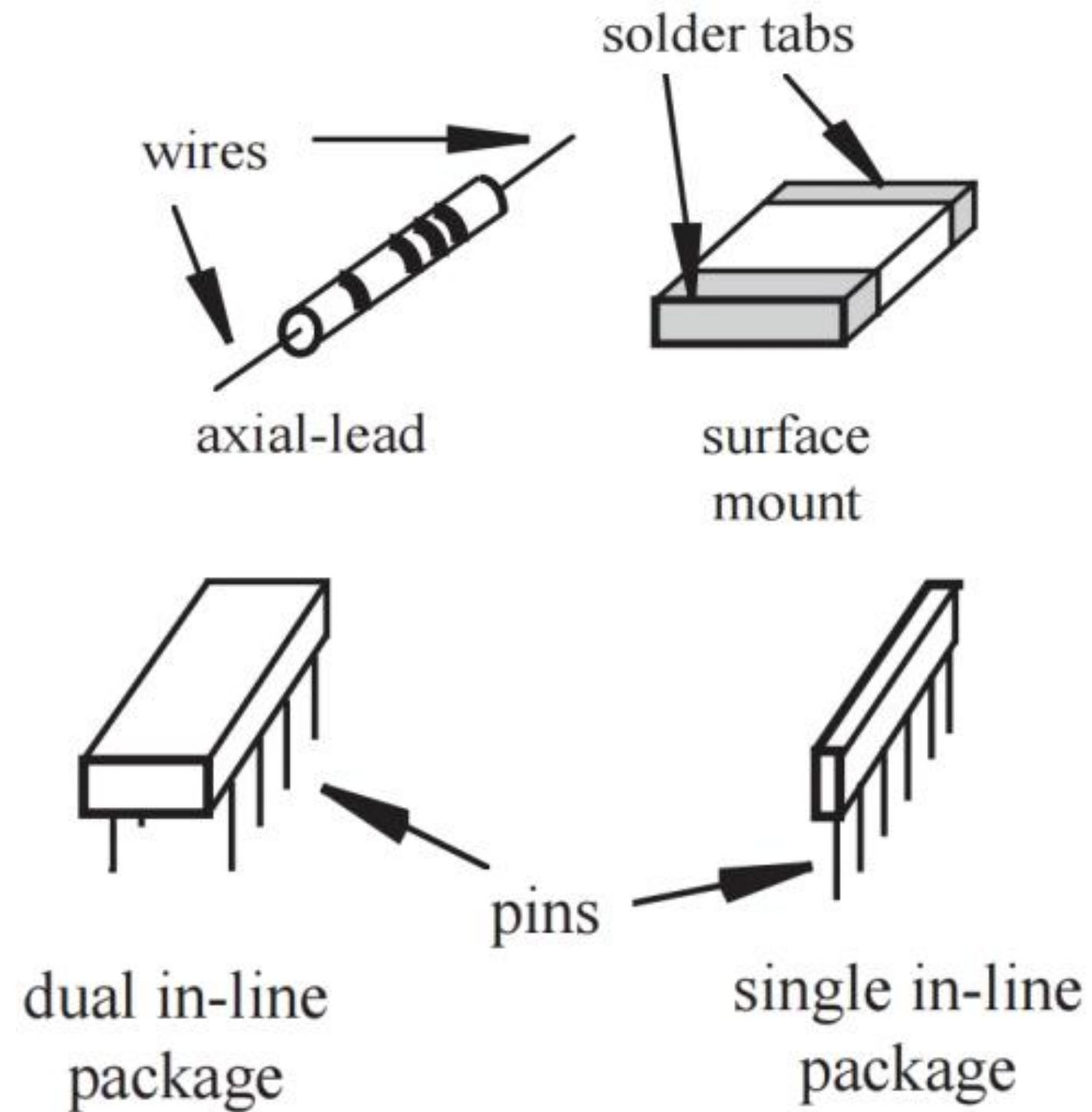
Conductor Material	Resistivity (Ohm meters @ 20 °C)
Silver	1.64×10^{-8}
Copper	1.72×10^{-8}
Aluminum	2.83×10^{-8}
Tungsten	5.50×10^{-8}
Nickel	7.80×10^{-8}
Iron	12.0×10^{-8}
Constantan	49.0×10^{-8}
Nichrome II	110×10^{-8}

BASIC ELECTRICAL ELEMENTS

A car headlight filament is made of tungsten and has a cold resistance of $0.350\ \Omega$. If the filament is a cylinder 4.00 cm long (it may be coiled to save space), what is its diameter?

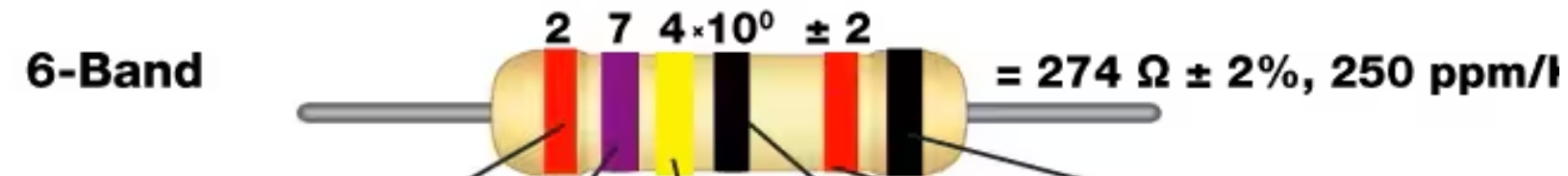
Conductor Material	Resistivity (Ohm meters @ 20 °C)
Silver	1.64×10^{-8}
Copper	1.72×10^{-8}
Aluminum	2.83×10^{-8}
Tungsten	5.50×10^{-8}
Nickel	7.80×10^{-8}
Iron	12.0×10^{-8}
Constantan	49.0×10^{-8}
Nichrome II	110×10^{-8}

BASIC ELECTRICAL ELEMENTS

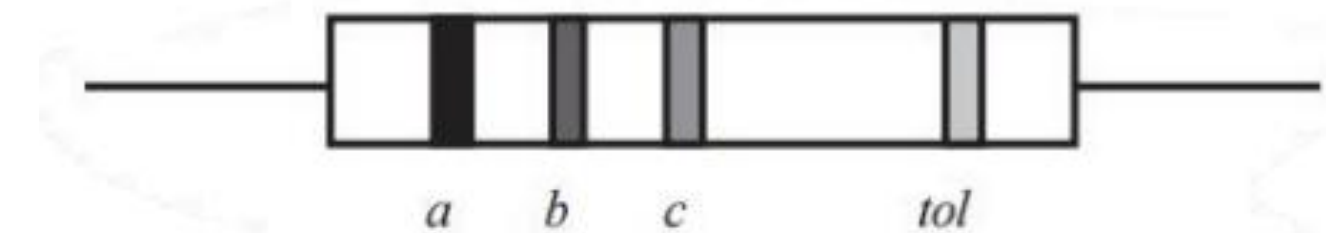


Actual resistors used in assembling circuits are packaged in various forms including **axial-lead components**, **surface mount components**, and the **dual in line package(DIP)** and the **single inline package(SIP)**.

BASIC ELECTRICAL ELEMENTS



Color	1st Digit	2nd Digit	3rd Digit	Multiplier	Tolerance	Temperature Coefficient
Black	0	0	0	1 Ω		250 ppm/K
Brown	1	1	1	10 Ω	$\pm 1\%$	100 ppm/K
Red	2	2	2	100 Ω	$\pm 2\%$	50 ppm/K
Orange	3	3	3	1k Ω		15 ppm/K
Yellow	4	4	4	10k Ω		25 ppm/K
Green	5	5	5	100k Ω	$\pm 0.5\%$	20 ppm/K
Blue	6	6	6	1M Ω	$\pm 0.25\%$	10 ppm/K
Violet	7	7	7		$\pm 0.1\%$	5 ppm/K
Grey	8	8	8			1 ppm/K
White	9	9	9			
Gold				0.1 Ω	$\pm 5\%$	
Silver				0.01 Ω	$\pm 10\%$	



Axial-lead resistor color bands.

BASIC ELECTRICAL ELEMENTS

Conductance is defined as the reciprocal of resistance.

It is a measure of how easily an element conducts current as opposed to how much it resists it.

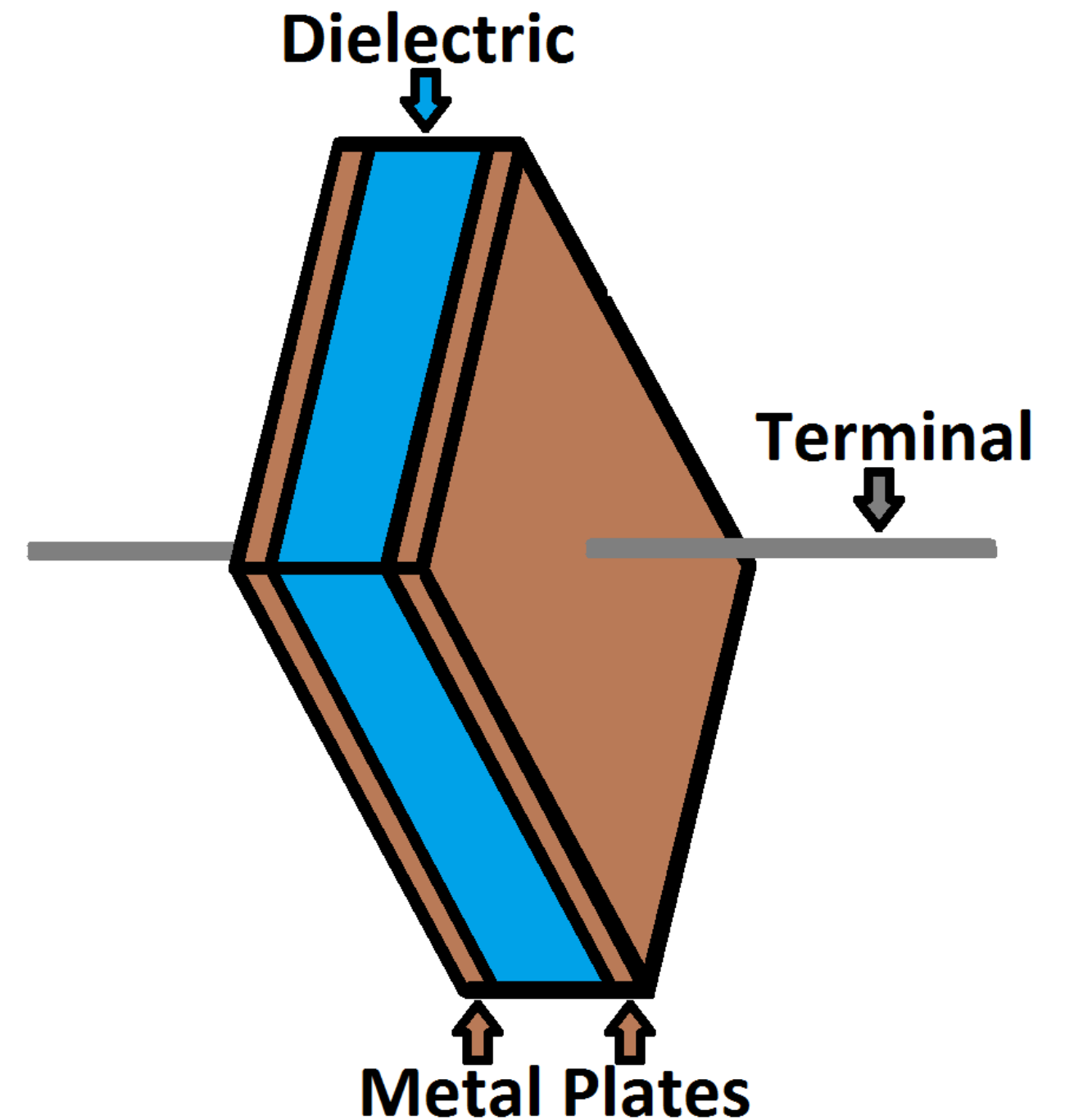
The unit of conductance is the siemen ($S = 1/\Omega = \text{mho}$).

BASIC ELECTRICAL ELEMENTS

Capacitors are passive elements that store energy in the form of an electric field.

The simplest capacitor consists of a pair of parallel conducting plates separated by a dielectric material.

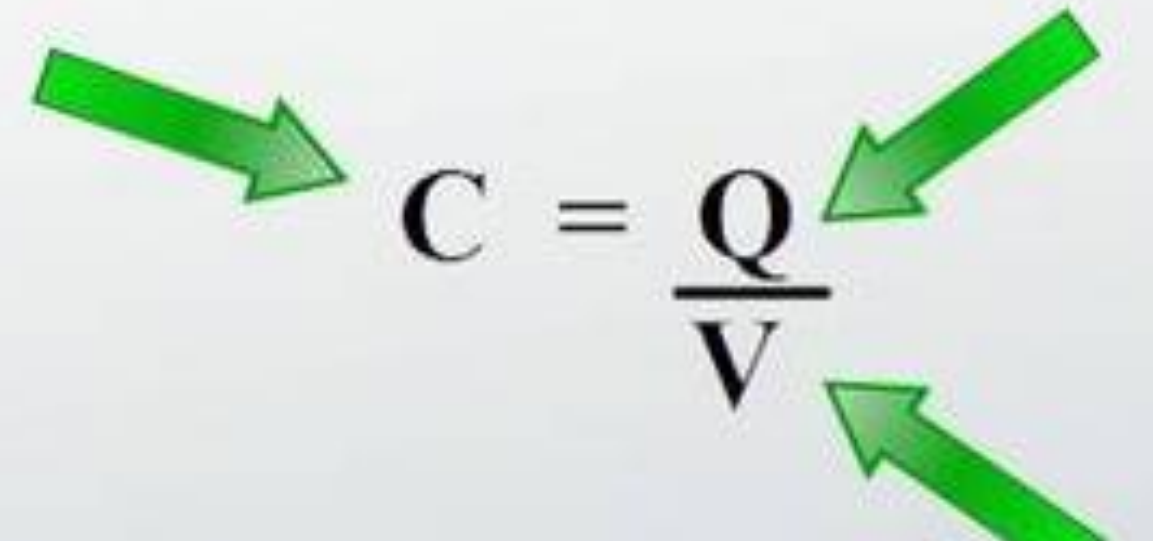
Dielectric material is an insulator that increase the capacitance as a result of permanent or induced electric dipoles in the material.



BASIC ELECTRICAL ELEMENTS

*capacitance of capacitor,
measured in farads (F)*

*charge stored,
measured in coulombs (C)*


$$C = \frac{Q}{V}$$

*voltage across capacitor,
measured in volts (V)*

BASIC ELECTRICAL ELEMENTS

Determine the amount of charge stored on either plate of a capacitor (4×10^{-6} F) when connected across a 12 volt battery.

BASIC ELECTRICAL ELEMENTS

The diagram illustrates the formula for capacitance, $C = \frac{\epsilon_0 A}{d}$, with color-coded labels and arrows pointing to the variables:

- Capacitance** (purple text) points to C with a purple arrow.
- Electrostatic constant** (orange text) points to ϵ_0 with an orange arrow. The value (8.85×10^{-12}) is written next to it.
- Area** (green text) points to A with a green arrow.
- Distance Between plates** (red text) points to d with a red arrow.

$$C = \frac{\epsilon_0 A}{d}$$

BASIC ELECTRICAL ELEMENTS

If the plate separation for a capacitor is 2.0×10^{-3} m, determine the area of the plates if the capacitance is exactly 1 F.

BASIC ELECTRICAL ELEMENTS

A parallel plate capacitor is constructed of metal plates, each with an area of 0.2 m^2 . The capacitance is 7.9 nF . Determine the plate separation distance

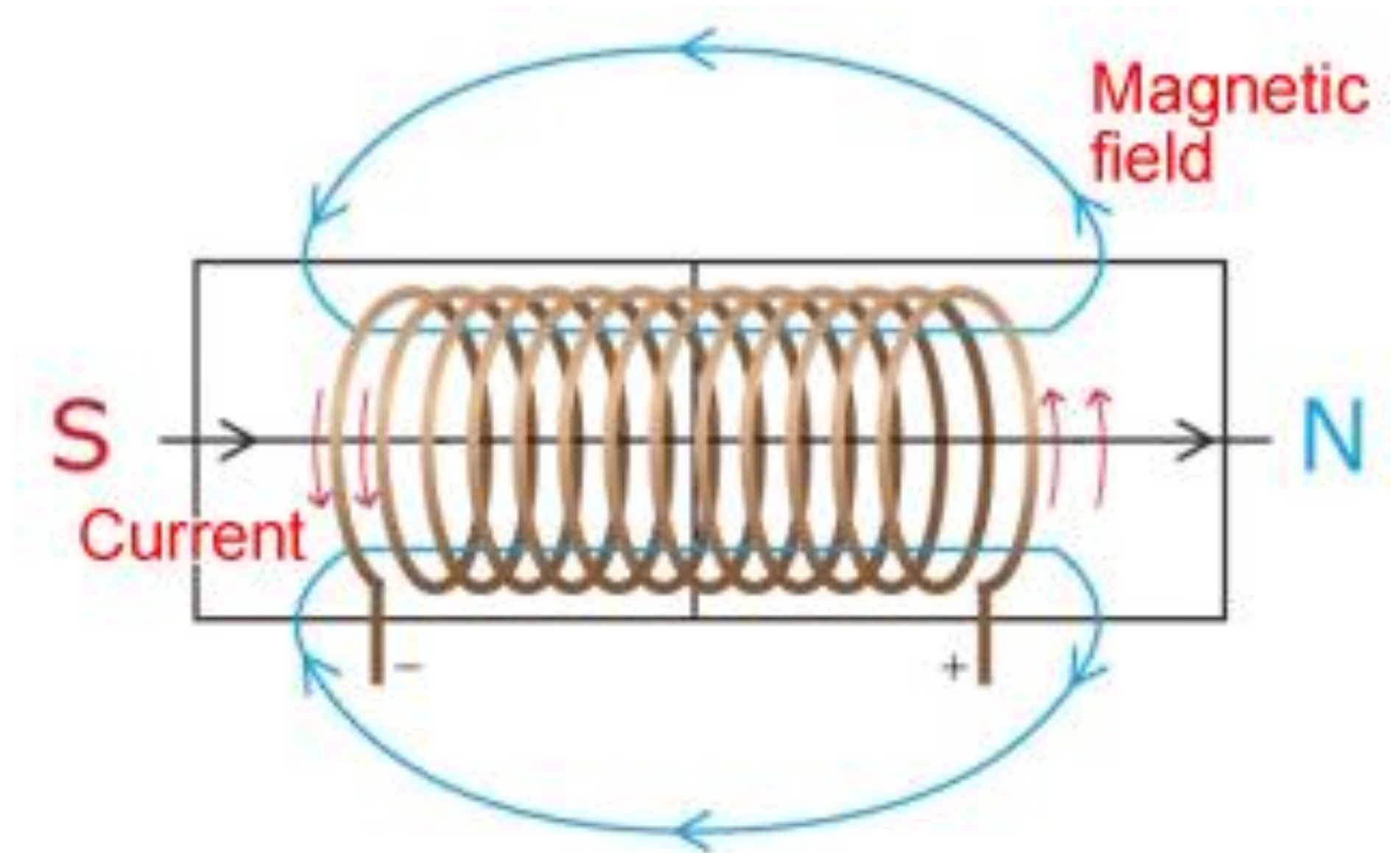
BASIC ELECTRICAL ELEMENTS

Calculate the voltage of a battery connected to a parallel plate capacitor with a plate area of 2.0 cm^2 and a plate separation of 2 mm if the charge stored on the plates is 4.0 pC .

BASIC ELECTRICAL ELEMENTS

Inductors are passive energy storage elements that store energy in the form of a magnetic field.

When current flows through an inductor with conductors wrapped around it in the same direction, the magnetic field generated around the wire is bound together and becomes an electromagnet



ANALOG SIGNAL PROCESSING USING OP AMPS

The transducer output is usually described as an **analog signal**, which is continuous and time-varying.

Usually these signals come from transducers, which convert physical quantities (*e.g., temperature, strain, displacement, flow rate*) into currents or voltages.

ANALOG SIGNAL PROCESSING USING OP AMPS

Often the signals from transducers are not in the form we would like them to be. They may

- Be too small, usually in the millivolt range
- Be too “noisy,” usually due to electromagnetic interference
- Contain the wrong information, sometimes due to poor transducer design or installation
- Have a DC offset, usually due to the transducer and instrumentation design

ANALOG SIGNAL PROCESSING USING OP AMPS

Many of these problems can be remedied, and the desired signal information extracted through appropriate **analog signal processing**.

The simplest and most common form of signal processing is **amplification**, where the magnitude of the voltage signal is increased.

Other forms include:

- signal inversion
- differentiation
- integration
- addition
- subtraction
- comparison

ANALOG SIGNAL PROCESSING USING OP AMPS

Since computers and microprocessors require digital signals, any application involving computer measurement or control requires

analog-to-digital conversion

ANALOG SIGNAL PROCESSING USING OP AMPS

AMPLIFIERS

Amplifier increases the amplitude of a signal without affecting the phase relationships of different components of the signal.

ANALOG SIGNAL PROCESSING USING OP AMPS

OPERATIONAL AMPLIFIERS

Operational amplifier, or op amp, is a low-cost and versatile integrated circuit consisting of many internal transistors, resistors, and capacitors manufactured into a single chip of silicon.

ANALOG SIGNAL PROCESSING USING OP AMPS

OPERATIONAL AMPLIFIERS

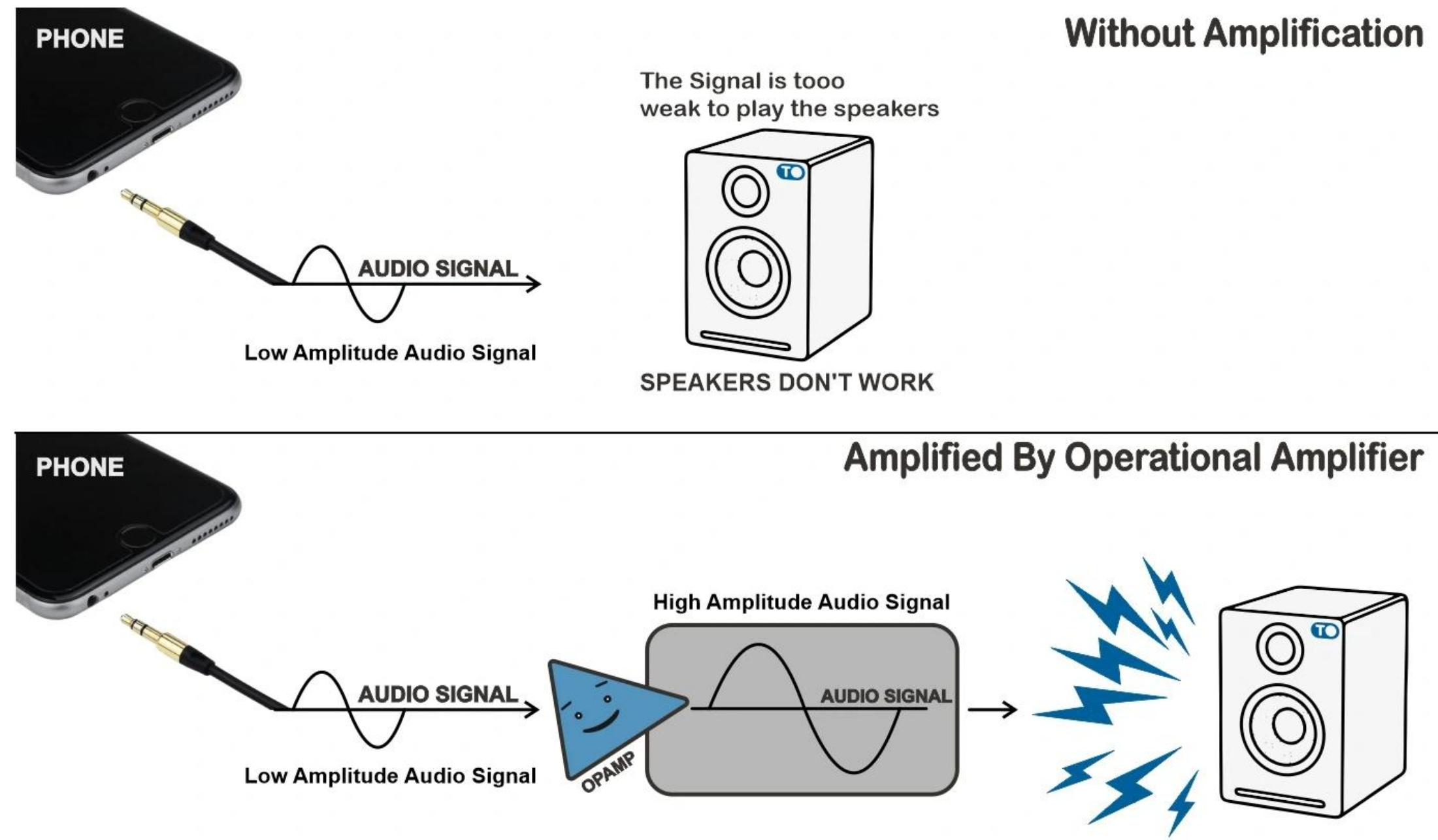
The op amp is the basic building block for:

- Amplifiers
- Integrators
- Summers
- Differentiators
- Comparators
- A/D and D/A converters
- Active filters
- Sample and hold amplifiers

ANALOG SIGNAL PROCESSING USING OP AMPS

APPLICATIONS OF OPERATIONAL AMPLIFIERS

Amplification of
an input signal

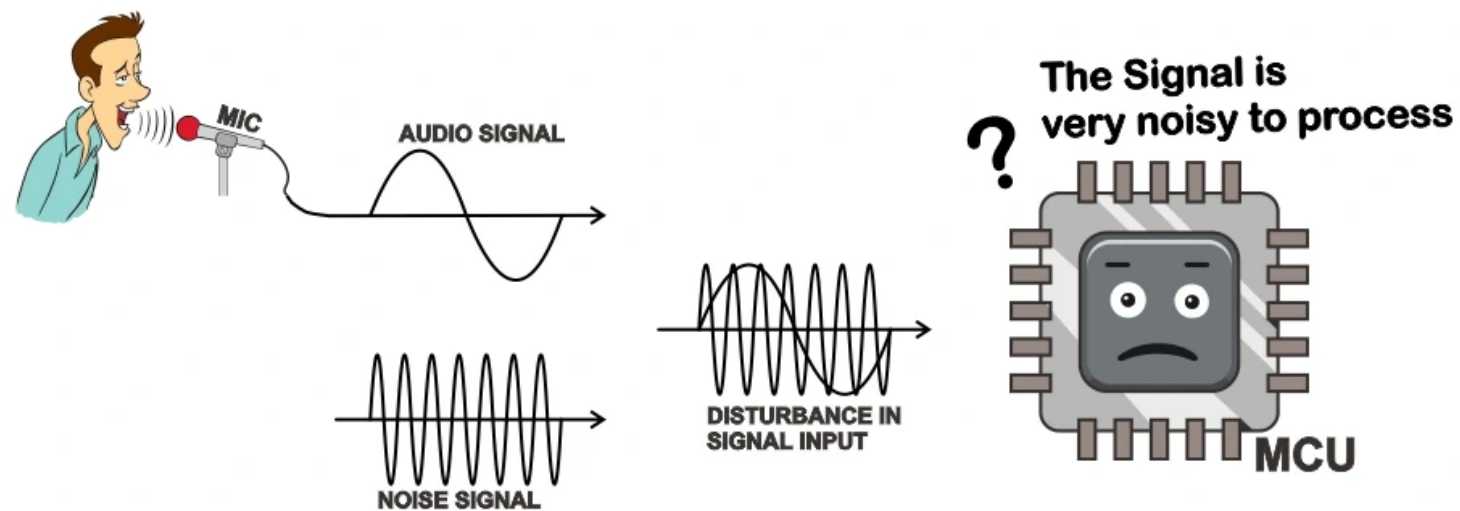


ANALOG SIGNAL PROCESSING USING OP AMPS

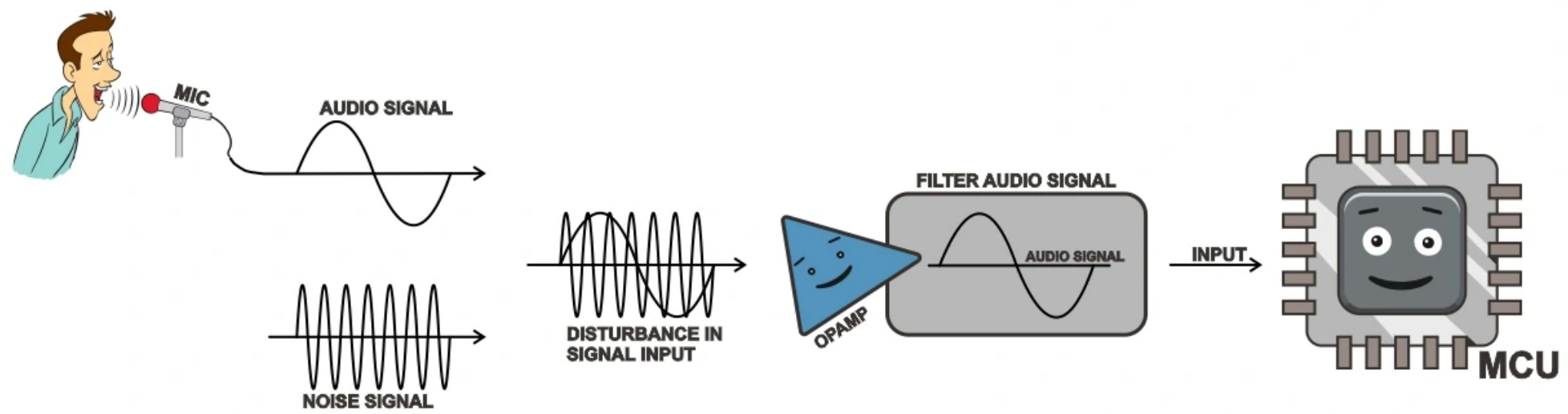
APPLICATIONS OF OPERATIONAL AMPLIFIERS

Filtering the
input signal

Without Filtering



Using an operational amplifier as a filter

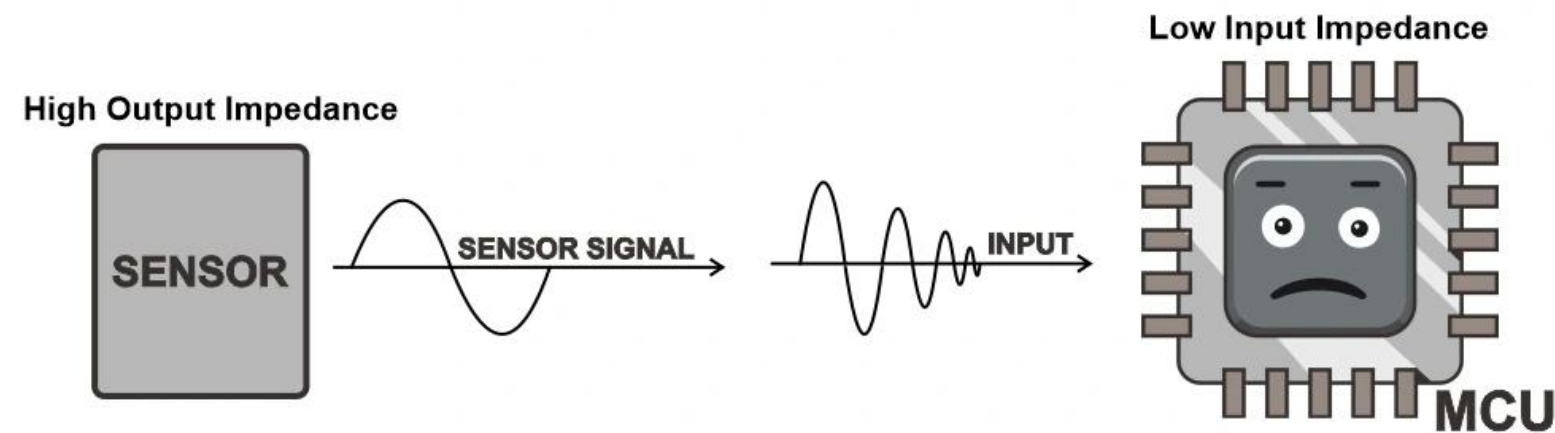


ANALOG SIGNAL PROCESSING USING OP AMPS

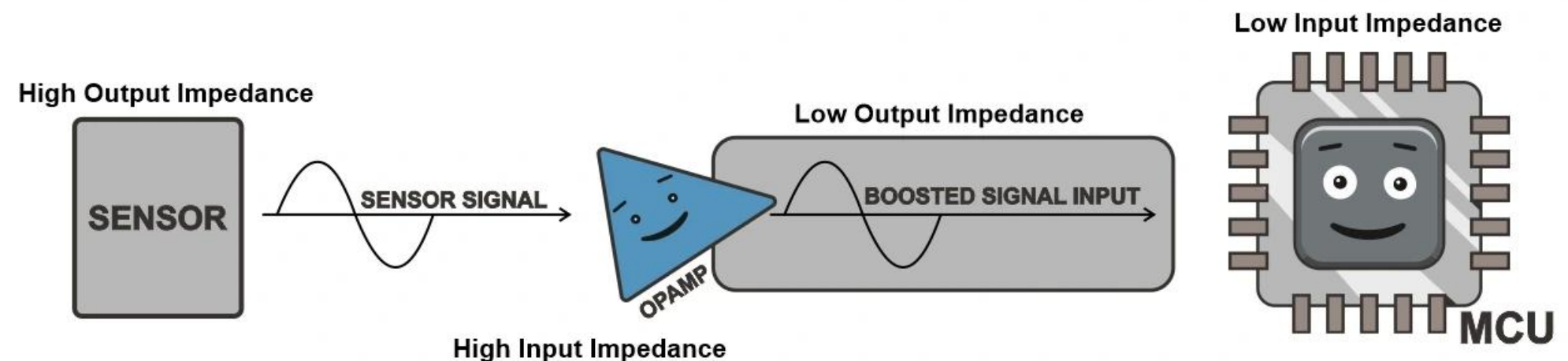
APPLICATIONS OF OPERATIONAL AMPLIFIERS

Boosting current
of input signal

Without Buffer



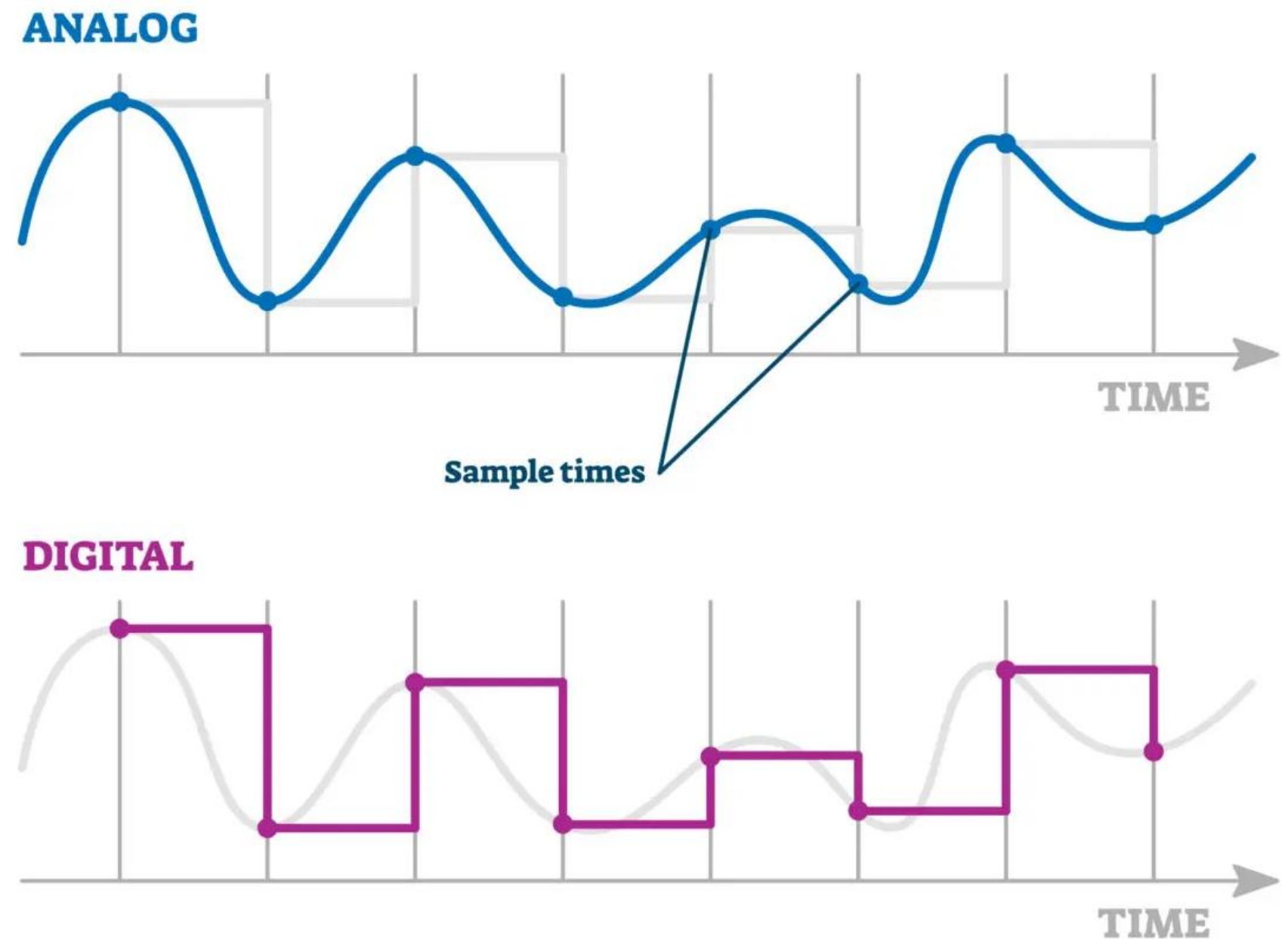
Operational Amplifier as a Buffer



DIGITAL CIRCUITS

Digital circuits have only two states: **on and off**, also referred to as **1 and 0**. This is opposed to analog circuits, which support a continuous range of values.

Digital circuits and the associated mathematics of Boolean algebra are the backbones of smartphones, computers, Wi-Fi, and the Internet.



ANALOG CIRCUITS VS. DIGITAL CIRCUITS

Analog signals are characterized by their continuous, smoothly varying nature. They can take an infinite number of values within a specified range.

Analog signals are like waves, representing information in a continuous and nuanced manner.

Digital signals are discrete and comprised of specific, distinct values, typically represented in binary code (0s and 1s).

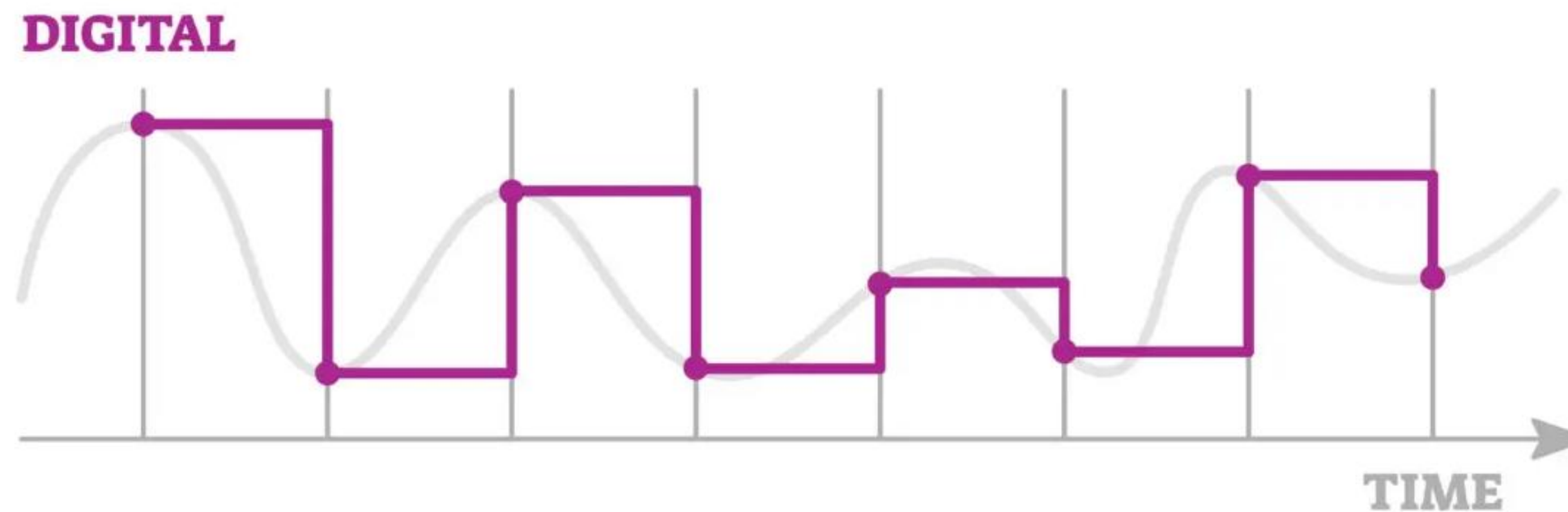
Digital signals convey information through a series of steps or levels, allowing for precise representation and transmission of data.

DIGITAL CIRCUITS

Most digital signals have only two states: high and low

A system using two-state signals allows the application:

1. Boolean logic
2. Binary number representations



DIGITAL CIRCUITS

Digital devices are categorized according to their function:

1. combinational logic
2. sequential logic

Digital devices convert digital inputs into one or more digital outputs. The difference between the two categories is based on signal timing.

DIGITAL CIRCUITS

combinational logic devices- outputs depend only on the instantaneous values of the inputs

sequential logic devices- the timing, or sequencing history, of the input signals plays a role in determining the output

NUMBER SYSTEMS

Numbering Systems

1. Binary₂
2. Octal₈
3. Decimal₁₀
4. Hexadecimal₁₆

Numbering System Conversions:

1. Binary to Decimal
2. Decimal to Binary
3. Hexadecimal to Decimal
4. Decimal to Hexadecimal
5. Hexadecimal to Binary
6. Binary to Hexadecimal
7. Octal to Decimal
8. Decimal to Octal
9. Octal to Binary
10. Binary to Octal
11. Octal to Hexadecimal
12. Hexadecimal to Octal

NUMBER SYSTEMS

Combinational Logic

- Converts binary inputs into binary outputs based on the rules of mathematical logic
 - Logic gates control the flow of input signals to single output as expressed in their truth tables
- *Logic Gates are special type of amplifier circuit designed to accept and generate voltage signals corresponding to binary 1's and 0's.
- Boolean expressions represent the equivalent equation derived from the truth table

LOGIC GATES

1. AND gate

- Also known as the all or nothing gate
- Requires all input signals to be TRUE for the output to be TRUE

A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

Gate	Operation	Symbol	Expression
AND gate	AND logic		$C = A \cdot B$

LOGIC GATES

2. OR gate

- Also known as the any or all gate
- Requires at least one of the input signals to be TRUE for the output to be TRUE

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1

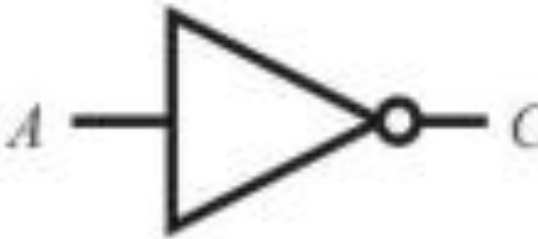
Gate	Operation	Symbol	Expression
OR gate	OR logic		$C = A + B$

LOGIC GATES

3. NOT gate

- Also known as the inverter gate
- Output signal is the inverse of the input signal
- Inversion is denoted by a ' symbol, or a bar above the letter (e.g. A' or $\bar{A}$)

Input	Output
A	Y
0	1
1	0

Gate	Operation	Symbol	Expression
Inverter (INV, NOT)	Invert signal (complement)		$C = \bar{A}$

LOGIC GATES

4. NAND gate

- NOT AND or inverted AND function.
- Output signal is the inverted output of an AND gate

A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

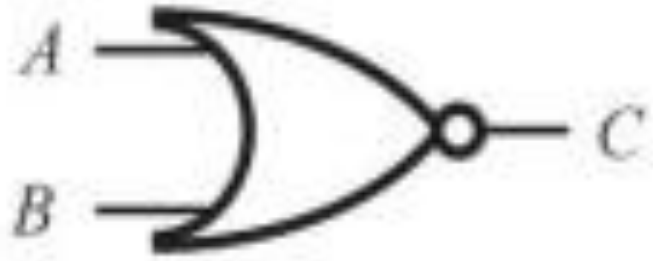
Gate	Operation	Symbol	Expression
NAND gate	Inverted AND logic		$C = \overline{A \cdot B}$

LOGIC GATES

5. NOR gate

- NOT OR or inverted OR function.
- Output signal is the inverted output of an OR gate

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0

Gate	Operation	Symbol	Expression
NOR gate	Inverted OR logic		$C = \overline{A + B}$

LOGIC GATES

6. XOR gate

- Exclusive OR or anything but all gate.

A	B	Output
0	0	0
1	0	1
0	1	1
1	1	0

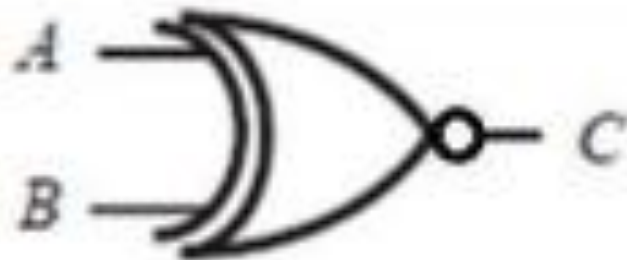
Gate	Operation	Symbol	Expression
XOR gate	Exclusive OR logic		$C = A \oplus B$ $= A \cdot \bar{B} + \bar{A} \cdot B$

LOGIC GATES

7. XNOR gate

- Exclusive NOR
- Output is the inverted output of XOR gate

A	B	Output
0	0	1
1	0	0
0	1	0
1	1	1

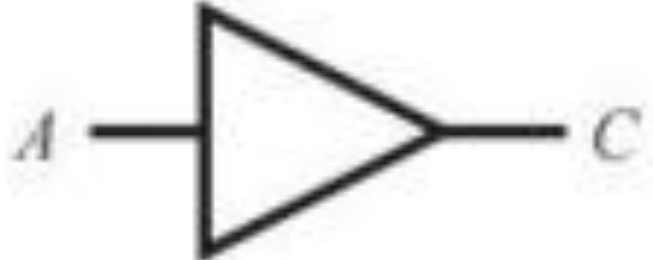
Gate	Operation	Symbol	Expression
XNOR gate	Exclusive NOR logic		$\overline{(A \oplus B)} = C$ $(A \oplus B)' = C$

LOGIC GATES

8. Buffer Gate

- used to increase the current supplied at the output while retaining the digital state

Input	Output
0	0
1	1

Gate	Operation	Symbol	Expression
Buffer	Increase output signal current		$C = A$

BOOLEAN ARITHMETIC

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 1$$

$$0 + 1 + 1 = 1$$

$$1 + 1 + 1 = 1$$

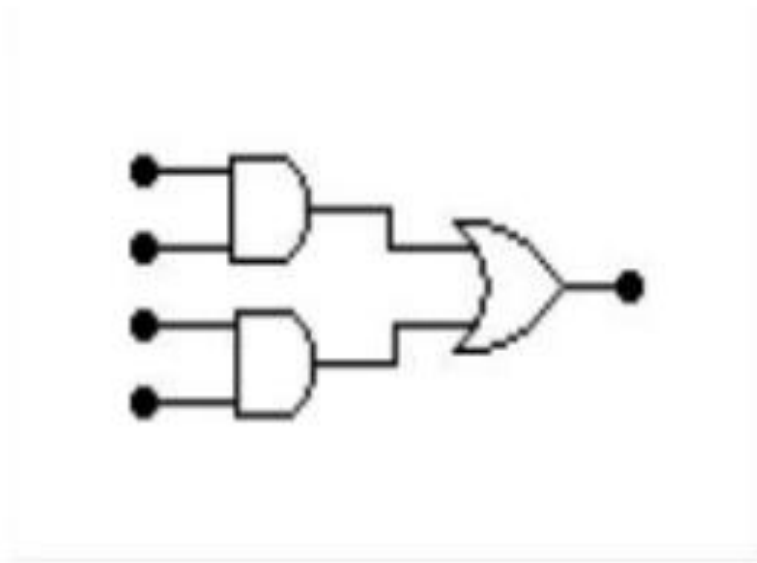
$$0 + 1 + 1 + 1 = 1$$

$$1 + 0 + 1 + 1 + 1 = 1$$

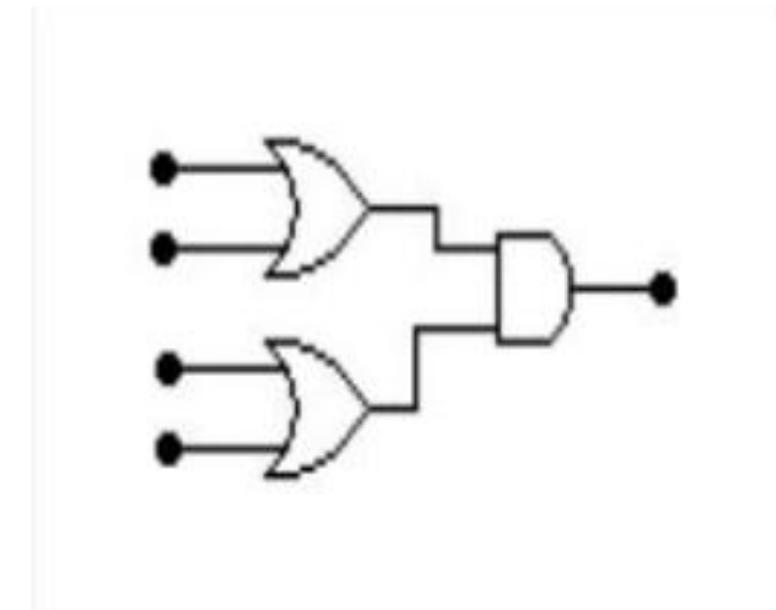
contradict the principles of addition for real numbers, but not for Boolean numbers.

Combinational Logic Circuits

- Can be expressed as sum of products (AND-OR pattern of gates) or products of sum (OR-AND pattern of gates)



AND-OR pattern



OR-AND pattern

Combinational Logic Circuits

Sum of Products

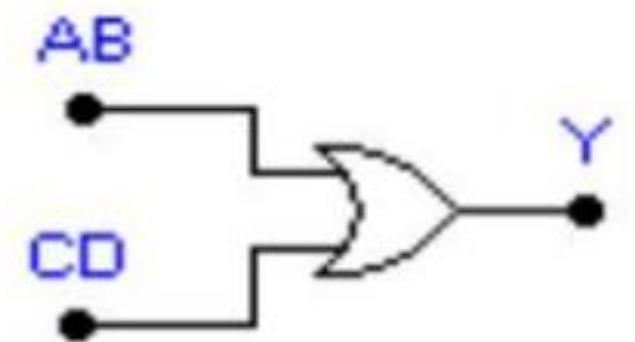
- Also known as MINTERM
- For a Boolean expression:

$$AB + CD = Y$$

the sum of products logic circuit (AND-OR pattern)

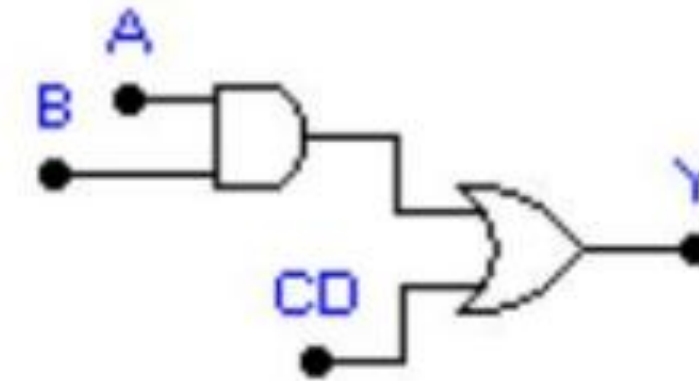
can be drawn as follows:

step 1. OR AB with CD

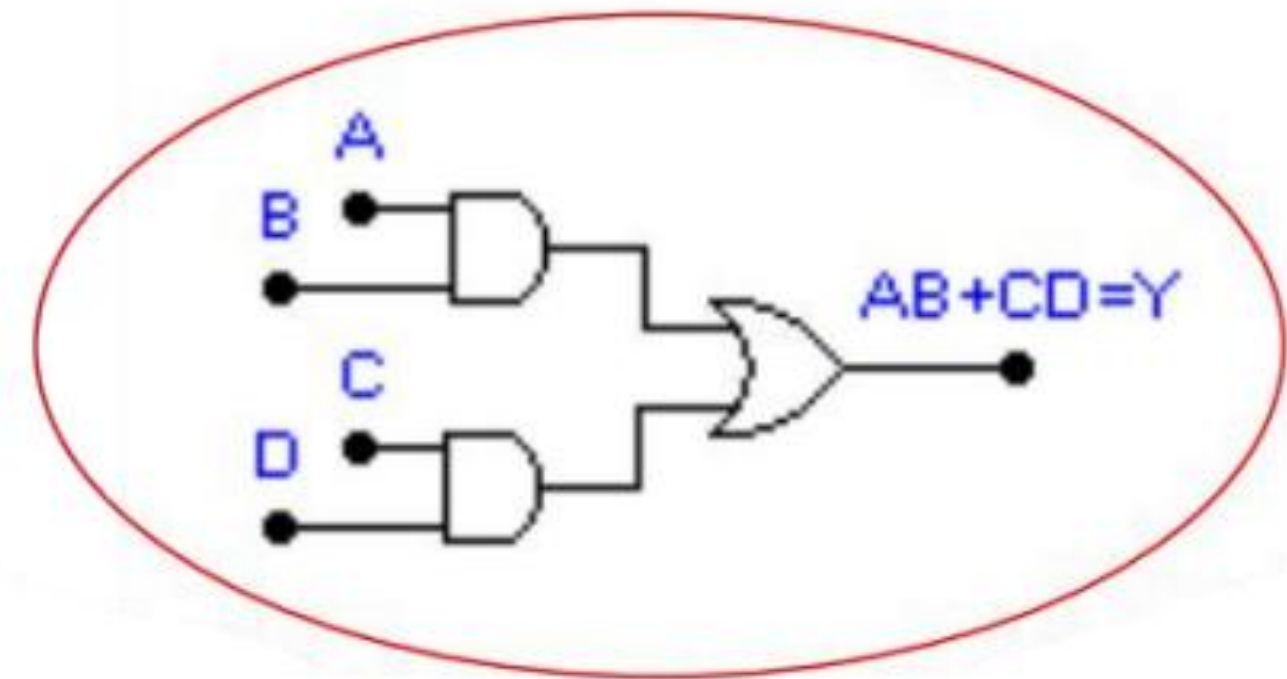


Combinational Logic Circuits

step 2. Add the AND gate for AB



step 3. Add the AND gate for CD



Combinational Logic Circuits

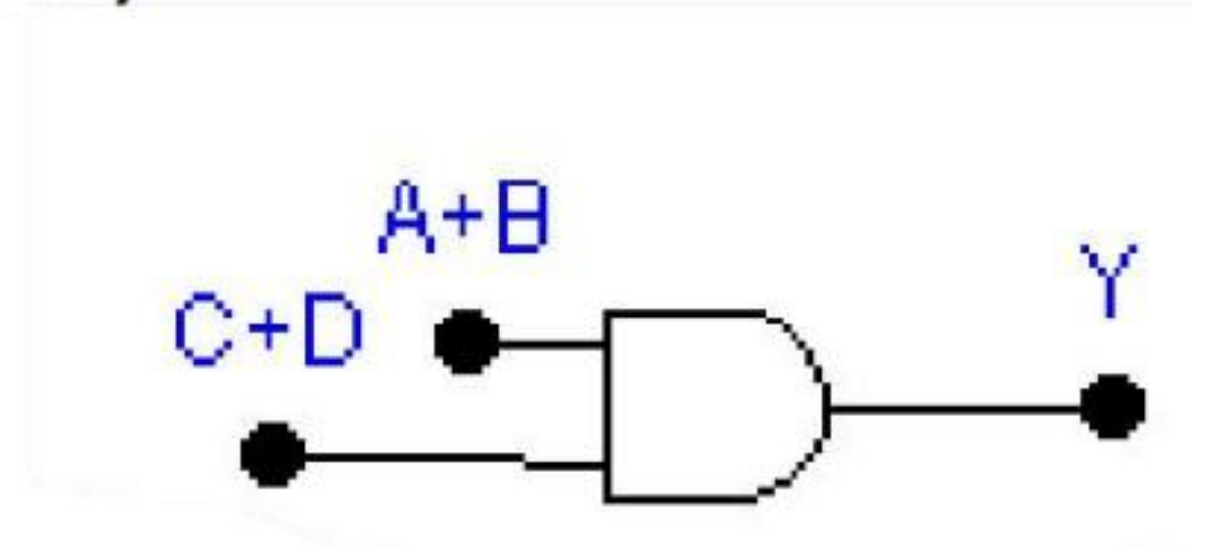
- Also known as MAXTERM
- For a Boolean expression:

$$(A+B)(C+D) = Y$$

the product of sums logic circuit (OR-AND pattern)

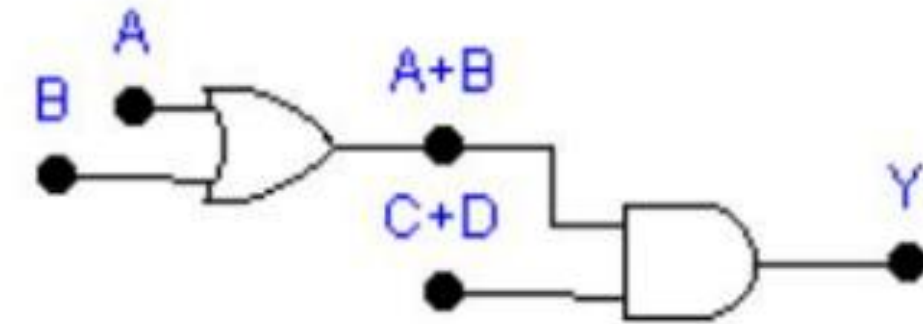
can be drawn as follows:

step 1. AND $A+B$ with $C+D$

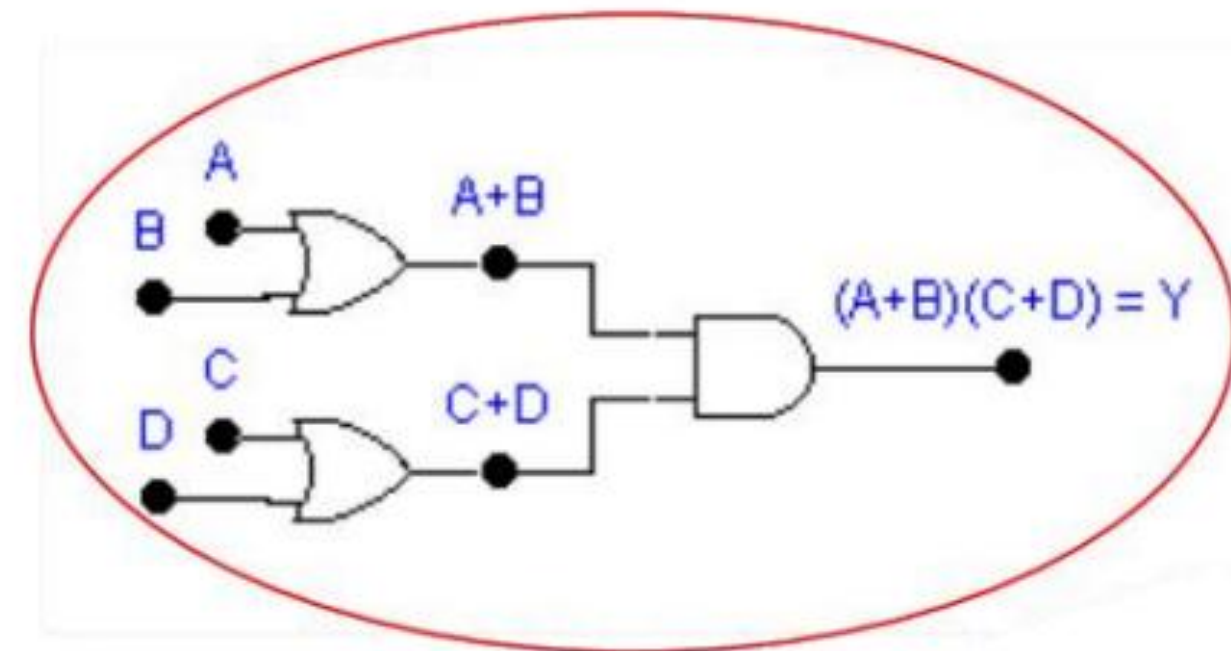


Combinational Logic Circuits

step 2. Add the OR gate for $A+B$



step 3. Add the OR gate for $C+D$



BOOLEAN ALGEBRA

- There are three laws of Boolean Algebra that are the same as ordinary algebra.
 - The Commutative Law
 - The Associative Law
 - The Distributive Law

Commutative Law

- ADDITION

$$A + B = B + A$$

(In terms of the result, the order in which variables are ORed makes no difference.)

- MULTIPLICATION

$$AB = BA$$

(In terms of the result, the order in which variables are ANDed makes no difference.)

Associative Law

- ADDITION

$$A + (B + C) = (A + B) + C$$

(When ORing more than two variables, the result is the same regardless of the grouping of the variables.)

- MULTIPLICATION

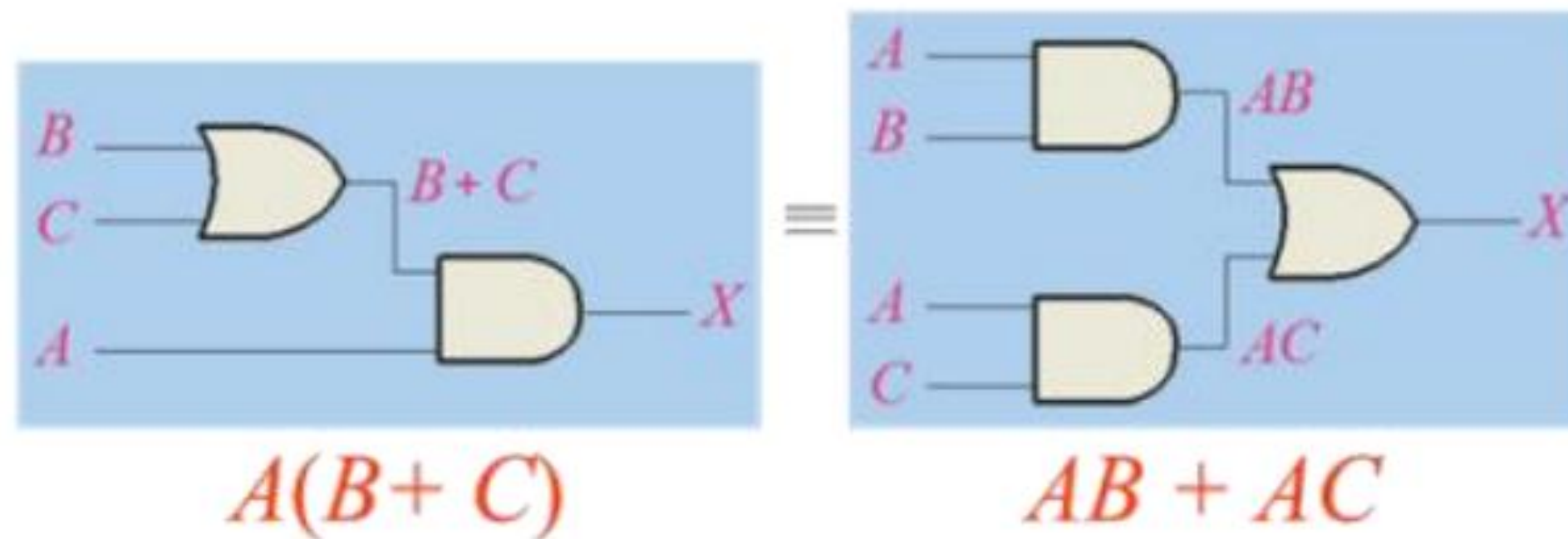
$$A(BC) = (AB)C$$

(When ANDing more than two variables, the result is the same regardless of the grouping of the variables.)

Distributive Law

- The distributive law is the factoring law.
- A common variable can be factored from an expression just as in ordinary algebra.

$$A(B + C) = AB + AC$$



BASIC RULES OF BOOLEAN ALGEBRA

Reference	Rule or Law
Rule 1	$A + 0 = A$
Rule 2	$A + 1 = 1$
Rule 3	$A \cdot 0 = 0$
Rule 4	$A \cdot 1 = A$
Rule 5	$A + A = A$
Rule 6	$A + A' = 1$
Rule 7	$A \cdot A = A$
Rule 8	$A \cdot A' = 0$
Rule 9	$(A')' = A$

BASIC RULES OF BOOLEAN ALGEBRA

Simplify the following expressions and draw the logic diagram:

1. $(A+B)(A+C)$

2. $AB + A(B + C) + B(B + C)$

BASIC RULES OF BOOLEAN ALGEBRA

Simplify the following expressions and draw the logic diagram:

1. $(A+B)(A+C)$

$$= AA + AC + AB + BC$$

(distributive law)

$$= A + AC + AB + BC$$

(rule #5)

$$= A(1 + C + B) + BC$$

(distributive law)

$$= A(1) + BC$$

(rule #4)

$$= A + BC$$

(rule #2)

BASIC RULES OF BOOLEAN ALGEBRA

Simplify the following expressions and draw the logic diagram:

$$\begin{aligned} 1. \quad & AB + A(B + C) + B(B + C) \\ &= AB + AB + AC + BB + BC && \text{(Distributive law)} \\ &= (AB + AB) + AC + BB + BC && \text{(Associative law)} \\ &= AB + AC + BB + BC && \text{(rule \#6)} \\ &= AB + AC + B + BC && \text{(rule \#5)} \\ &= B(A + 1 + C) + AC && \text{(Distributive law)} \\ &= B(1) + AC && \text{(rule \#4)} \\ &= B + AC && \text{(rule \#2)} \end{aligned}$$

MICROCONTROLLER PROGRAMMING

Microcontroller programming involves writing software that enables microcontrollers to control and execute tasks within an embedded system. It is a computer system that is integrated into a device or product to perform specific functions.

A **microcontroller** is a compact integrated circuit designed to perform specific tasks within an embedded system.

When a microprocessor is packaged on a printed circuit board (PCB) with other components, such as interface and memory chips, the resulting assembly is referred to as a **microcomputer or single-board computer**.

HISTORY OF MICROCONTROLLER

1971

Intel 4004 was released, first microprocessor

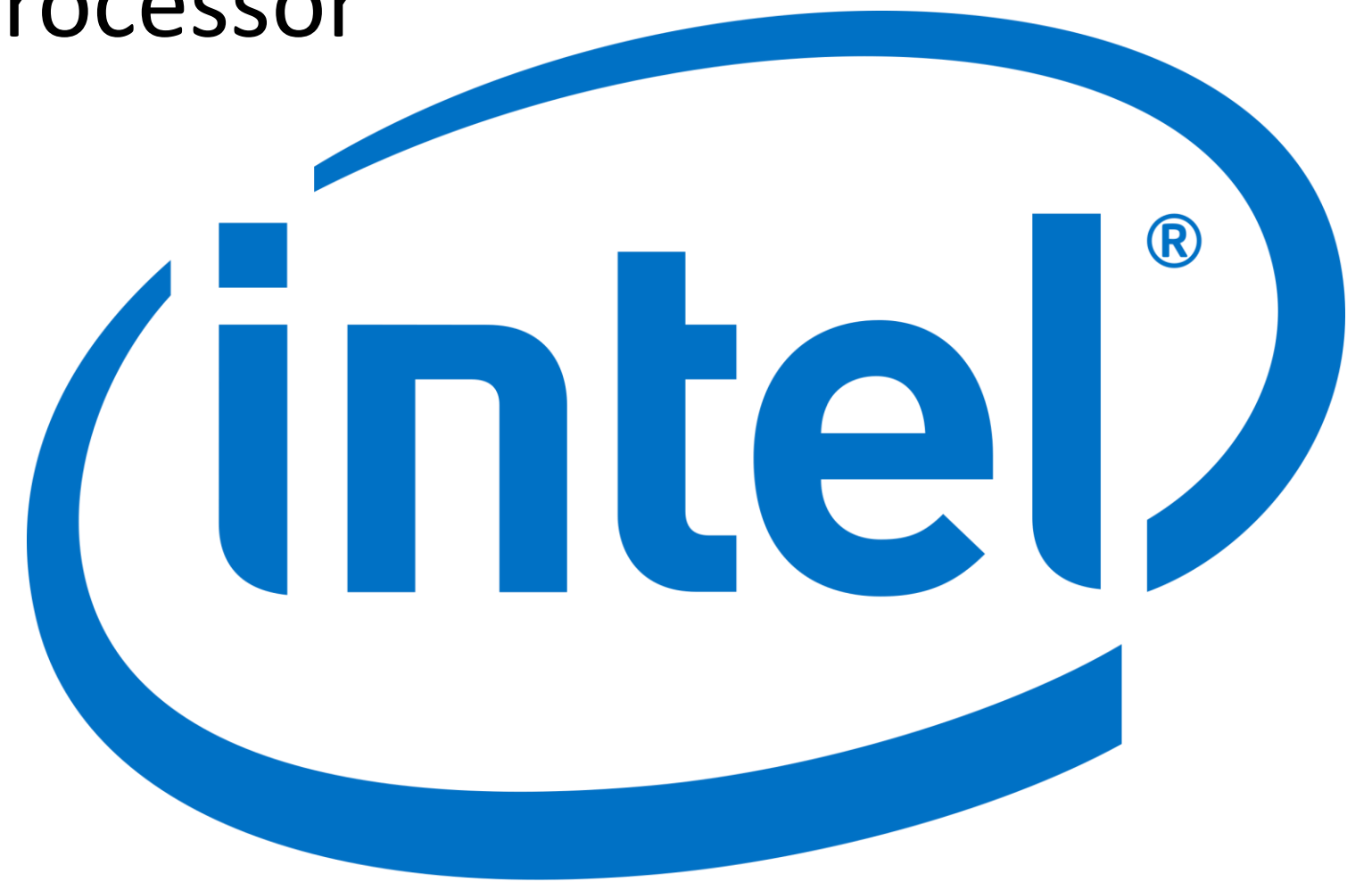
1972

8008 was released and then 8080

1975

Intel 8048 was released

Intel President, Luke J. Valenter



HISTORY OF MICROCONTROLLER

1976

Zilog, founded by Federico Faggin

Z80

Used in Gameboy

1979

Introduced Z8



HISTORY OF MICROCONTROLLER

1993

Microchip

PIC16x84 was released

First with EEPROM program memory



1993

Atmel

First microcontroller with Flash memory



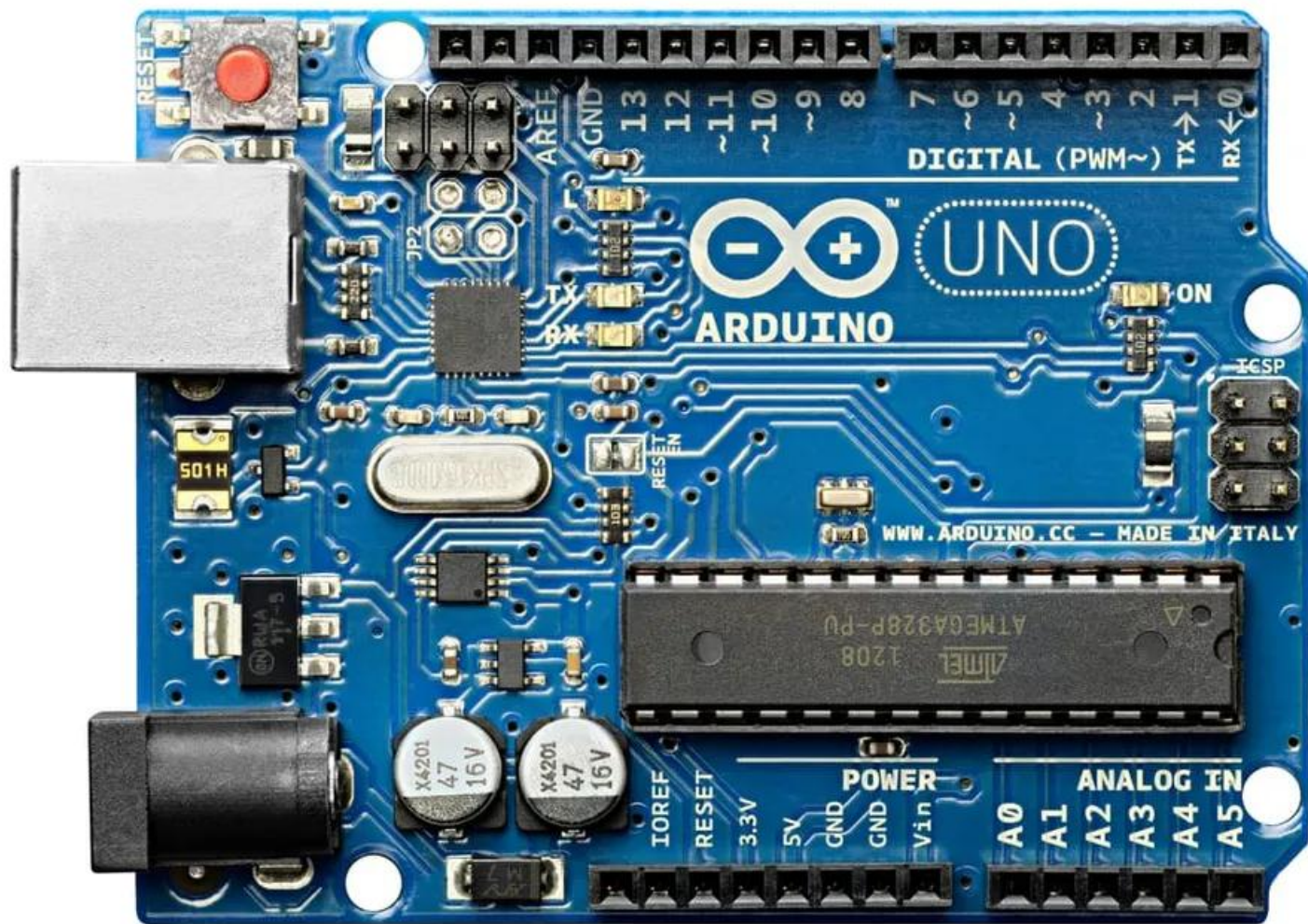
At present

Companies follow what Atmel has started

MICROCONTROLLER PROGRAMMING

Popular Microcontroller Platforms

Arduino -user-friendly interface and extensive community support.



Advantages

- its open-source nature, extensive library support, IDE environment and ease of use.

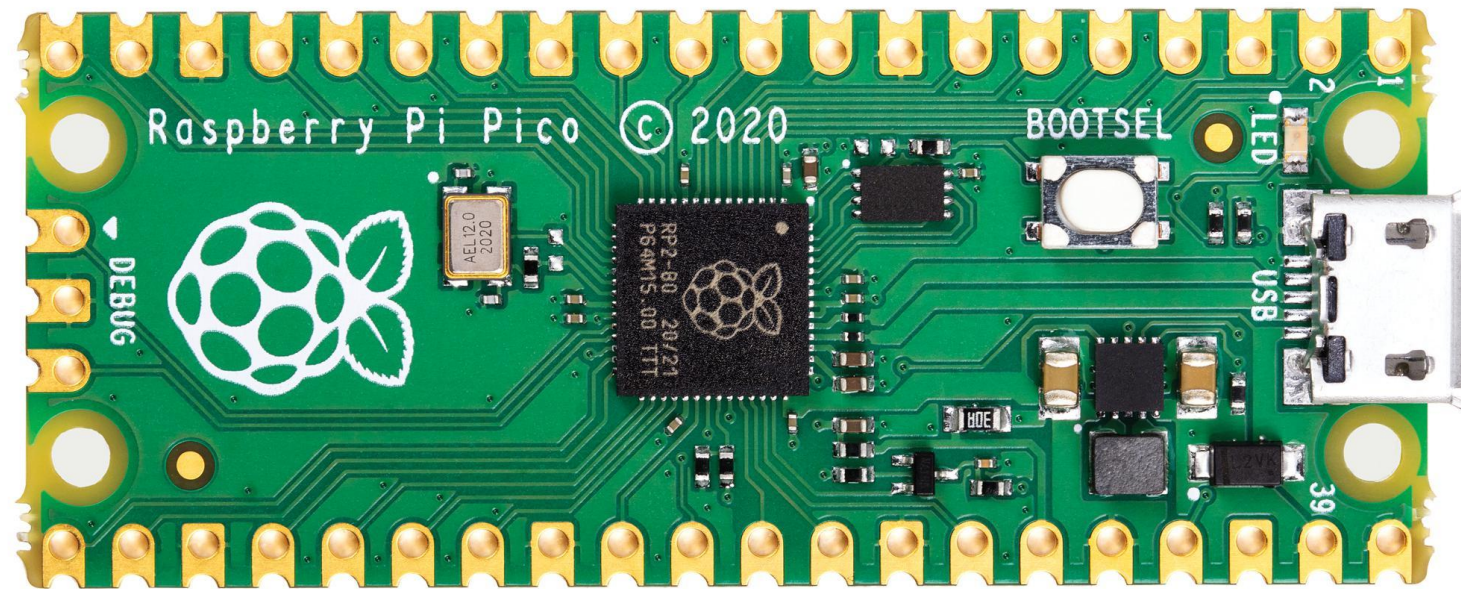
Disadvantages

- may not be suitable for all applications due to their limited processing power and memory storage.

MICROCONTROLLER PROGRAMMING

Popular Microcontroller Platforms

Raspberry Pi -known for its versatility and powerful processing capabilities.



Advantages

- powerful processing capabilities and large community support
- has its own operating system known as “Raspbian”

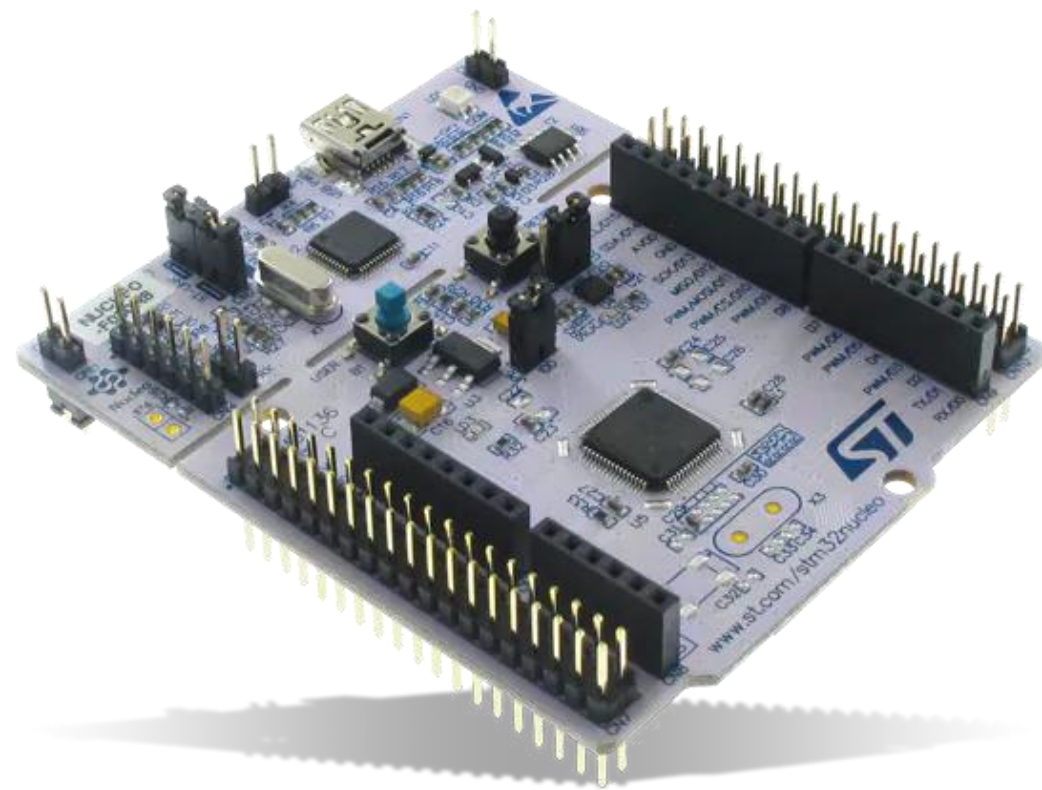
Disadvantages

- complexity, storage limitation and higher power consumption compared to other microcontroller platforms.

MICROCONTROLLER PROGRAMMING

Popular Microcontroller Platforms

STM32 - known for its high-performance microcontrollers, which offer a range of features and capabilities suitable for various applications like printers and complex circuit boards.



Advantages

- high processing power, energy efficiency, and extensive peripheral support

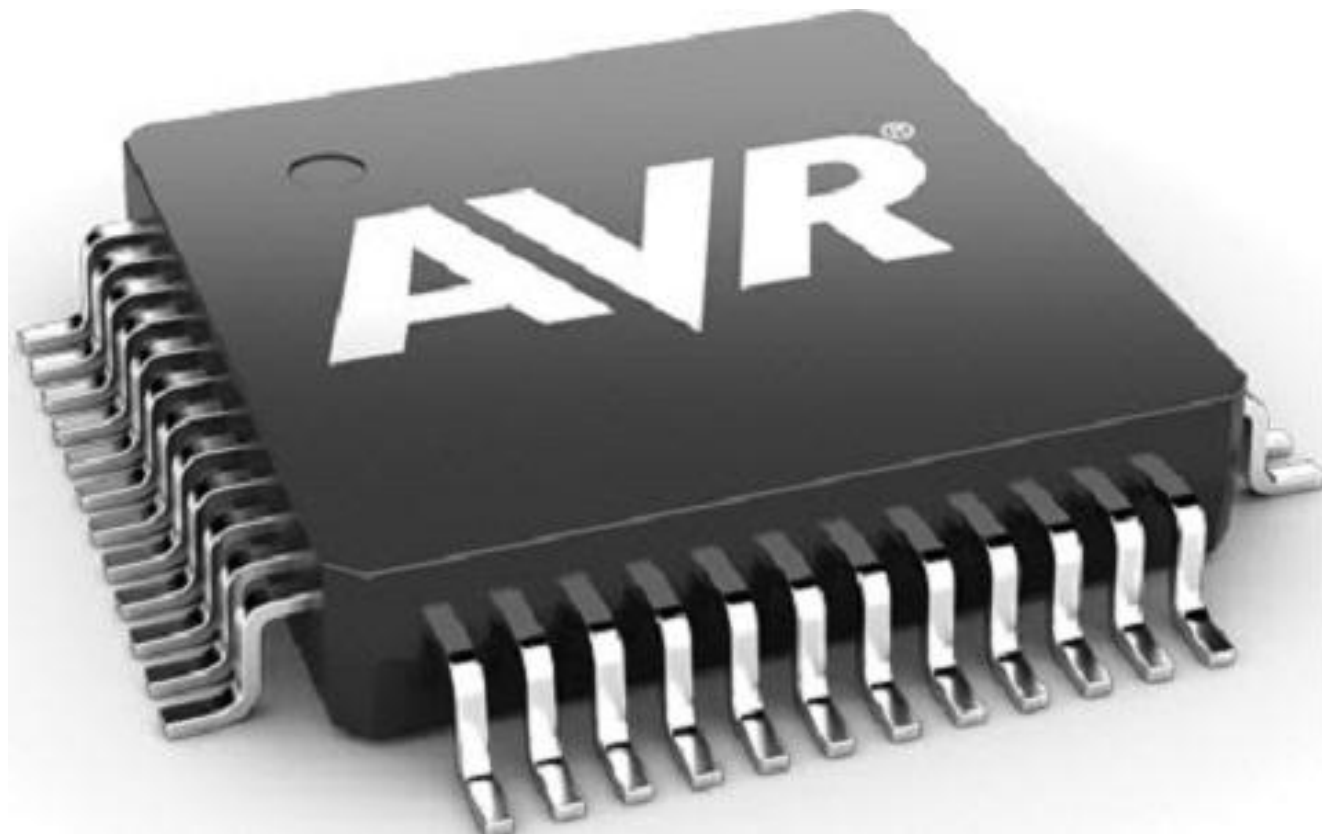
Disadvantages

- steeper learning curve compared to Arduino and Raspberry Pi
- more challenging for beginners

MICROCONTROLLER PROGRAMMING

Popular Microcontroller Platforms

AVR - best choice for hobbyists and engineers to execute firmware and hardware projects with ease



Advantages

- include 8-16 bit timers, 8-bit and 16-bit timers, internal oscillators, 10-12 bit ADC and DAC, etc

Disadvantages

- offer low-power operation with a reduced clock speed ranging from 0 to 20 MHz

MICROCONTROLLER PROGRAMMING

Microcontroller Programming Languages

C/C++

Advantages: high performance, extensive library support, and compatibility with various MCU platforms.

Disadvantages: code is too long for the compiler to execute

Python

Advantages: error-handling capabilities, simplicity, ease of use, high readability, low maintenance and extensive library support

Disadvantages: not as performant as C/C++; may not be suitable for applications that particularly require low-level high performance.

MICROCONTROLLER PROGRAMMING

Terminologies

The **microprocessor**, also called the central processing unit (CPU), is where the primary computation and system control operations occur.

The **arithmetic logic unit** (ALU) within the CPU executes mathematical functions on data structured as binary words. A word is an ordered set of bits, usually 8, 16, 32, or 64-bits long.

The **bus** is a set of shared communication lines that serves as the central nervous system of the computer. Data, address, and control signals are shared by all system components via the bus.

MICROCONTROLLER PROGRAMMING

Terminologies

Different types of memory

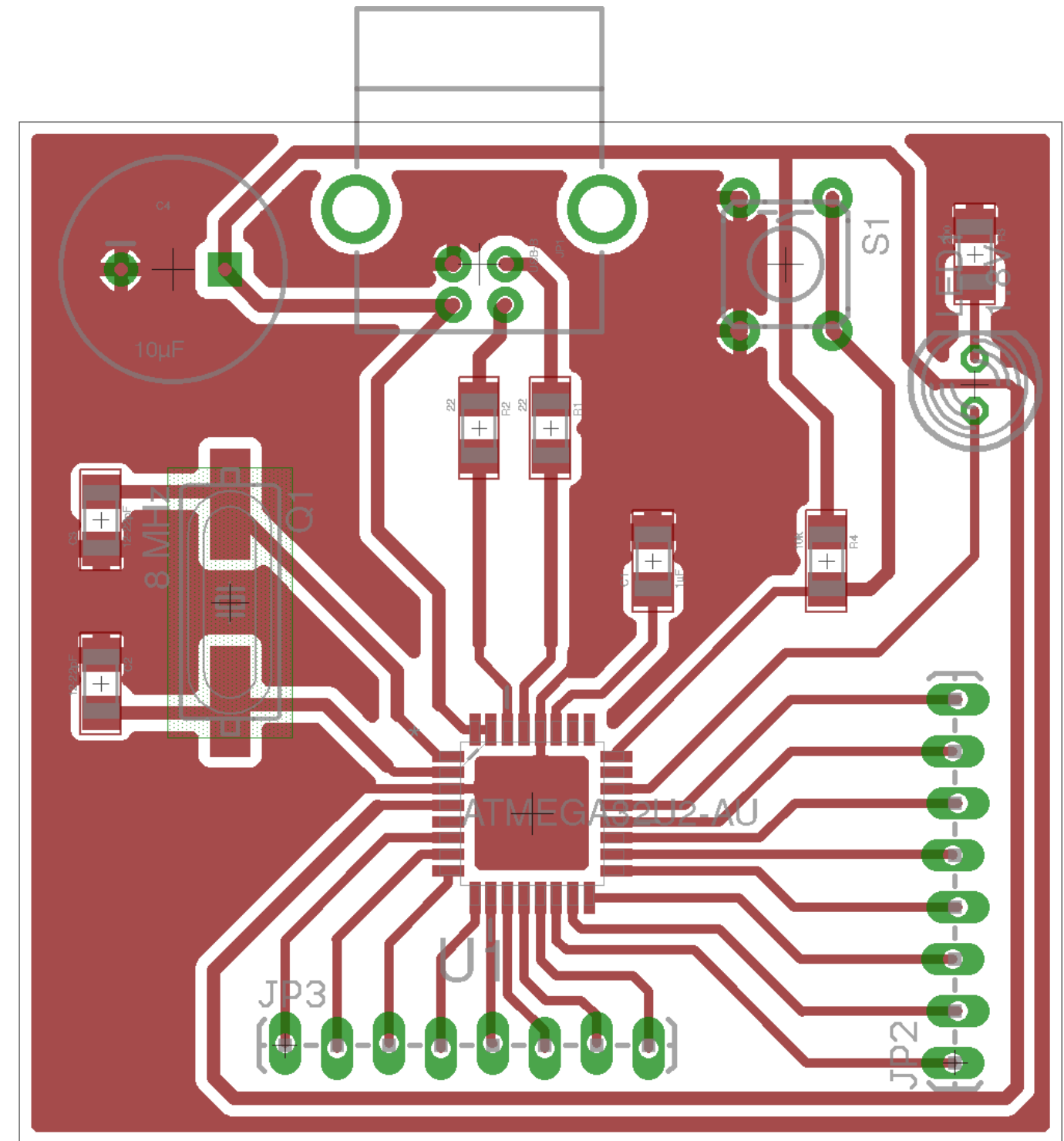
ROM (Read-Only Memory) is used for permanent storage of data that the CPU can read, but the CPU cannot write data to ROM. ROM does not require a power supply to retain its data and therefore is called non-volatile memory.

RAM (Random Access Memory) can be read from or written to at any time, provided power is maintained. The data in RAM is considered volatile because it is lost when power is removed.

MICROCONTROLLER PROGRAMMING

Caveats with using Microcontrollers

- Though the microcontroller is cheap, initial cost is **HIGH**
- Microcontroller **programmers** needed
- **Creating the PCB** for the circuit is necessary



ARDUINO

Hernando Barragan

Colombian student at Interaction Design Institute Ivrea (Ivrea, Italy)

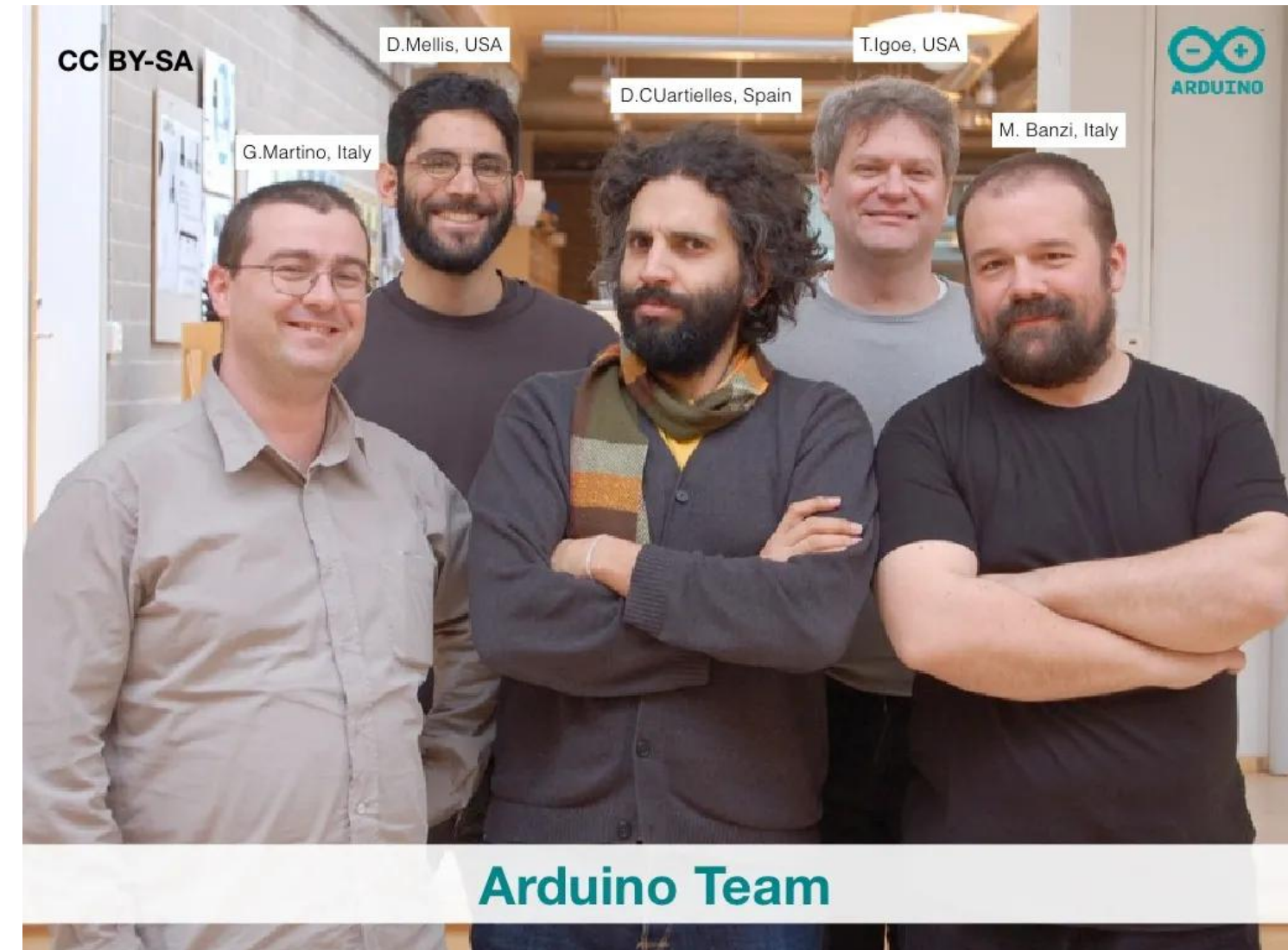
Thesis: “Arduino – The Revolution of Open Hardware” (2005)

Developed the programming platform: **Wiring**

ARDUINO

Developed a new prototype board, later coined as: “Arduino”

Massimo Banzi
David Cuartielles
Tom Igoe
Gianluca Martino
David Mellis

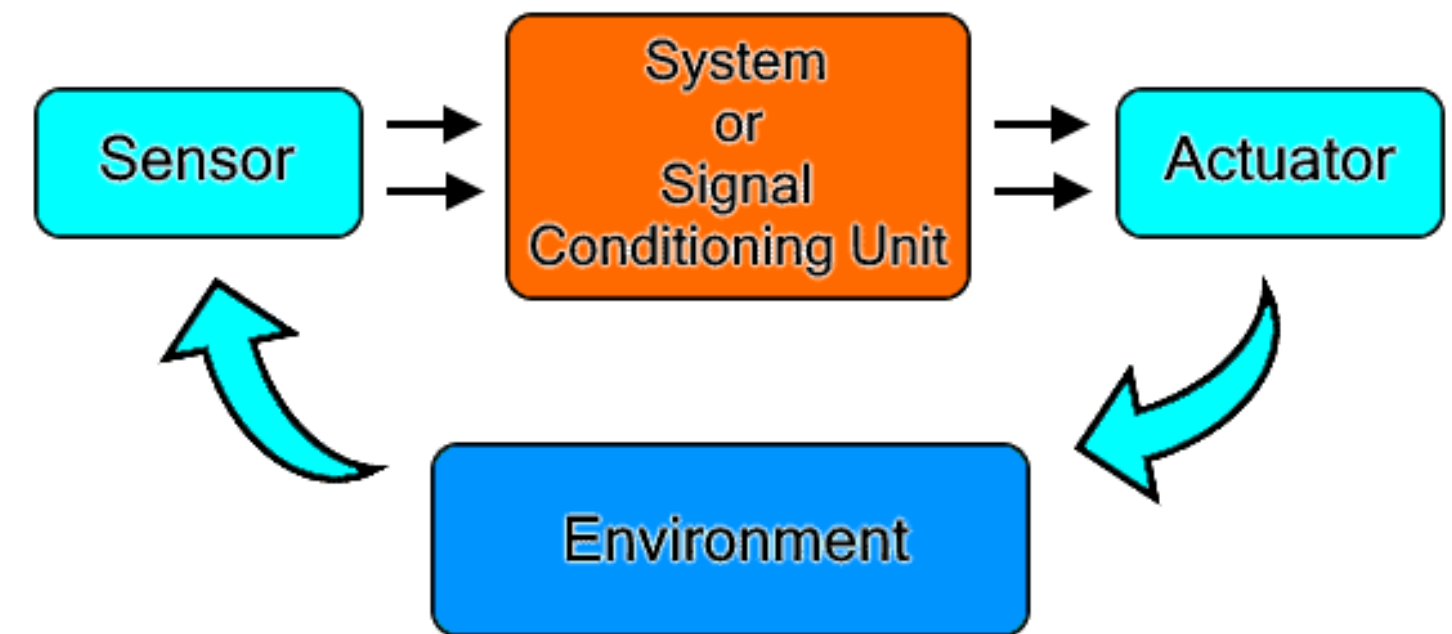


Named after a local bar in Ivrea

"Arduino" is an Italian masculine first name, meaning "strong friend"

ARDUINO

- “Interactive Device” pattern
- Circuit that is able to sense the environment using components called “sensors” and processing the information through “behaviour” implemented as software
- Interact back with the world using “actuators”



ARDUINO

Arduino Derivatives (Clones)

- Boarduino
- Funduino
- Gizduino
- Aceduino
- Tinkduino

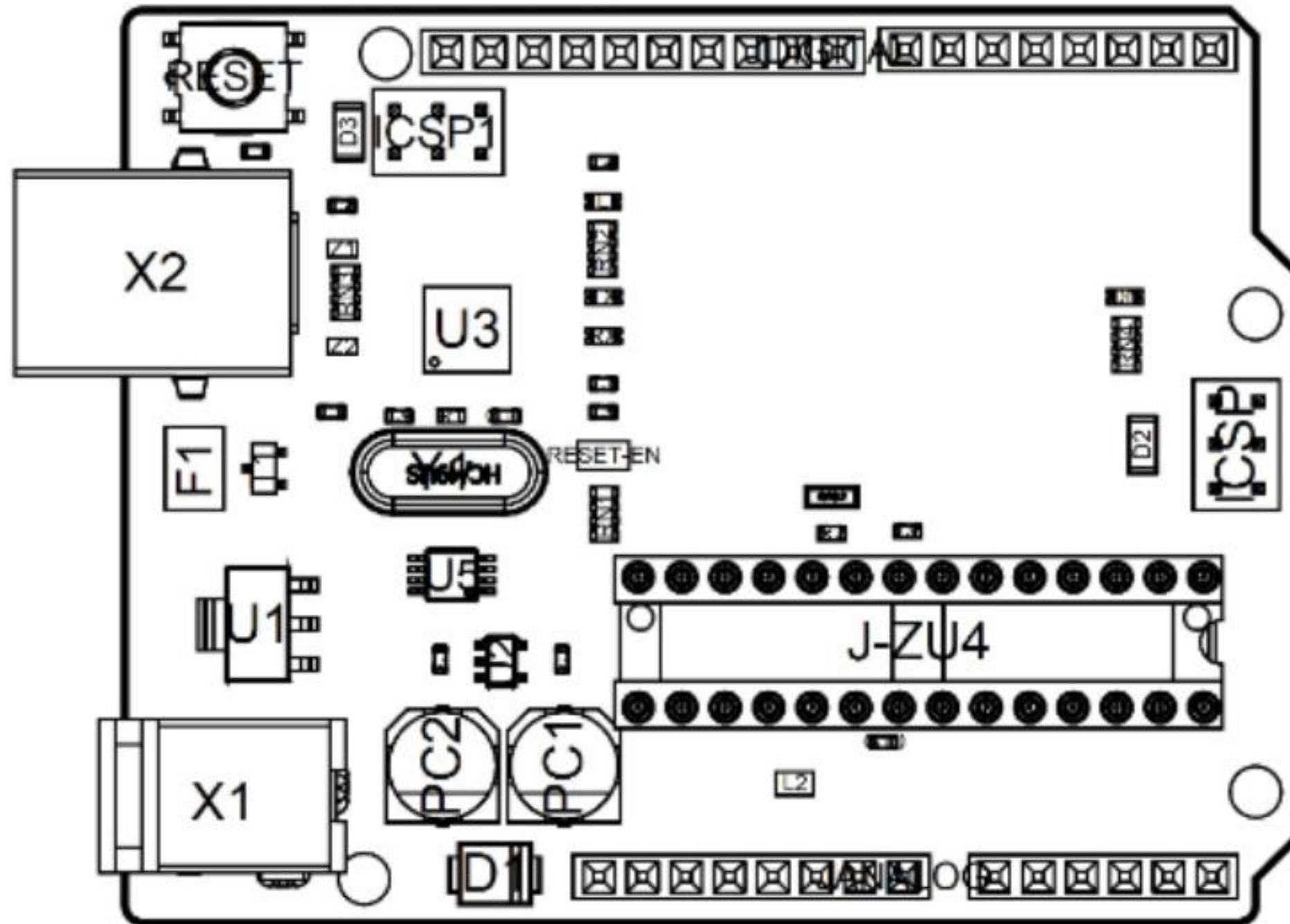
ARDUINO

ADVANTAGES

OPEN SOURCE

- Files are free!
 - Go to Arduino website and get what you need
 - Instructions
 - Schematics
- Large community worldwide which supports open-source
 - Sharing of technology and projects
- DIY : everyone has an opportunity to experience building their ideas into reality

ARDUINO



Board topology

Ref.	Description
X1	Power jack 2.1x5.5mm
X2	USB B Connector
PC1	EEE-1EA470WP 25V SMD Capacitor
PC2	EEE-1EA470WP 25V SMD Capacitor
D1	CGRA4007-G Rectifier
J-ZU4	ATMEGA328P Module
Y1	ECS-160-20-4X-DU Oscillator

Ref.	Description
U1	SPX1117M3-L-5 Regulator
U3	ATMEGA16U2 Module
U5	LMV358LIST-A.9 IC
F1	Chip Capacitor, High Density
ICSP	Pin header connector (through hole 6)
ICSP1	Pin header connector (through hole 6)

ARDUINO



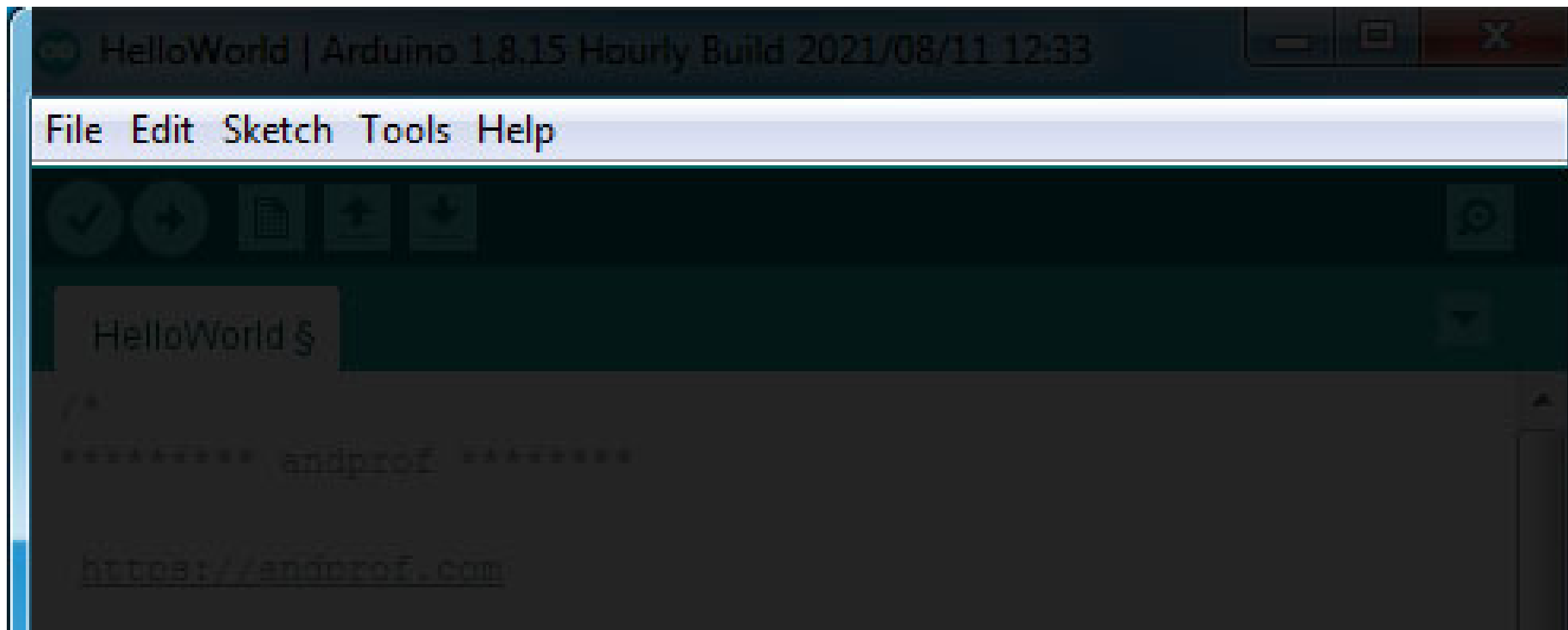
The arduino software (IDE) is an open source software, which is used to program the Arduino boards, and is an integrated development environment, developed by arduino.cc. It allows writing and uploading of code to the Arduino boards and it consists of many libraries and a set of examples of mini projects.

ARDUINO



Arduino software interface

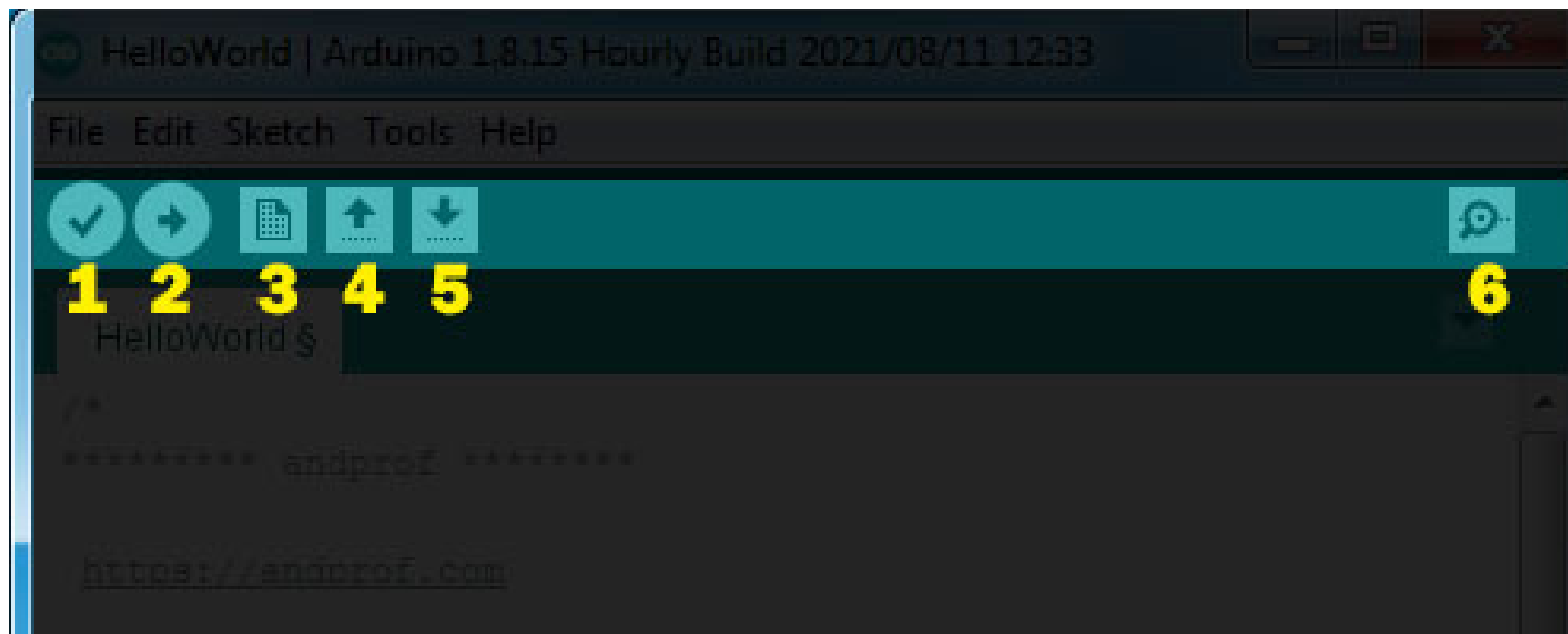
ARDUINO



Menus section

ARDUINO

Toolbar section



1. **Verify**: this button is used to review the code, or make sure that is free from mistakes.
2. **Upload**: this button is used to upload the code on the Arduino board.

3. **New**: this button is used to create new project, or sketch
4. **Open**: is used when you want to open the sketch from sketchbook.
5. **Save**: save the current sketch in the sketchbook.
6. **Serial monitor**: shows the data which have been sent from Arduino.

DATA ACQUISITION SYSTEMS

Data Acquisition

- Process of collecting and storing data using a microprocessor or computer
- Provides more compact method storage of data
- Can result in greater data accuracy, allows use of the data in a real time control system
- Enables data processing long after the events have occurred
- Transforms analog data into coded digital values for it to be stored in a digital circuit or microprocessor

SIGNAL CONDITIONING

- Manipulation of a signal in a way that prepares it for the next stage of processing

Filtering

- unwanted vibration frequencies are removed from the signal being monitored. This ensures that only important information is being sent to the machine operator.

SIGNAL CONDITIONING

TYPES OF FILTERS

Low-pass filters remove vibration frequencies above a particular frequency, while passing the frequencies below the filter.

High-pass filters remove vibration frequencies below a certain frequency, while passing the higher frequencies above the filter.

A **band pass filter** takes advantage of both the high and low pass filters as described above and passes the band in between these two frequencies.

SIGNAL CONDITIONING

Amplification

- carried out by either increasing the resolution of the input signal or by increasing the signal to noise ratio

Isolation

- used in hazardous areas to separate the sensor in the hazardous area from the monitoring equipment that is usually in a safe area.

SIGNAL CONDITIONING

Excitation

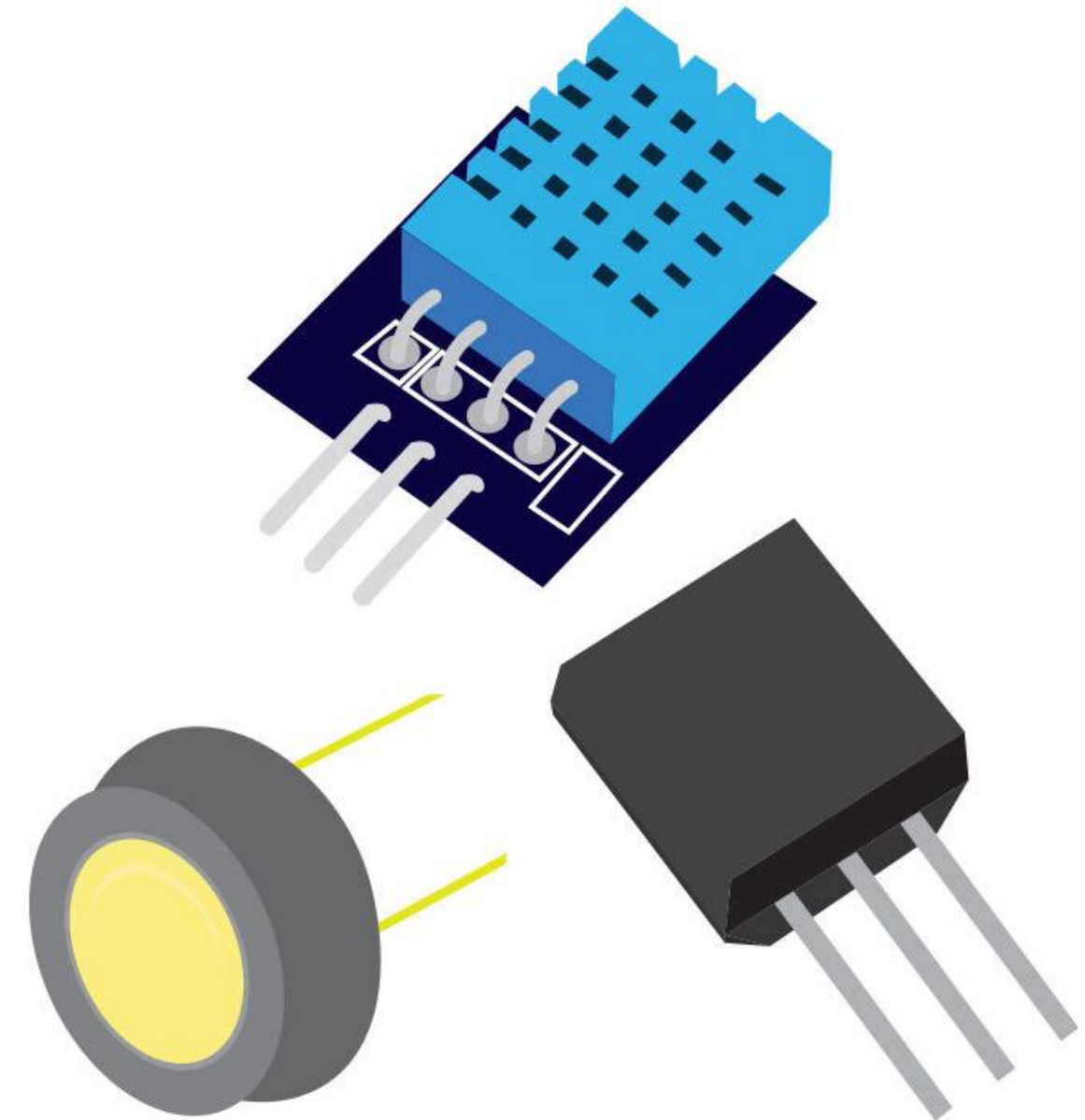
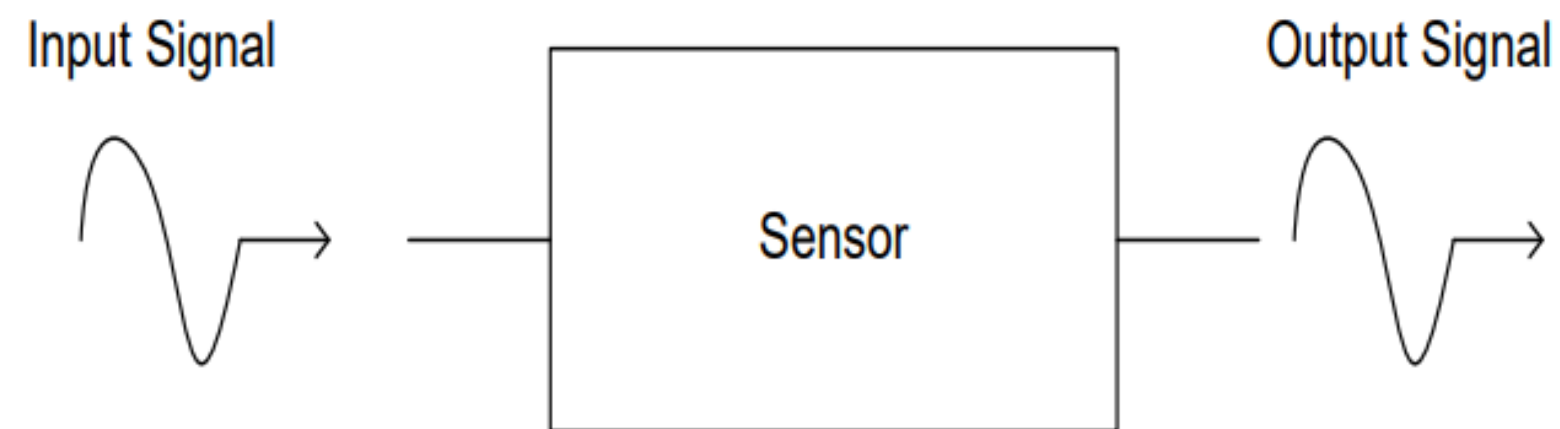
- process of delivering power to the sensor.

Linearization

- software linearizes the voltage levels from transducers, so the voltages can be scaled to measure physical phenomena.

SENSORS

A **sensor** is a device, module, machine, or subsystem that detects events or changes in its environment and sends the information to other electronics, frequently a computer processor.

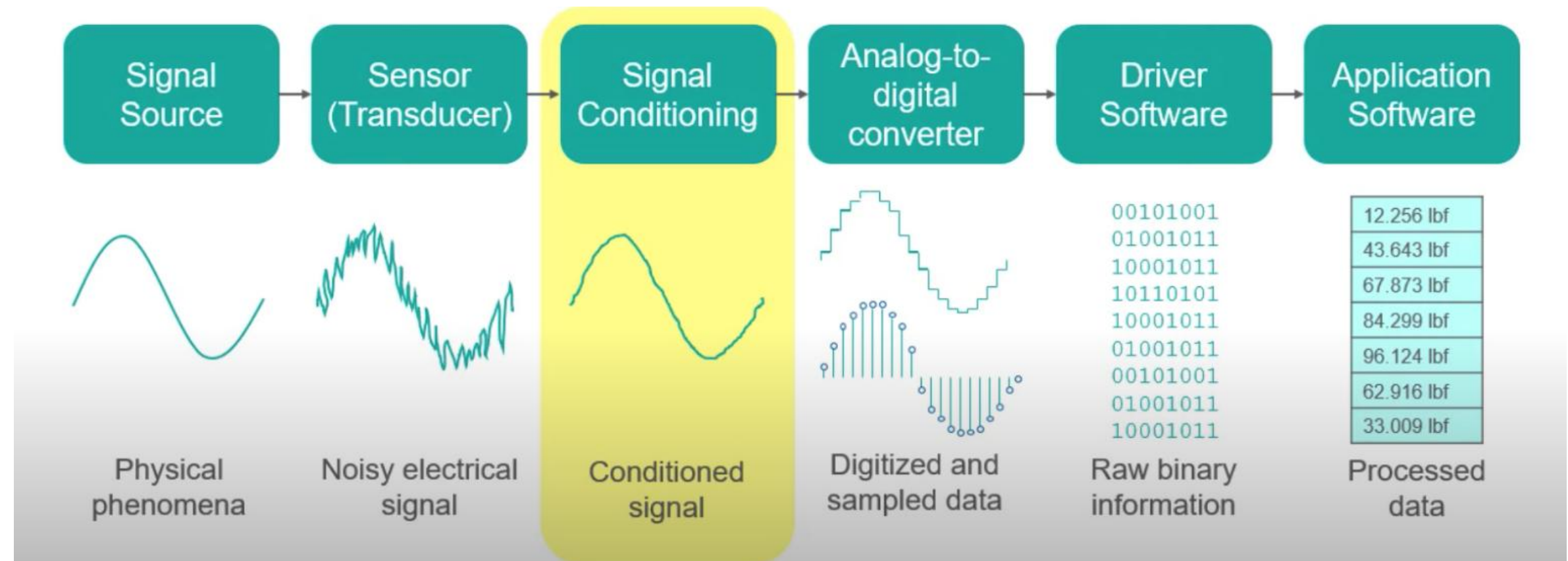


SENSORS

HOW DO SENSORS WORK?

Sensors work on the principle of transduction, which involves converting one form of energy to another.

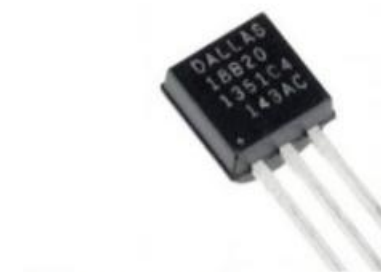
1. Sensing
2. Transduction
3. Signal Conditioning
4. Output



SENSORS

Sensing

detect and measure some physical effect, providing this information to the control system.



Temperature Sensor



Humidity Sensor



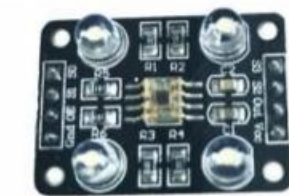
Proximity Sensor



Light Sensor



Metal Sensor



Color Sensor

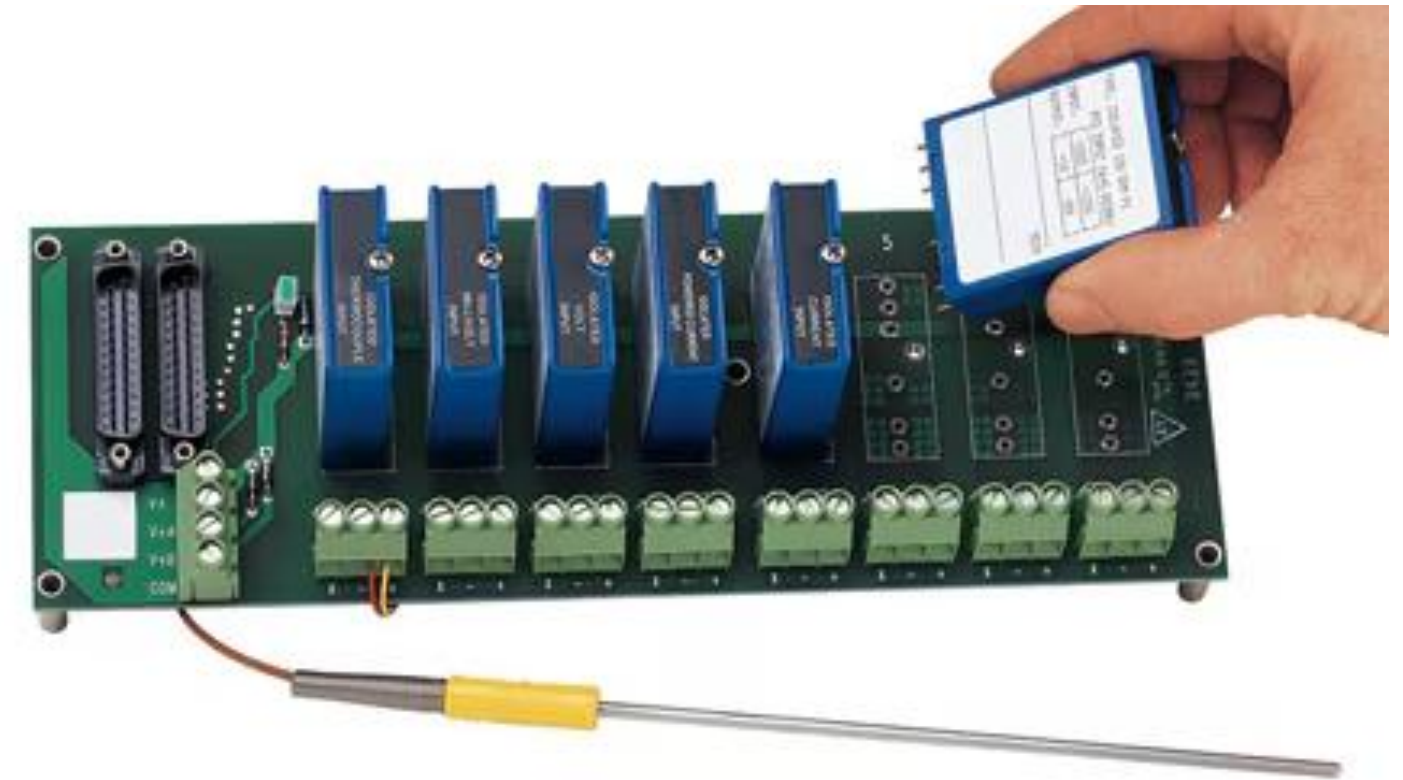
Transduction

The sensed signal is converted into another measurable form, suitable for processing and analysis.

SENSORS

Signal Conditioning

The electric signal is amplified, processed or filtered to enhance its accuracy, stability and compatibility with data acquisition systems

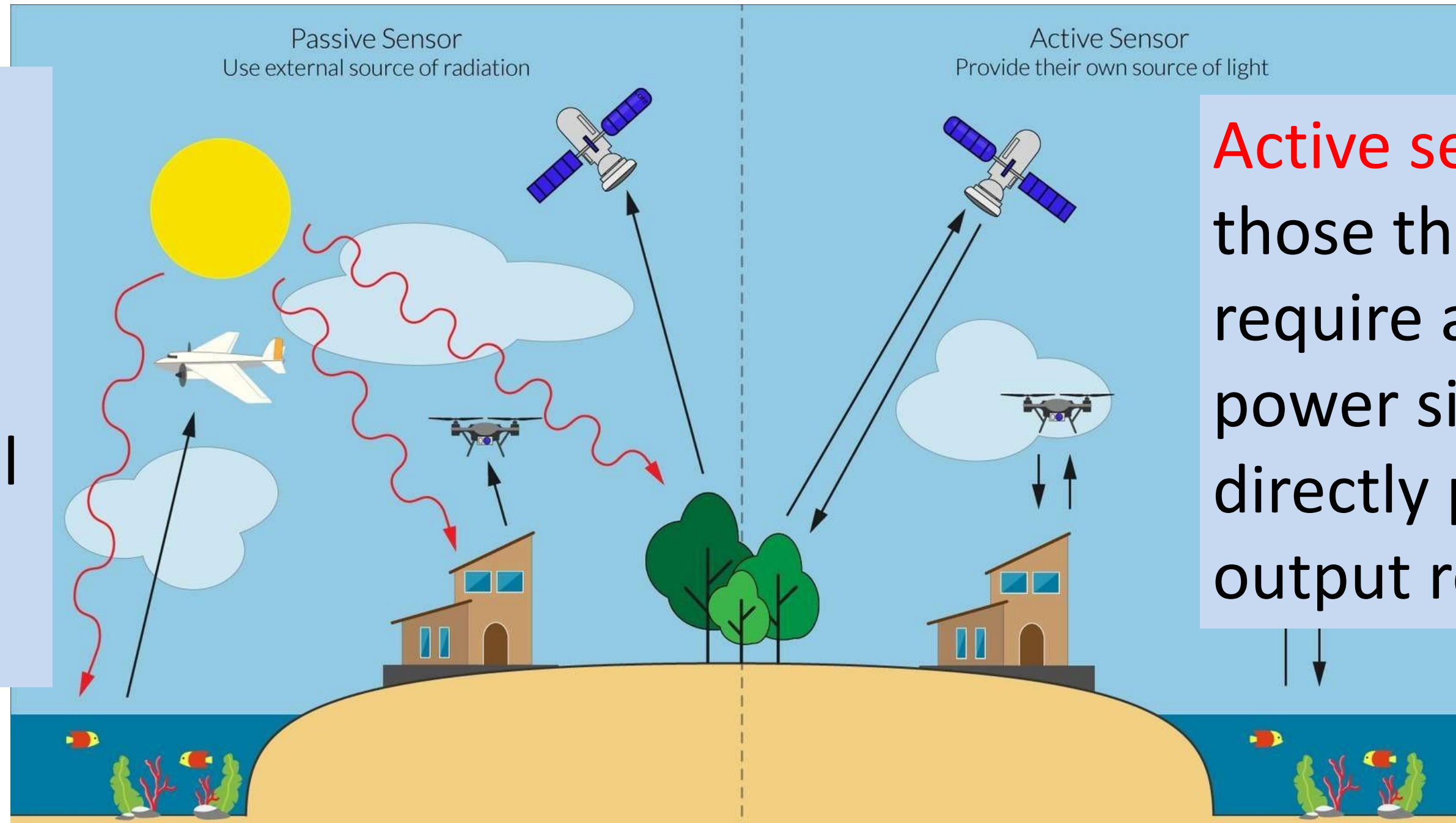


Output

The processed signal is used for further analysis, visualization and for control purposes.

Classification of Sensors

Passive sensors are sensors needing an external excitation signal or a power signal.



Active sensors are those that do not require any external power signal and directly produce an output response.

Examples of Sensors

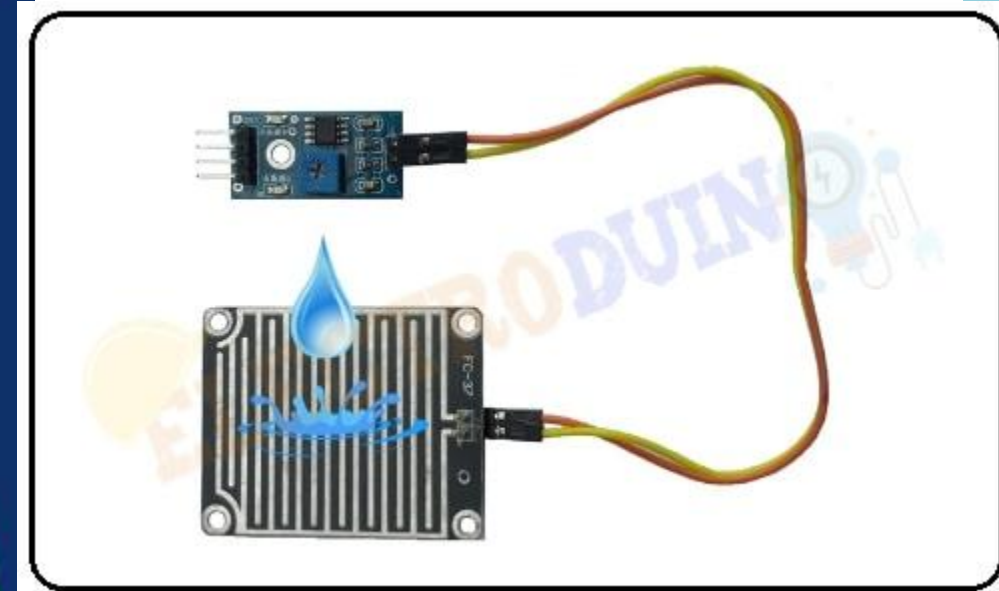
1. Environmental Sensors
2. Crops Sensors
3. Soil Sensors
4. Livestock Sensors



Examples of Sensors

Environmental Sensors

1. Temperature Sensors
2. Light Sensors
3. Humidity Sensors
4. Rainfall Sensors
5. Wind Sensors



Examples of Sensors

Soil Sensors

1. Moisture Sensors
2. pH Sensors
3. Nutrient Sensors



Examples of Sensors

Crop Sensors

1. Chlorophyll Sensors
2. Canopy Sensors
3. Leaf area meters
4. Fruit maturity Sensors



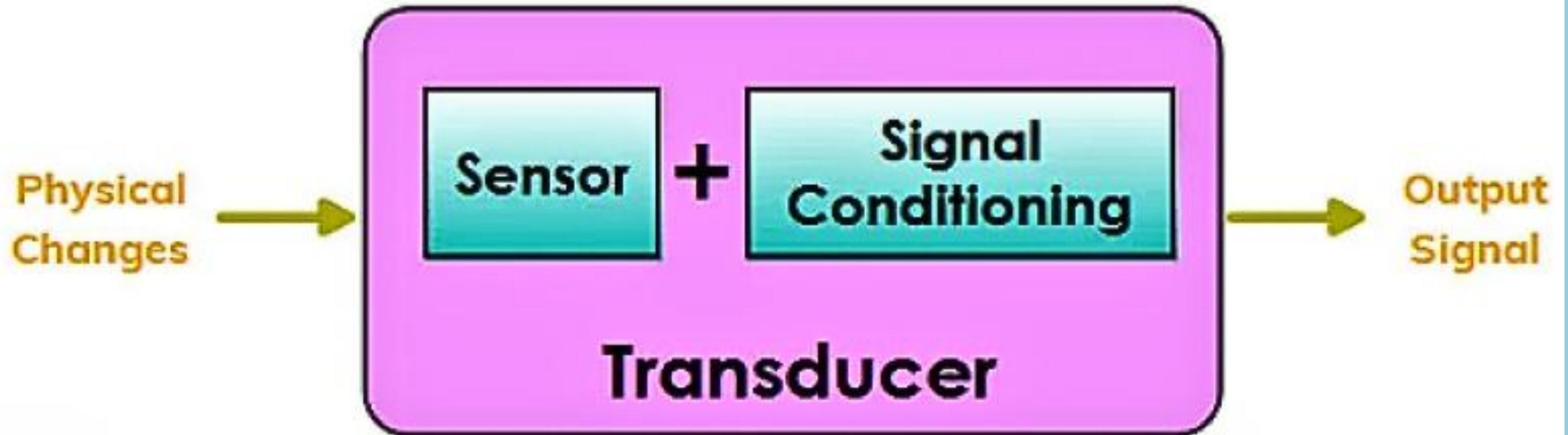
Examples of Sensors

Livestock Sensors

1. Temperature Sensors
2. Heart rate monitor
3. Activity sensors
4. pH Sensors



SENSORS



01



WIRELESS SENSOR NETWORKS (WSN)

- Integration of sensors with wireless communication for remote monitoring
- Improved data collection efficiency and scalability
- Reduced installation and maintenance costs

02



INTERNET OF THINGS (IOT)

- Interconnectivity of sensors, devices, and systems for seamless data sharing
- Real-time data analytics and decision-making capabilities
- Enhanced automation and control in agricultural and biosystems

SENSOR TECHNOLOGY ADVANCEMENTS

01



REMOTE SENSING AND IMAGING

- Use of drones, satellites, and aerial platforms for data acquisition
- Multispectral and hyperspectral imaging for detailed crop analysis
- Thermal imaging for stress detection and water management

02



DATA ANALYTICS AND ARTIFICIAL INTELLIGENCE (AI)

- Big data analytics to extract valuable insights from sensor data
- Machine learning algorithms for predictive modeling and decision support
- Automated data-driven recommendations for improved system performance

SENSOR TECHNOLOGY ADVANCEMENTS

Actuators

- Devices used to produce motion or action
- Produce physical changes such as linear and angular displacement
- Modulate the rate and power associated with changes

Some of the most important actuators:

1. solenoids
2. electric motors
3. hydraulic cylinders
4. rotary motors
5. pneumatic cylinders

APPLICATIONS OF AUTOMATION SYSTEMS FOR AGRICULTURAL SAFETY

1. Automated machinery and equipment
2. Robotic systems for hazardous tasks and repetitive activities
3. Autonomous vehicles for field operations and transportation
4. Precision agriculture technologies for precise and controlled operations
5. Unmanned aerial vehicles (UAVs) for monitoring and data collection

BENEFITS OF AUTOMATION SYSTEMS FOR AGRICULTURAL SAFETY

1. Reduction in human exposure to hazardous environments
2. Increased accuracy and precision in farm operations
3. Enhanced worker safety and well-being

CONTROLLED ENVIRONMENT AGRICULTURE

Controlled-Environment Agriculture (CEA) is the practice of growing plants indoors using engineering, plant science, and computer-managed greenhouse control technologies to optimize plant growth, quality, and production efficiency.



CONTROLLED ENVIRONMENT AGRICULTURE

Types of Controlled Environment Agriculture Practices

Greenhouses: Greenhouses provide a controlled environment with regulated temperature, humidity, and light. They protect crops from extreme weather and pests, extending the growing season and improving yield quality.

Hydroponics: This method involves growing plants in nutrient-rich water solutions without soil. Hydroponic systems can be designed to use less water and allow for faster plant growth.

CONTROLLED ENVIRONMENT AGRICULTURE

Types of Controlled Environment Agriculture Practices

Aeroponics: In aeroponics, plants are grown in an air or mist environment without the use of soil or an aggregate medium. Nutrients are delivered via a mist, which can improve nutrient uptake and efficiency.

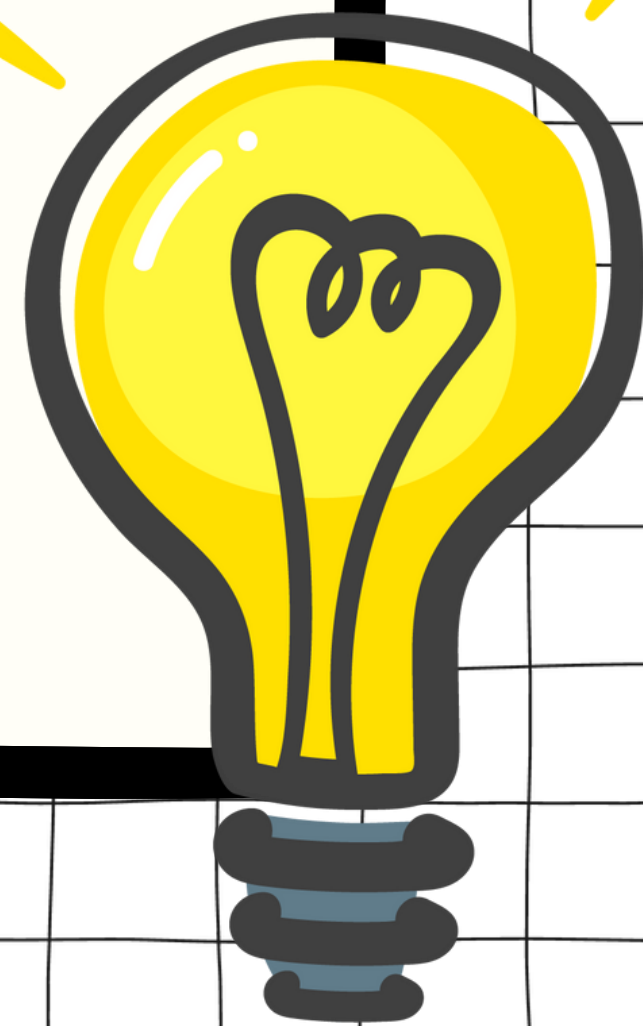
Aquaponics: Combining aquaculture (raising fish) with hydroponics, aquaponics creates a symbiotic environment where fish waste provides an organic nutrient source for the plants, and the plants help filter and clean the water for the fish.

CONTROLLED ENVIRONMENT AGRICULTURE

Types of Controlled Environment Agriculture Practices

Vertical Farming: Vertical farming stacks layers of crops in a controlled environment, making it possible to grow more food in less space. This approach is particularly effective in urban areas where land is limited.

Metrology



The word comes from Greek μέτρον (*metron*) which means "measure" and "λόγος" (*logos*), amongst others meaning "speech, oration, discourse, quote, study, calculation, reason".

In Ancient Greek the term μετρολογία (metrologia) meant "theory of ratios".

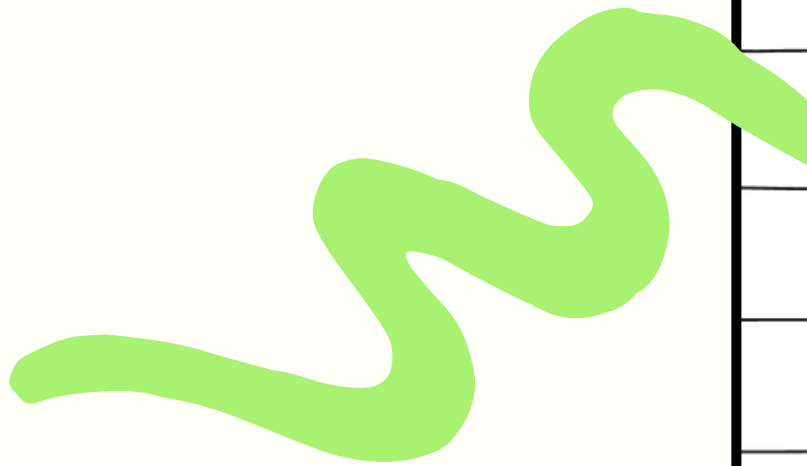
Metrology covers three main activities:

1. The definition of internationally accepted units of measurement
2. The realization of units of measurement
3. The establishment of traceability chains



Metrology is the scientific study of measurement. It is a field that establishes a common understanding of units, methods, and systems of measurement, ensuring accuracy and consistency across various applications.

Metrology is defined by the International Bureau of Weights and Measures (BIPM) as "the science of measurement, embracing both experimental and theoretical determinations at any level of uncertainty in any field of science and technology."



3 Subfields of Metrology

1. **Scientific metrology** is the basis of all subfields, and concerns the development of new measurement methods, the realization of measurement standards, and the transfer of these standards to users.
2. **Applied metrology** – measurement science is developed toward manufacturing and other processes, ensuring the suitability of measurement instruments, their calibration, and quality control.
3. **Legal metrology** concerns regulatory requirements of well-established measurements and measuring instruments for the protection of consumers and fair trade.

...

The _____ is the national metrology institute of the Philippines tasked by law to be responsible for establishing and maintaining the national measurement standards for physical quantities such as mass, temperature, pressure, voltage, resistance, luminous intensity and time interval and their dissemination to Filipino users.

- a. National Metrology Laboratory
- b. National Institute of Standards and Technology
- c. Philippine National Standards for Metrology
- d. None of the above

Answer: a. National Metrology Laboratory

The **National Metrology Laboratory (NML)** is the national metrology institute of the Philippines tasked by law to be responsible for establishing and maintaining the national measurement standards for physical quantities such as mass, temperature, pressure, voltage, resistance, luminous intensity and time interval and their dissemination to Filipino users.

The NML operates under the Industrial Technology Development Institute (ITDI), one of the agencies of the Department of Science and Technology (DOST).

DOST is the executive department of the Philippine Government responsible for the coordination of science and technology-related projects in the Philippines and for the formulation of policies and projects in the fields of science and technology in support of national development.

[Republic Act No. 9236]

"The National Metrology Act of 2003"

AN ACT ESTABLISHING A NATIONAL MEASUREMENT INFRASTRUCTURE SYSTEM (NMIS) FOR STANDARDS AND MEASUREMENTS, AND FOR OTHER PURPOSES

It is hereby declared the policy of the State to facilitate the development of scientific and technical knowledge and progress in the national economy by encouraging the standardization and modernization of units and standards of measurements to adapt to the needs of the times, thereby complying with international standards and protecting the health, interest and safety of every consumer and his environment from the harmful effects of inaccurate or false measurements

[Republic Act No. 9236]

"The National Metrology Act of 2003"

Definition of Terms

- a) Accredited Laboratory** – refers to a laboratory that has been evaluated and has complied with the requirements of the International Standardization Organization (ISO) 17025 Standard "General Requirements for the Competence of Testing and Calibration Laboratories," and accredited by the national accrediting body.
- b) Board-Authorized Units (BAU)** – are units prescribed by the International System of Units (SI) and other units of measurement authorized by the Board.
- c) Calibration** – set of operations establishing under specified condition, relationship between values indicated by a measuring instrument or measuring system, or values represented by material measure, and its corresponding known values of measure.

[Republic Act No. 9236]

"The National Metrology Act of 2003"

Definition of Terms

- d) Measurement** – set of operations having the object of determining the value of a quantity.
- e) Measurement Standard** – a measuring instrument or system intended to define, realize, conserve or reproduce a unit or one or more known values of a quantity in order to transmit them to the measuring instrument by comparison.
- f) Metrology** – the field of science concerned with weights and measures which includes all aspects of measurement in whatever level of accuracy and in any field of science and technology.

[Republic Act No. 9236]

"The National Metrology Act of 2003"

Definition of Terms

- g) Metrology Controls** – refer to type approval, verification, calibration and other processes and means of checking the accuracy and reliability of measurement standards and measuring equipment.
- h) Regulated Areas of Application** – refers to field of endeavors or areas which are critical to citizens as they affect health, safety, welfare, physical conditions, trade and commerce, legal transactions, environment and other areas as may be determined by the Board.
- i) Type Approval on Measuring Equipment** – the approved evaluation of conformity based on one or more specimens of a product.

[Republic Act No. 9236]

"The National Metrology Act of 2003"

Definition of Terms

- j) **Primary Standard** – a standard which has highest metrological quality in a specified field.
- k) **Secondary Standard** – one which value is fixed by comparison with primary standard.
- l) **Traceability** – the property of a result of a measurement relating to appropriate standards, generally national or international through an unbroken chain of comparison.
- m) **Verification** – a confirmation by examination of evidence that the measuring equipment fulfills specified requirements.

...

The **measurement** is the set of operations whose purpose is to determine an experimental value – taking into account the environment – called *measurand* by comparing it directly or indirectly to a standard which is the material representation of the unit in which the measurand will be expressed.

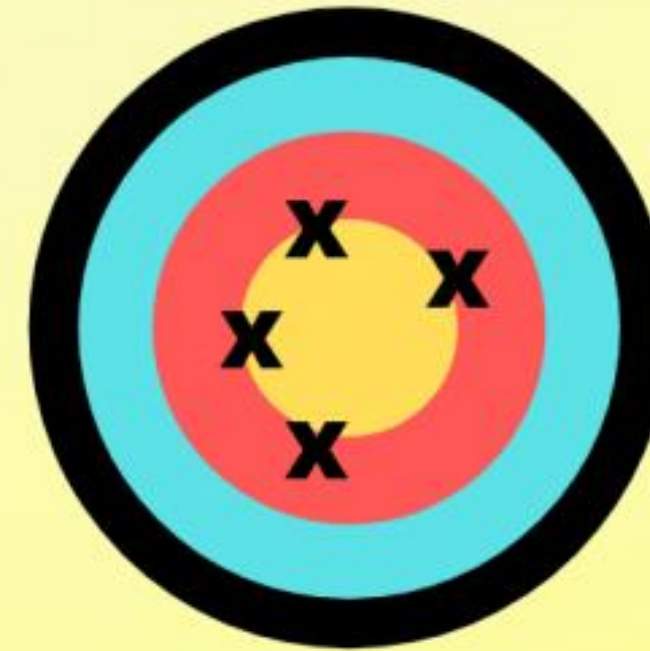
The **measurand**, or measured quantity, is the physical quantity in metrology, subject to measurement

Accuracy and Precision

**Accurate
Precise**



**Accurate
Not Precise**



**Not Accurate
Precise**



**Not Accurate
Not Precise**



Accuracy is a measure of how closely your experimental measurements agree with known values. The closer your measurements are to the known value, the more accurate the measurement.



Precision is a measure of how close your experimental measurements agree with each other. The closer each measurement is to the other measurements, the more precise your measurement.



Which Is More Important? Accuracy or Precision?

Accuracy is more important when trying to hit a target. You either hit your target, or you don't. It does no good to be precise if you miss all your shots. You can improve accuracy in future measurements by factoring in a correction factor. If your shots are all going to the left, alter your aim to the right. Accuracy is something you can fix in future measurements.

Precision is more important in calculations. When using a measured value in a calculation, you can only be as precise as your least precise measurement. This is the main idea behind the topic of significant figures in calculations. You can improve precision by using a better measuring tool and/or improving your skill at using the tool.

...

Errors inevitably occur in any measuring system:

- the measuring system is never perfect since it is generally more or less sensitive to the environment (temperature, pressure, humidity...).**
- the standards used to calibrate the instrumentation are only an imperfect materialization of the definition of the unit that they are responsible for representing**
- the wrong definition of the quantity is itself a source of error.**

Measurement Errors

All measurements have errors associated with them.
These errors fall into two categories:

- a. Systematic errors (determinate errors) – affect the **accuracy** of the measurement, or the closeness of the result to the “true” value;
- b. Random errors (indeterminate errors) – affect the **precision** of the measurements, or the closeness of the results to each other.

Systematic errors are characterized by measurements that are consistently higher than the true value (positive systematic bias) or lower than the true value (negative systematic bias).

a. Sampling errors arise when a collected sample does not represent the environment being sampled.

b. Method Errors result from not being able to accurately measure the sample's concentration or mass. This usually occurs because another substance present in the sample matrix is interfering in the measurement process.

c. Instrument Errors involve using the proper glassware/equipment for analyses and knowing the errors involved in its use;

d. Personal Errors are also known as human errors and arise negligence like over-titrating an endpoint or using a dirty cuvette for analysis.

Ways to minimize systematic errors

1. Collect representative samples
2. Analyze standard reference materials whose concentrations are known
3. Analyze blank samples
4. Use multiple methods to make measurements
5. Participate in a round-robin study with other labs
6. Vary the sample size

Random errors are always present during any measurement and can't be controlled for. They arise from minor differences in sampling between different people, from how people read a buret or interpret an endpoint color, and even from electrical noise.

The contribution of random error to a measurement can be estimated through error propagation methods, by measuring the standard deviation of a series of measurements that show a normal distribution, and by monitoring the behavior of a piece of equipment over time.

In general, the magnitude of systematic errors greatly exceeds those of random errors, so that is where analysts typically focus their attention to minimize overall error in a process.

The **International System of Units (SI)** defines *seven quantities and their units of measure* as a basic set from which all other SI units are derived.

Table 1. The seven SI base quantities & their units

quantity	base unit	unit symbol
mass	kilogram	kg
length	metre	m
time, duration	second	s
electric current	ampere	A
thermodynamic temperature	kelvin	K
amount of substance	mole	mol
luminous intensity	candela	cd

The **International System of Units (SI)** defines *seven quantities and their units of measure* as a basic set from which all other SI units are derived.

Table 2. Commonly used SI derived units

quantity	derived unit	unit symbol
area	square metre	m ²
volume	cubic metre	m ³
speed	metre per second	m/s
density	kilogram per cubic metre	kg/m ³
frequency	hertz	Hz
force	newton	N
pressure	pascal	Pa
power	watt	W
temperature	degree Celsius	°C

The **International System of Units (SI)** defines *seven quantities and their units of measure* as a basic set from which all other SI units are derived.

Table 3. Some of the non-SI units accepted by SI

quantity	non-SI unit	unit symbol
area	hectare	ha
time	minute	min
	hour	h
	day	d
volume	litre	L or l
mass	tonne	t
pressure	bar	bar
plane angle	degree	°
	minute	'
	second	"

...

Which metrology tool is most appropriate for determining the moisture content of harvested rice before storage?

- a. Thermometer**
- b. Moisture meter**
- c. Weighing scale**
- d. pH meter**

Answer: b. Moisture meter

...

A digital caliper is primarily used in agriculture to:

- a. Weigh fertilizers before application**
- b. Measure the thickness or diameter of small machine parts**
- c. Monitor ambient greenhouse temperature**
- d. Measure the flow rate of irrigation water**

Answer: b. Measure the thickness or diameter of small machine parts

...

Which of the following instruments is essential in ensuring safe and effective composting?

- a. pH meter**
- b. GPS**
- c. Thermometer**
- d. Caliper**

Answer: c. Thermometer